PROBABILITY

It is a concept of mathematics which measures the degree of certainty or uncertainty of the occurrence of events.

- If any event can happen in m ways and fails in n ways and each of the $(m+n)$ ways are equally likely to occur, then probability of the happening of the events is defined as the ratio, $\frac{m}{m+n}$ and that of its failing as $\frac{n}{m+n}$.
- If probability of the happening is denoted by p and not happening by q, then $p + q = 1$.
- If event is certain to happen, its probability is unity.
- If happening is impossible, then its probability is zero.

SAMPLING

A small section selected from the population is called a *sample* and the process of drawing a sample is called *sampling*.

Random Sampling

It is essential that the sample must be a random selection so that each member of the population has the same chance of being included in the sample. Thus the fundamental assumption underlying theory of sampling is *random sampling*

Simple Sampling

A special case of random sampling in which each event has the same probability p of success and the chance of success of different events are independent whether previous trials have been made or not, is called *simple sampling*.

Parameters

The statistical constants of the population such as mean (μ), standard deviation (σ) etc. are called *parameters*.

Statistics

Constants for the sample drawn from the given population, i.e. mean ($\bar{x}$), standard deviation (S) etc. are called *statistic*. The population parameters are in general, not known and their estimates given by the corresponding sample statistic are used. Greek letters are used to denoted population parameters and Roman letters for sample statistic.

Objectives of Sampling

Sampling aims at gathering the maximum information about the population with the minimum effort, cost and time. The object of sampling studies is to obtain the best possible value of the parameters under specific conditions. Sampling determines reliability of these estimates. The logic of the sampling theory is the logic of induction in which we pass from a particular (sample) to general (population). Such a generalization from sample to population is called *statistical inference*.

Sampling Distribution

Consider all possible sample of size n which can be drawn from a given population at random. For each sample, mean can be computed. The mean of the samples will not be identical. If these different means are grouped according of their frequencies, the frequency distribution so formed is called *sampling distribution of the mean*. Similarly sampling distribution of the standard deviation etc. can be obtained.

When drawing each sample, the previous sample is put back so that the parent population remains the same. This is called *sampling with replacement* and all subsequent formulae will pertain to sampling with replacements.

Standard Error

It is used to assess the difference between expected and observed values. Standard deviation of the sampling distribution is called *standard error* **(S.E.)**. Thus standard error of the sampling distribution of means is called *standard error of means*. The reciprocal of the standard error is called *precision*.

If $n \geq 30$, a sample is called **large** otherwise **small**. The sampling distribution of large samples is assumed to be normal.

Population

Collection of all possible samples is population. It is impractical to collect data on each sample of a population. Statistics helps to determine best estimate of the population parameter from randomly selected samples. Because random erros are involved in determination of parameters, the estimate will represent a parameter with a given probability only.

Given an event B such that $P(B) > 0$ and any other event A, we define conditional probability of A given by

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Simple Multiplication Rule

$$P(A \cap B) = P(B)P(A|B) = P(A)P(B|A)$$

Example. Let a balanced die is tossed once. Use the definition to find the probability of getting 1, given that an odd number was obtained.

Solution.

Probability of A given that the event B has occurred,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{But} \quad P(B) = \frac{1}{2},$$

$$\therefore \quad P(A|B) = \frac{P(A\cap B)}{P(B)} = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{3}$$

CONDITIONAL PROBABILITY

Consistency of the definition with the relative frequency concept of probability can be obtained from the following construction.

Let an experiment is repeated a large number, of times, N resulting in

both A and B, $A\cap B$, n_{11} times;

A, not B, $A\cap\bar{B}$, n_{21} times;

B, not A, $\bar{A}\cap B$, n_{12} times;

and neither A nor B, $\bar{A}\cap\bar{B}$, n_{22} times.

These results are:

	A	$\bar{A}$	
B	n_{11}	n_{12}	$n_{11}+n_{12}$
$\bar{B}$	n_{21}	n_{22}	$n_{21}+n_{22}$
	$n_{11}+n_{21}$	$n_{12}+n_{22}$	N

$n_{11} + n_{12} + n_{21} + n_{22} = N$.

Then it follows that with large N

$$P(A) \approx \frac{n_{11}+n_{21}}{N}, \quad P(B) \approx \frac{n_{11}+n_{12}}{N},$$

$$P(A\cup B) \approx \frac{n_{11}+n_{12}+n_{21}}{N}$$

and conditional probabilities

$$P(B|A) \approx \frac{n_{11}}{n_{11}+n_{21}} \quad \text{and} \quad P(A\cap B) \approx \frac{n_{11}}{N}$$

$$P(A/B) \approx \frac{n_{11}}{n_{11}+n_{12}}$$

Here definitions of P(B | A) and P(A | B) is verified.

Example. Let a box contains r red balls labeled 1, 2, 3, ..., r and b black balls labeled 1, 2, 3, ..., b. Assume that the probability of drawing any particular ball is $\frac{1}{b+r}$

If a ball from the box is known to be red, what is the probability it is the red ball labeled 1?

Solution.

First find probability that it is red (event A).

$$P(A) = \frac{r}{b+r}$$

Then compute probability that the ball is the red ball with the number 1 on it.

$$P(A\cap B) = \frac{1}{b+r}$$

where B is the event that the ball has a 1 on it.

Then probability that the ball is red and labeled 1 given that it is red is given by

$$P(B|A) = \frac{P(A\cap B)}{P(A)} = \frac{\frac{1}{b+r}}{\frac{r}{b+r}} = \frac{1}{r}$$

This differs from the unconditional probability of B, which is given by

$$P(B) = \frac{2}{b+r}$$

Summation Rule

Let $B_1, B_2, \ldots, B_n$ be (pairwise) disjoint events of positive probability whose union is $\cap$.

Then $$A = \bigcup_{j=1}^{n}(A\cap B_j)$$

Using multiplication rule, we have

$$P(A) = P\left(\bigcup_{j=1}^{n}(A\cap B_j)\right) = \sum_{j=1}^{n} P(A\cap B_j)$$

$$= \sum_{j=1}^{n} P(A|B_j)P(B_j)$$

e.g. Consider rolling a die. Let the sample space be given by $\Omega = \{1, 2, 3, 4, 5, 6\}$

Now consider a partition B_1, B_2, B_3, given by

$B_1 = \{1, 2\}$

$B_2 = \{3, 4\}$

$B_3 = \{5, 6\}$

Let A be the event that the number on the die is equal to one or greater than or equal to four. This gives

$A = \{1, 4, 5, 6\}$

$$P(A) = \frac{4}{6} = \frac{2}{3}.$$

This can also be computed using the formula as follows :

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3)$$

$$= \left(\frac{1}{2}\right)\left(\frac{1}{3}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{3}\right) + (1)\left(\frac{1}{3}\right)$$

$$= \frac{1}{6} + \frac{1}{6} + \frac{1}{3} = \frac{2}{3}$$

Bayes Rule

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{P(A)}$$

Now, if $B_1, B_2, \ldots, B_n$ are disjoint events of positive probability whose union is Ω, then

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_{j=1}^{n} P(A|B_j)P(B_j)}$$

Consider same example as before.

Probability for B_1,

$$P(B_1|A) = \frac{P(A|B_1)P(B_1)}{P(A)} = \frac{\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)}{\frac{2}{3}} = \frac{\frac{1}{6}}{\frac{2}{3}} = \frac{1}{4}$$

Probability for B_2,

$$P(B_2|A) = \frac{P(A|B_2)P(B_2)}{P(A)} = \frac{\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)}{\frac{2}{3}} = \frac{\frac{1}{6}}{\frac{2}{3}} = \frac{1}{4}$$

Probability for B_3,

$$P(B_3|A) = \frac{P(A|B_3)P(B_3)}{P(A)} = \frac{(1)\left(\frac{1}{3}\right)}{\frac{2}{3}} = \frac{\frac{1}{3}}{\frac{2}{3}} = \frac{1}{2}$$

Conditional Probability for given Multiple events

Conditional probability of A given $B_1, B_2, \ldots B_n$ is written $P(A|B_1, B_2, \ldots, B_n)$ and is defined as

$$P(A|B_1, B_2, \ldots, B_n) = P(A|B_1\cap B_2\cap\ldots\cap B_n)$$

for any events A, $B_1, B_2, \ldots, B_n$ such that

$$P(B_1\cap B_2\cap\ldots\cap B_n) > 0$$

More Complex Multiplication Rule

$$P(B_1\cap B_2\cap B_3\ldots\cap B_n)$$

$$= P(B_1)P(B_2|B_1)P(B_3|B_1B_2)\ldots P(B_n|B_1,B_2,\ldots B_{n-1})$$

ELEMENTS OF PROBABILITY

1. **Experiment**

   It is a process in which a certain work is repeated under the same conditions, the outcomes (or results) of which need not be the same.

   e.g. tossing a coin or rolling a die are experiments

2. **Sample space**

   The set S of all possible outcomes of a given experiment is called *sample space* for the experiment. An outcome an element of S, is called a *sample point*.

   e.g. for experiment of tossing a fair (or unbiased) coin

   sample space S = (H, T)

   where H, T refer to head, tail respectively.

3. **Event**

   A subset of the sample space S is called an *event*. The set (a) consisting of a single sample point a in S is called *elementary event*.

   Since empty set ϕ and S are always subsets of S, they are also events ϕ is called *impossible event* or *null event*, and S is called *sure* or *certain event*.

   Given two events A, B. form New events can be formed using the set operations of union, intersection and complementation.

   (*i*) $A\cup B$ is the event that occurs if and only if A occurs or B occurs (or both).

(*ii*) $A\cap B$ is the event that occurs if and only if both A and B occurs.

(*iii*) A^c or $\bar{A}$, the complement of A, is the event that occurs if and only if A does not occur.

Mutually exclusive events

Two events A and B are said to be mutually exclusive if and only if they cannot occur simultaneously, i.e. if $A\cap B = \phi$.

Three or more events are called *mutually exclusive*, if every two of them are mutually exclusive.

Example. Find total number of possible events that can occur for an experiment.

Solution

Let sample space S consist of n sample points. Then total number of possible events

= total number of subsets of S

$= |p(S)|$

where $p(S)$ is power set of $S = 2^n$.

Example. In an experiment, a fair coin is tossed 4 times. Describe the sample

Solution

The sample space S consists of $16 (= 2^4)$ sample points.

S = {HHHH, HHHT, HHTH, HTHH, THHH, HHTT, HTTH, TTHH, THTH, HTHT, THHT, HTTT, THTT, TTHT, TTTH, TTTT}

Finite Sample Space

A sample space S is said to be a finite sample space, if S is a finite set.

Finite Probability Space

Let $S = (a_1, a_2, \ldots\ldots, a_n)$ be a finite sample space. S is said to be a finite probability space or probability model, if each sample point a_i in S we can assign a real number p_i, called probability of a_i such that

(*i*) $p_i \geq 0$, for each i

(*ii*) $p_1 + p_2 + \ldots\ldots + p_n = 1$

If A is an event, then probability of A, denoted by P(A) is sum of the probabilities of the sample points in A. For the elementary event (a_i) we write $P(a_i)$ instead of $P(a_i)$.

Equiprobable Space

A finite probability space S in which each sample point has the same probability is called an *equiprobable space*.

If E is an event, then

$$P(E) = \frac{\text{number of elements in E}}{\text{number of elements in S}} = \frac{|E|}{|S|}$$

$$= \frac{\text{number of outcomes favourable to E}}{\text{total number of possible outcomes}}$$

- The word "*at random*" is used only when dealing with a equiprobable space. By the statement, "*a ball is drawn at random from a bag condisining 10 balls*" we mean that each ball in the bag has the same probability of being chosen.

- If p = probability of happening of an event $E = P(E)$,

  q = probability of not happening of event $E = P(\bar{E})$

  then $q = P(\bar{E}) = \frac{|\bar{E}|}{|S|} = \frac{|S| - |E|}{|S|} = 1 - \frac{|E|}{|S|} = 1 - p$

  $\Rightarrow \quad p + q = 1$

DEPENDENT EVENTS

Events A and B in the space S are said to be independent if occurence of one of them does not influence the occurence of the other

i.e. B is dependent of A if $P(B/A) = P(B)$.

Now, $P(A \cap B) = P(A).P(B|A) = P(A).P(B)$

Thus events A and B are independent, if

$$P(A \cap B) = P(A).P(B)$$

Mutual Independence

Let $(E_k : k$ is a positive integer) be a collection of events. These events are said to be mutually independent if for each finite non-empty subset $(E_1, E_2, \ldots\ldots E_n)$

$$P(E_1 \cap E_2 \cap \ldots\ldots \cap E_n) = P(E_1).P(E_2) \ldots\ldots P(E_n)$$

Pairwise Independence

Let $(E_k : k$ is a positive integer) be a collection of events. These events are said to be pairwise independent if

$$P(E_i \cap E_j) = P(E_i).P(E_j) \text{ for all } i \neq j.$$

Possible event

An event E is said to be a possible event, if $P(E) > 0$.

- Possible independent events are not mutually exclusive.
- Possible mutually exclusive events are not independent.

Example. Let p be the probability that a man aged y years will meet with an accident in a year. What is the probability that a man among n men all aged y years will meet with an accident first ?

Solution

Probability that a man aged y years will not meet with an accident $= 1 - p$

P (none meets with an accident)

$= (1-p)(1-p)\ldots\ldots(1-p)$ (n times) $= (1-p)^n$

$\therefore$ P (atleast one man meets with an accident)

$= 1 - (1-p)^n$

So, P (atleast one man meets with anaccident

[a person is chosen) $= \frac{1}{n} \times [1 - (1-p)^n]$

Example. A box contains 4 white, 3 blue and 5 green balls. Four balls are chosen. What is the probability that all three colours are represented ?

Solution

Total number of balls in the box is 12.

Hence total number of ways in which 4 balls can be chosen

$$= {}^{12}C_4 = \frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2 \times 1} = 495$$

Each colour will be represented in the following mutually exclusive ways :

	White	Blue	Green
(*i*)	2	1	1
(*ii*)	1	2	1
(*iii*)	1	1	2

Hence number of ways of drawing four balls in the above fashion

$$= {}^4C_2 \times {}^3C_1 \times {}^5C_1 + {}^4C_1 \times {}^3C_2 \times {}^5C_1 + {}^4C_1 \times {}^3C_1 \times {}^5C_2$$
$$= 90 + 60 + 120 = 270$$

$\therefore$ Required probability $= \frac{270}{495}$.

LAW OF TOTAL PROBABILITY

Let $B_1, B_2, \ldots\ldots B_n$ be mutually exclusive and let an event A occur only if anyone of B_i occurs. Then

$$P(A) = \sum_{i=1}^{n} P(A/B_i)P(B_i)$$

Baye's Theorm

Let $E_1, E_2, \ldots\ldots E_n$ be mutually exclusive events such that $P(E_i) > 0$ for each i.

Then for any event $A \subseteq \bigcup_1^n E_i$ such that $P(A) > 0$, we have

$$P(E_i/A) = \frac{P(A/E_i)P(E_i)}{\sum_{i=1}^{n} P(A/E_i)P(E_i)}, \quad i = 1, 2, \ldots, n$$

Example. Three urns A, B and C have 1 white , 2 black, 3 red balls, 2 white 1 black, 1 red balls and 4 white, 5 black, 3 red balls respectively. One urn is chosen at random and two balls are drawn. They happen to be white and red balls. What is the probability that they came from urn B ?

Solution

Let E_1, E_2 and E_3 be the events of choosing urns A, B and C respectively.

Then $\quad P(E_1) = P(E_2) = P(E_3) = \frac{1}{3} \qquad \ldots (i)$

Let X be the event of choosing two balls, white and red. To find $P(E_2/X)$.

By Baye's theorem, (E_2/X)

$$= \frac{P(X/E_2)P(E_2)}{P(X/E_1)P(E_1) + P(X/E_2)P(E_2) + P(X/E_3)P(E_3)}$$

$$= \frac{P(X/E_2)}{P(X/E_1) + P(X/E_2) + P(X/E_3)} \qquad \text{[from } (i)\text{]}$$

Now, $\quad P(X/E_1) = \frac{{}^1C_1 \times {}^3C_1}{{}^6C_2} = \frac{1}{5}$;

$P(X/E_2) = \frac{{}^2C_1 \times {}^1C_1}{{}^4C_2} = \frac{1}{3}$ and $P(X/E_3)$

$$= \frac{{}^4C_1 \times {}^3C_1}{{}^{12}C_2} = \frac{2}{11}$$

From (ii), $\quad P(E_2/X) = \frac{\frac{1}{3}}{\frac{1}{5} + \frac{1}{3} + \frac{2}{11}} = \frac{55}{118}$

MEASURES OF CENTRAL TENDENCY

1. **Mean** (Arithmetic Mean A.M. or the average),
2. **Median** (positional average)
3. **Mode** (value of the variable which occurs most, frequently)
4. **Geometric Mean** (G.M. used extensively in finding the rate of population growth)
5. **Harmonic Mean** (H. M.)

1. Arithmetic mean $(\bar{x})$

Let $x_1, x_2, \ldots\ldots\ldots\ldots x_n$ be the observed values and let f_i be the frequency of x_i (t_i) value of x_i repeated f_i times.

i.e., if the data is 3, 1, 3, 3, 2, 1, 3, 2, 2, then $x_1 = 1$, $x_2 = 2$ and $x_3 = 3$, and $f_1 = f_2 = 3$ and $f_3 = 4$.

$$\bar{x} = \frac{1}{n}(x_1 + x_2 + \ldots\ldots + x_n) = \frac{1}{n}\sum_{i=1}^{n} x_i$$

and for frequency distribution

$$\bar{x} = \frac{1}{N}(f_1x_1 + f_2x_2 + \ldots\ldots + f_nx_n)$$

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{n} f_ix_i \text{ where } N = \sum_{i=1}^{n} f_i$$

2. Median

Median is the middle score. If number of events are even then average of the two middles are taken. The median is better for describing the typical value.

Example. At a ski rental shop, data was collected on number of rentals on each of ten consecutive Saturdays:

44, 50, 38, 96, 42, 47, 40, 39, 46, 50.

To find sample mean, add them and divide by 10:

$$\frac{44+50+38+96+42+47+40+39+46+50}{10} = 49.2$$

Note: Mean value is not a value of the sample.

To find the median, first sort the data:

38, 39, 40, 42, 44, 46, 47, 50, 50, 96

There are two middle numbers 44 and 46. To find the median, we take average of the two.

$$\text{Median} = \frac{44+46}{2} = 45$$

Also mean is larger than all but three of the data points. The mean is influenced by outliers while the median is robust.

$$\text{Median} = L + \left(\frac{\frac{N}{2} - F_{i-1}}{f_m}\right)h$$

where, L = lower limited of Median class

N = Total Number of data items

h = width of Median class

f_m = frequency of Median class.

3. Mode

Mode is the value which occurs most frequently.

$$\text{Mode} = a + \frac{C(f_i - f_{i-1})}{2f_i - f_{i-1} - f_{i+1}}$$

where a = modal class

f_i = maximum frequency

C = constant difference for each class.

Example. Find median of the following data.

Cost	10–20	20–30	30–40	40–50	50–60
Items in a group	4	5	3	6	3

Solution

Cost	Number of items in the group	Cumulative frequency
10 – 20	4	4
20 – 30	5	9
30– 40	3	12
40 – 50	6	18
50 – 60	3	21

Here, $N = 21$.

$\frac{N}{2} = 10.5$

∴ The median class is 30 – 40.

From formula, Median = $30 + \frac{10}{12}(10.5 - 9) = 30 + 1.25 = \mathbf{31.25}$

Example. Find mode for the following distribution:

Class interval	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80
Frequency	5	9	8	12	28	20	12	11

Solution

Maximum frequency = 28

∴ Modal class = 40 – 50

$$\text{Mode} = a + \frac{C(f_i - f_{i-1})}{2f_i - f_{i-1} - f_{i+1}} = 40 + \frac{10(28-12)}{(2\times 28) - 12 - 20} = 40 + 6.666 = \mathbf{46.67}$$

Example. The expenditure of 100 families is given in the following frequency table.

Monthly Expenditure (in hundreds of rupees)	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Number of families	14	?	27	?	15

If mode for the distribution is 24, calculate the missing frequencies.

Solution

Let missing frequencies for the classes 10 – 20 and 30 – 40 be f_1 and f_2 respectively.

Expenditure	Number of families	Cumulative frequency
0 – 10	14	14
10 – 20	f_1	$14 + f_1$
20 – 30	27	$41 + f_1$
30 – 40	f_2	$41 + f_1 + f_2$
40 – 50	15	$56 + f_1 + f_2$

$$N = 100 = 56 + f_1 + f_2$$

$$\Rightarrow \quad f_1 + f_2 = 100 - 56 = 44 \qquad \ldots(i)$$

Mode is given to be 24, which lies in the class 20 – 30. So 20 – 30 is the modal class.

$$\text{Mode} = a + \frac{C(f_i - f_{i-1})}{2f_i - f_{i-1} - f_{i+1}}$$

$$\therefore \quad 24 = 20 + \frac{10(27 - f_1)}{(2\times 27) - f_1 - f_2}$$

$$\Rightarrow \quad 24 - 20 = \frac{270 - 10f_1}{54 - f_1 - f_2} = \frac{270 - 10f_1}{54 - 44} \qquad (\because f_1 + f_2 = 44)$$

$$\therefore \quad f_1 = 23$$

Substituting in (i), we get $f_2 = 21$.

Relation between Mean, Mode and Median

For a symmetrical distribution

Mean = Median = Mode

For a moderately asymmetrical distribution

Mean – Mode = 3 (Mean – Median)

4. Geometric mean (G)

$$G = \sqrt[n]{x_1 x_2 \ldots\ldots x_n} = (x_1 x_2 \ldots\ldots x_n)^{1/n}$$

For a frequency distribution,

$$G = \left(x_1^{f_1} x_2^{f_2} \ldots\ldots x_n^{f_n}\right)^{1/N}, \quad \text{where } N = \sum_{i=1}^{n} f_i$$

For both cases, taking log

$$\log G = \frac{1}{n}\sum_{i=1}^{n} \log(x_i)$$

For a frequency distribution

$$\log G = \frac{1}{N}\sum_{i=1}^{n} f_i \log(x_i)$$

- If n_1 and n_2 are sizes and G_1 and G_2 are geometric means of two series, then geometric mean of the combined series is given by

$$\log G = \frac{n_1 \log G_1 + n_2 \log G_2}{n_1 + n_2}$$

- Geometric mean is used to find the rate of population growth and the rate of interest and it is also used in the construction of index numbers.

5. Harmonic mean

It is the reciprocal of mean of reciprocals, i.e.

$$H = \frac{1}{\left(\frac{1}{n}\sum_{i=1}^{n}(1/x_i)\right)}$$

For a frequency distribution,

$$H = \frac{1}{\left(\frac{1}{N}\sum_{i=1}^{n}(f_i/x_i)\right)}; \text{ where } N = \sum_{i=1}^{n} f_i$$

Skewness

It is defined as lack of symmetry.

$$\text{Coefficient of skewness} = \frac{\text{Mean} - \text{Mode}}{\text{Standard deviation}}$$

Negatively skewed Positively skewed

MEASURES OF DISPERSION

The set of constant which would in a concise way explain the "*variability*" or "*spread*" in a data is called *measures of dispersion or variability*.

The average for two groups of the same number of measurements may be equal, but one group may be more variable than the other.

e.g. set of five values 5, 6, 7, 8, 9 has the mean as 7; while another set of five values 1, 6, 4, 10, 14 also has the same mean 7. The second set has, obviously more variability than the first.

Usually four measures of dispersion or variability are defined:

1. Range

This is the difference between two extreme observations in the data given.

In a frequency distribution ,

R = (largest x value) – (smallest x value).

Obviously, this is not a very satisfactory measure except in cases where the number of observations are small and a quick calculation for the scatter is needed. It is used in statistical quality control studies rather widely.

2. Quartile deviation (Q.D.)

Median bisects the distribution. If distribution divided into four parts, *quartiles* are obtained. *First quartile*, Q_1, would have 25 percent of the values below it and the rest above it;

Third quartile, Q_3 would have 75 percent of values below it and the rest above it. The method of calculation of quartiles is similar to that of the median with slight variations.

$$Q_1 = l + \frac{\frac{N}{4} - f_{Q_1}}{f} \times C; \qquad Q_3 = l + \frac{\frac{3N}{4} - f_{Q_3}}{f} \times C$$

where we locate Q_1 – class and Q_3 – class properly

l = lower limit of the quartile class

C = common factor

Quartile deviation is defined as

$$\text{Q.D.} = \frac{1}{2}(Q_3 - Q_1)$$

3. Average deviation (A)

If average chosen is A (say), then average deviation about A is average deviation

$$\text{A. D. (A)} = \frac{1}{n}\sum |(x_i - A)| \text{ for discrete data}$$

$$= \frac{1}{n}\sum f_i |x_i - A| \text{ for a frequency distribution.}$$

Usually, either median or the mean are takes as the average.

4. Standard deviation (Root mean square deviation)

Standard deviation,

$$\sigma = \sqrt{\frac{1}{n}\Sigma(x_i - \bar{x})^2} \quad \text{for discrete data}$$

$= \sqrt{\frac{1}{N}\Sigma f_i(x_i - \bar{x})^2}$ for frequency distribution.

Square of the standard deviation , σ^2 is defined as the *variance* (V).

$$\therefore \quad V = \sigma^2 = \frac{1}{N}\Sigma f_i(x_i - \bar{x})^2$$

Out of these measures, σ is widely used as a companion to $\bar{x}$ on which is based, when dealing with dispersion or scatter.

Calculation of V or σ

$$V = \left[\frac{\Sigma f_i d_1^2}{N} - \frac{(\Sigma f_i d_1^2)^2}{N}\right] \times C^2$$

Some important results

- Average deviation is least when taken from the median.
- Standarsd deviation is not less than the average deviation from the mean.
- For raw data $x_1, x_2, \ldots\ldots x_n$,

$$V = \frac{\Sigma x^2}{N} - \left(\frac{\Sigma x}{N}\right)^2 \quad \text{or} \quad V = \frac{\Sigma x^2}{n} - (\bar{x})^2$$

If n_1, n_2 are sizes of two groups, $\bar{x}_1$, $\bar{x}_2$ their means; σ_1, σ_2 their standard deviations, then standard deviation of the combined group is determined from

$$(n_1 + n_2)\sigma_2 = n_1\left(\sigma_1^2 + d_1^2\right) + n_2\left(\sigma_2^2 + d_2^2\right)$$

where, $d_1 = \bar{x} - \bar{x}_1$ and $d_2 = \bar{x} - \bar{x}_2$,

$\bar{x}$ being the combined mean.. This result can be extended to more than two groups.

Determination of Variance and Standard Deviation

Steps

1. Calculate the mean, $\bar{x}$.
2. Write a table that subtracts the mean from each observed value.
3. Square each of the differences.
4. Add this column.
5. Divide by n –1, where n is number of items in the sample This is the *variance*.
6. To get the *standard deviation*, take square root of the variance.

Example. The owner of a restaurant is interested in how much people spend at the restaurant. He examines 10 randomly selected receipts for parties of four and writes down the following data :

44, 50, 38, 96, 42, 47, 40, 39, 46, 50

Find mean, standard deviation and variance.

Solution

Mean is calculated by adding and dividing by 10

Mean, $\bar{x}$ = 49.2

Following table is used for getting standard deviation :

p	x–49.2	$(x-49.2)^2$
44	–5.2	27.04
50	0.8	0.64
38	11.2	125.44
96	46.8	2190.24
42	–7.2	51.84
47	–2.2	4.84
40	–9.2	84.64
39	–10.2	104.04
46	–3.2	10.24
50	0.8	0.64
Total		**2600.4**

Now $V = \left(\frac{2600.1}{10-1}\right) = 288.7 \approx 289$

Hence variance is 289 and standard deviation is square root of 289 = 17.

Since standard deviation can be thought of measuring how far the data values lie from the mean, we take mean and move one standard deviation in either direction. The mean for this example was about 49.2 and the standard deviation was 17.

We have, 49.2 – 17 = 32.2

and 49.2 + 17 = 66.2

What is means is that most of the patrons probably spend between 32.20 and 66.20.

Coefficients to Dispersion

When two series of measurements have to be compared, then averages may be different and the units which the measurements are recorded may also be different. Hence, coefficients of dispersion are provided as

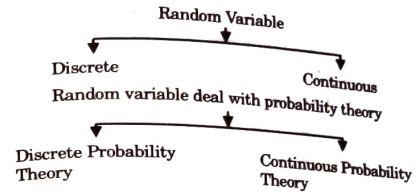
- $\frac{\text{Range}}{A+B}$, *where A is largest and B is smallest of the values.*
- $\frac{Q_3 - Q_1}{Q_3 + Q_1}$
- $\frac{\text{Average deviation about A}}{A}$
- *Each of these is free from units of measurement* and is a pure number.

Coefficient of variation (C.V.) $= \frac{\sigma}{x} \times 100$

Thus coefficient of variation is a percentage.

RANDOM VARIABLE

Many variables of interest in a system are not expressible on explicit function of time, all such variable on explicit function of time, all such variable are classified as random variable. These are implying uncertainty about their exact nature.



Let X be a random variable.

- Function $F(x) = P(X \le x)$ is called *distribution function of X.*
- Mean or Expectation of $X = \mu = E(X)$

$$= X(a_1)P(a_1) + \ldots\ldots\ldots\ldots + X(a_n)P(a_n)$$

$$= \sum_{i=1}^{n} X(a_i)P(a_i),$$

where S is probability space $(a_1, a_2, \ldots\ldots a_n)$

- Variance of $X = \text{Var}(X) = \sigma^2 = E(X^2) - [E(X)]^2$

Standard deviation of $X = \sigma$

Example. Find expectation of the number of points when a fair die is rolled.

Solution

Let X be the random variable showing number of points.

Then X = 1, 2, 3, 4, 5, 6

a_i	$P(X = a_i) = P(a_i)$	Product
1	$\frac{1}{6}$	$\frac{1}{6}$
2	$\frac{1}{6}$	$\frac{2}{6}$
3	$\frac{1}{6}$	$\frac{3}{6}$
4	$\frac{1}{6}$	$\frac{4}{6}$
5	$\frac{1}{6}$	$\frac{5}{6}$
6	$\frac{1}{6}$	$\frac{6}{6}$
		$E(X) = \frac{21}{6} = \frac{7}{2}$

$$\therefore \quad \text{Expectation} = \frac{7}{2}.$$

Example. What is the expectation of the number of failure preceding the first success in an infinite series of independent trials with constant probability of success p in each trial ?

Solution

Let p = probability of succes

$\therefore$ q = probability of failure = 1 – p

Let X be the random variable representing the number of failure preceding the first success.

Since an infinite series of independent trials are conducted, X = 0 , 1, 2

P(X = x) = probability that there are x failures preceding the first success.

$$= q \times q \times \ldots\ldots \times q \times p = q^x p$$

$$\therefore \quad E(X) = \sum_{x=0}^{\infty} xp(X = x) = \sum_{x=0}^{\infty} xq^x p$$

$$= 1\,qp + 2q^2p + 3q^3p + \ldots\ldots$$

$$= qp(1 + 2q + 3q^2 + \ldots\ldots)$$

$$= qp(1-q)^{-2} \quad [\because (1-q)^{-2} \text{ by Binomial theroem}]$$

$$= \frac{qp}{(1-q)^2} = \frac{qp}{p^2} = \frac{q}{p}$$

Example. The Bernouilli probability law with parameter p in which $0 \le P \le 1$ is defined by a random variable X taking the value 1 with probability P and the value 0 with probability q. Find the mean and variance of X.

Solution

a_i	$P(a_i)$	Product
1	p	p
0	q	0
		$E(X) = p$

$\therefore$ Mean = E (X) = p.

Now to find Var (X) :

a_i	a_i^2	$P(a_i)$	Product $[a_i^2 \times P(a_i)]$
1	1	p	p
0	0	q	0
			$E(X^2) = p$

$$\therefore \quad \text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= p - p^2 = (1-p)\,p = pq$$

Absolute Frequency

Let a value x_i is repeated f_i times, then f_i is called *frequency or absolute frequency.*

Let N = sum of all frequencies f_i, then

$\frac{f_i}{N}$ is called *relative frequency* , i = 1, 2 .....n.

Sum of all relative frequencies

$$= \frac{f_1}{N} + \frac{f_2}{N} + \ldots\ldots\ldots + \frac{f_n}{N}$$

$$= \frac{f_1 + f_2 + \ldots\ldots\ldots + f_n}{N} = \frac{N}{N} = 1.$$

PROBABILITY DISTRIBUTION

Let a random experiment is repeated n times. If an event A occurs n_A times, then its probability is defined as

$$p(A) = p(x_A) = \lim_{n\to\infty} \frac{n_A}{n}$$

= Probability of outcome A.

where, n_A = times occur A

n = total time experiment

- This limit may not exist.
- This definition presumes that all outcomes are equally likely to occur.

$$p(A + B) = p(A) + p(B) - p(AB)$$

Here, p(AB) = joint probability.

Type of Probability Distribution

The are of two types :

1. Discrete Distribution

It is of two types.

(i) Binomial Distribution

It is concerned with trials of a repetitive nature in which only the occurrence or non-occurrence, success or failure, acceptance or erejection, yes or no of a particular event is of interest.

If a series of independent trials are perfomed such that for each trial, p is the probability of success and q that of a failure, then probability of r successes in a series of n trials is given by ${}^nC_r p^r q^{n-r}$, where r takes any integral value from 0 to n.

The probabilities of 0, 1, 2,.....r,.....,n successes are therefore, given by

q^n, ${}^nC_1\, pq^{n-1}$, ${}^nC_2\, p^2q^{n-2}$,....${}^nC_r p^r q^{n-r}$,.....,p^n

Probability of the number of successes so obtained is called **binomial distribution** because probabilities are successive terms in the expansion of the binomial $(q + p)^n$.

Hence sum of the probabilities

$$= q^n + {}^nC_1\, pq^{n-1} + {}^nC_2 p^2q^{n-2} + ...+ p^n$$
$$= (q + p)^n = 1$$

Constants of Binomial Distribution

Moment generating function about the origin,

$$M_0(t) = E(e^{tx}) = \sum {}^nC_1\, p^x q^{n-x} e^{tx}$$
$$= \sum^n C_x (pe^t)^x\, q^{n-x} = (q + pe^t)^n$$

Differentiating with respect to t and putting $t = 0$, we get mean given by

$$M_m(t) = e^{-npt}\,(q + pe^t)^n = (qe^{-pt} + pe^{qt})^n$$

$$= 1 + pq\frac{t^2}{2!} + pq(q^2 - p^2\frac{t^3}{3!}$$

$$+ pq(q^3 + p^3)\frac{t^4}{4!} + \ldots$$

$$\Rightarrow 1 + \mu_1 t + \mu_2\frac{t^2}{2!} + \mu_3\frac{t^3}{3!} + \mu_4\frac{t^4}{4!} + ...$$

$$= 1 + npq\frac{t^2}{2!} + npq(q - p)\frac{t^3}{3!}$$

$$+npq\,[1 + 3(n - 2)\,pq]\frac{t^4}{4!} + ...$$

Equating coefficients of like power of t on either side, we have

$$\mu_2 = npq,\ \mu_3 = npq(q-p),\ \mu_4 = npq\,[1+ 3(n - 2)pq]$$

Also, $\beta_1 = \dfrac{\mu_3^2}{\mu_2^3} = \dfrac{(q-p)^2}{npq} = \dfrac{(1-2p)^2}{npq}$

$$\beta_1 = \frac{\mu_4}{\mu_2^2} = 3 + \frac{1 - 6pq}{npq}$$

$\therefore$ Mean = np

Standard deviation = $\sqrt{(npq)}$

$$\text{Skewness} = \frac{1-2p}{\sqrt{npq}}$$

Kurtosis = β_2

Obs. Skewness is

positive for $p < \left(\frac{1}{2}\right)$

negative for $p > \left(\frac{1}{2}\right)$

zero for $p = \left(\frac{1}{2}\right)$, i.e. when probability curve of binomial distribution will be symmetrical (bell-shaped)

As number of trials n increase indefinitely $\beta_1 \to 0$ and $\beta_2 \to 3$

Binomial Frequency Distribution

If n independent trials constitute one experiment and this experiment be repeated N times, then frequency of r successes = $N^n C_r\, p^r\, q^{n-r}$.

The possible number of successes together with these expected frequencies constitute binomial frequency distribution.

Applications of Binomial Distribution

This distribution is applied to problems concerning

(a) number of defectives in a sample from production line.

(b) estimation of reliability of system.

(c) number of rounds fired from a gun hitting a target.

(d) rader detection.

Bernouilli Trials

Independent repeated trials of an experiment with two outcomes only are called *Bernouilli trials*, call One of the outcomes is called *success* and the other outcome is called *failure*.

Let p = probability of success in a Bernouilli trial

q = probability of failure = $1 - p$.

A binomial experiment consisting of a fixed number n of trials is denoted by B (n, p).

Probability of r success in the experiment B (n, p) is given by

$$P(r) = {}^nC_r\, p^r\, q^{n-r}$$

The function P(r) for $r = 0, 1, 2$ ....., n for B (n, p) is called **binomial distribution.**

Example. If four coins are tossed , find the chance that there should be two heads and two tails.

Solution.

p = probability of a head = $\frac{1}{2}$

q = probability of a tail = $1 - p = \frac{1}{2}$

Hence probability of getting 2 heads (and 2 tails) when four coins are tossed

$$= P(2) = {}^4C_2 p^2 q^{4-2}$$

$$= 6 \times \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{2}\right)^2 = \frac{3}{8}$$

(ii) Poisson Distribution

It is a distribution related to the probabilities of events which are extremely rare, but which have a large number of independent opportunities for occurrence.

e.g. number of persons born blind per year in a large city is the phenomena, in which this law is followed.

This distribution can be derived as limiting case of the binomial distribution by making n very large and p very small keeping np fixed (= m, say)

The probability of r successes in a binomial distribution

$$P(r) = {}^nC_r p^r q^{n-r}$$

$$= \frac{n(n-1)(n-2)...(n-r+1)\,p^r q^{n-r}}{r!}$$

$$= \frac{np(np-p)(np-2p)..(np-r-1p)(1-p)^{n-r}}{r!}$$

As $n \to \infty$, $p \to 0$ $(np = m)$,

$$p(r) = \frac{m^r}{r!}\,\underset{n\to\infty}{\text{Lt}}\,\frac{(1 - m/n)^2}{(1 - m/n)^r} = \frac{m^r}{r!}\,e^{-m}$$

so that probabilities of 0, 1, 2, ..., r,..... success in poisson distributio is given by

$$e^{-m},\ me^{-m},\ \frac{m^2 e^{-m}}{2!},\ ....\ \frac{m^r e^{-m}}{r!},\ .....$$

Sum of these probabilities is unity as it should be.

2. Continuous Probability Distribution

When a variate X takes every value in an interval, it gives rise to continuous distribution of X. e.g. distributions defined by the variates like heights or weights are continuous distributions.

A major conceptual difference, however, exists between discrete and continuous probabilities. When thinking in discreate terms, the probability associated with an event is meaningful. With continuous event, however, where number of events is infinitely large, the probability that a specific event will occur is practically zero. For this reason, continuous probability statements must be worded somewhat, differently from discrete ones. Instead of finding the probability that z equals some value, we find the probability of x falling in a small interval.

Thus probability distribution of a continuous variate x is defined by a function f(x) such that probability of the variate x falling in the small interval

$$[x - \frac{1}{2}\,dx] \text{ to } [x + \frac{1}{2}\,dx] \text{ is } f(x)\,dx.$$

Let X be a continuous random variable. If $f(x)$ is probability density functions of X, then

- $P(X \le a) = \int_{-\infty}^{a} f(x)dx$
- $P(a \le X \le b) = \int_{a}^{b} f(x)dx$
- Expectation of X = E (X) = Mean = $\int_{-\infty}^{\infty} xf(x)dx$
- Variance of X = y (X) = $\int_{-\infty}^{b} [x - E(x)^2 f(x)dx]$
- $\int_{-\infty}^{\infty} f(x)dx = 1$

Types of Continuous Probability Distribution

(i) Uniform Distribution

A continuous random variable X is said to follow uniform distribution or rectangular distribution if its probability density function is given by

$$f(x) = \frac{1}{b-a},\ a \le x \le b = 0, \text{ otherwise}$$

The uniform distribution generally arises in the study of round off errors where measurements are recorded upto a certain level of accuracy.

Cumulative distribution function of X is given by

$$f(x) = p\,(X \le x).$$

$$= \int_{-\infty}^{x} f(x)dx = \begin{cases} 0 & \text{if } x \le a \\ \dfrac{x-a}{b-a} & \text{if } a \le x \le b \\ 1 & \text{if } x > b \end{cases}$$

$$\text{Mean} = \frac{a+b}{2},\ \text{and Variance} = \frac{(b-a)^2}{12}$$

Example. The melting point X of a certain specimen be assumed to be a continuous random variable which is uniformly distributed over the interval [110, 120]. Find density function of X, mean of X, variance of X and P (112 ≤ x ≤ 115.).

Solution

Here, a = 100, b = 120

$$\therefore \quad f(x) = \begin{cases} \frac{1}{120-110} & 110 \le x \le 120 \\ 0, & \text{otherwise} \end{cases}$$

$$= \begin{cases} \frac{1}{10} & 110 \le x \le 120 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Mean of X} = \frac{a+b}{2} = \frac{110+120}{2} = 115$$

$$\text{Variance of X} = \frac{(b-a)^2}{12} = \frac{(120-110)^2}{12} = \frac{25}{3}$$

$$P(112 \le x \le 115) = \int_{112}^{115} f(x)dx = \int_{112}^{115} \frac{1}{10}dx = \frac{3}{10}$$

(ii) Exponential Distribution

A continuous random variable X is said to have the exponential distribution if its probability density function is given by

$$f(x) = \begin{cases} ae^{-ax}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

a being some positive number, called its *parameter*
This distribution arises in the study of life-length of electronic components.
Also the inter-arrival time and service time in queuing theory have exponential distributions.

$$\text{Mean} = \frac{1}{a}, \text{ Variance} = \frac{1}{a^2}$$

Example. The sales tax return of a salesman is exponentially distributed with parameter $\frac{1}{4}$. What is the probability that his sale will exceed ₹ 10,000 assuming that sales tax is charged at the rate of 5% on the sales ?

Solution

Here, $f(x) = \begin{cases} \frac{1}{4}e^{-\frac{x}{4}}, & x > 0 \\ 0 & \text{otherwise} \end{cases}$

Sales tax for the sale of ₹ 10,000

$$= 10000 \times \frac{5}{100} = ₹\ 500$$

Now, P (x > 500) = 1 P (x ≤ 500)

$$= 1 - \int_0^{500} f(x)dx = 1 - \frac{1}{4}\int_0^{500} e^{-x/4}dx$$

$$= 1 + \left(e^{-x/4}\right)_0^{500} = e^{-125}$$

(iii) Normal Distribution

A continuous random variable X is said to follow normal distribution if its probability density function is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

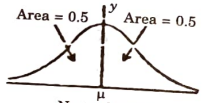
$$-\infty \le x \le \infty, \sigma > 0, -\infty < \mu < \infty$$

In such a situation, write X – N (μ, σ²)
The distribution involves two parameters μ and σ.

Properties.

(a) Distribution is symmetrical.

(b) Mean = μ, Variance = σ².



Normal curve

(c) Mean, median and mode coincide.

(d) $f(x) \ge 0$ for all x.

(e) $\int_{-\infty}^{\infty} f(x)dx = 1$, *i.e.* total area under the curve $y = f(x)$ bounded by the axis of x is 1.

Note:

Curve $y = f(x)$, called *normal curve* is a *bell-shaped curve.* It is symmetrical about $x = \mu$. The two tails on the left and right sides of the mean extend to infinity.

Put $z = \frac{x-\mu}{\sigma}$. Then z is called a standard *normal variate* and its probability density function is given by

$$p(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}, \quad -\infty \le z \le \infty.$$

Mean of the standard normal distribution is 0 and variance is 1. Write Z – N (0, 1).

Area under the standard normal curve

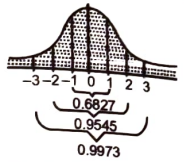
(a) Between $z = -1$ and $z = 1$ is 0.6827 (since total area under standard normal curve is 1)
i.e. $p(-1 < z < 1) = 0.6827$

(b) Between $z = -2$ and $z = 2$ is 0.9545
i.e., $p(-2 < z < 2) = 0.9545$

(c) Between $z = -3$ and $z = 3$ is 0.9973
i.e., $p(-3 < z < 3) = 0.9973$

In other words,

$p(\mu - \sigma < x < \mu + \sigma) = 0.6827$
$p(\mu - 2\sigma < x < \mu + 2\sigma) = 0.9545$
$p(\mu - 3\sigma < x < \mu + 3\sigma) = 0.9973$
$p(\mu - 1.96\sigma < x < \mu + 1.96\sigma) = 0.95$
$p(\mu - 2.58\sigma < x < \mu + 2.58\sigma) = 0.99$



Since distribution is symmetric, we consider only positive values of z.

Then $\phi(z) = \frac{1}{\sqrt{2\pi}}\int_0^z e^{-\frac{z^2}{2}} dz$

gives area under the normal curve between $z = 0$ and $z = z$.
For different values of z, $\phi(z)$ can be calculated.
These values of $\phi(z)$ have been tabulated.
There is another table called *table of ordinates.*
We read

$$p(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

corresponding to a value of z from this table.
Many distributions tend to a normal distribution in the limit.
When a variable is not normal, it can be made normal using some suitable transformation.
When the sample size is large, distributions of the sample mean, sample variance etc. approach normality. Thus distribution forms a basis for tests of significance.
Normal distribution is also called *distribution of errors.*

Normalized Gaussian random variable

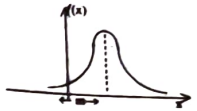
It mean of the standard normal distribution is 0 and variance is 1, write Z – N (0, 1).
The distribution function of X are plotted respectively.
Mean and variance of X are

$$\mu_x = E[X] = \mu$$
$$\sigma_x^2 = E[(X - \mu_x)^2] = \sigma^2$$

Notation $N(\mu; \sigma^2)$ are used to denote that X is normal (or Gaussian) with mean m and variance σ^2.
In particular, X = N (0 ; 1), i.e. X with zero mean and unit variance is defined as a *normalized Gaussian random variable.*

Central Unit Theorem

Normal (or Gaussian) distribution has played a significant role in the study of random phenomena in nature. Many naturally occurring random phenomena are approximately normal. *Central–limit theore* states that *sum of a large number of independent random variables, under certain conditions can be approximated by a normal distribution.*



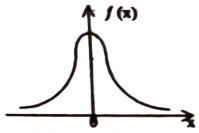
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp -\frac{(x-\mu)^2}{2\sigma^2}$$

If mean = 0, i.e. μ = 0,

then $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp -\frac{(x)^2}{2\sigma^2}$

Gaussian Process

$$f(y) = \frac{1}{\sqrt{(2\pi\sigma_y^2)}} \exp \frac{[-(y-\mu_y)^2}{2\sigma_y^2}$$



A Gaussian Pulse

where, μ_y = mean
σ_y^2 = variance

Properties of Gaussian process

(i) If a Gaussian process x(t) is applied to a stable linear filter, then random process y(t) developed. The output of the filter is also Gaussian.

(ii) Consider the set of random variables or samples $x(t_1), x(t_2), x(t_3)$ —— obtained by observing a random process x(t) at time t_1, t_2, ——.

Second moment or Mean square value

Consider random variable X which assume the possible value x_1, x_2, x_3 —— x_m. In a sequence of n experiment, let event x_1 occur n_1 time, x_2 occur n_2 times etc.
Arithmetic average or mean of x_k is

$$X_{avg} = \frac{n_1x_1 + n_2x_2 + \text{-----} + x_m n_m}{n} = \frac{1}{n}\sum_{k=1}^{m} x_k n_k$$

If $n \to \infty$, then $\frac{n_1}{n} \to p(x_1)$

$$\therefore \quad X_{avg} = p_1x_1 + p_2x_2 + \text{----} p_m x_m = \sum_{k=1}^{m} p_k x_k$$

and $$X^2_{avg} = \frac{n_1x_1^2 + n_2x_2^2 + \text{-----}}{N}$$

CORRELATION

In a divariate distribution, if the classes in one variable are associated by clases in the other, then variables are called *correlated*. If ratio of two variable deviations is constant, then correlation is called *perfect*.

Co-efficient of Correlation

Numerical measure of correlation is called *co-efficient of correlation* and is defined as

$$r = \frac{\Sigma XY}{n\sigma_x\sigma_y}$$

$$= \frac{\Sigma XY}{\sqrt{\Sigma X^2 \Sigma Y^2}} \quad \left[\because \sigma_x^2 = \frac{\Sigma x^2}{n}; \sigma_y^2 = \frac{\Sigma y^2}{n}\right]$$

where, X = deviation from mean, $= x - \bar{x}$

Y = deviation from mean, $= y - \bar{y}$

σ_x = standard deviation of x series

σ_y = standard of y series

n = number of values of the two variables

Method of Calculation (P)

Direct method : Substituting the value of σ_x and σ_y in the above formula, we get

$$r = \frac{\Sigma XY}{\sqrt{\Sigma X^2 \Sigma Y^2}},$$

or $$\frac{n\sum XY}{\sqrt{\left[n\sum x^2 - (\sum x)^2 \times \{n\sum y^2 - \sum x^2\}\right]}}$$

Rank Correlation

A group of n individuals may be arranged in order of merit respect to some characteristic. The same group would give different order for different characteristics. Consider order corresponding to two characteristic A and B. The correlation between these n pairs of ranks is called *rank correlation* in the *characteristics A and B* for that group of individuals.

Let x_i, y_i be the ranks of the i^{th} individuals in A and B respectively. Assuming that no two individuals are bracketed equal in either case, each of the variables taking the values 1, 2, 3,....., n, we have

$$\bar{x} = \bar{y} = \frac{1+2+3+\ldots+n}{n} = \frac{n(n+1)}{2n} = \frac{n+1}{2}$$

If X, Y be the deviation of x, y from their mean, then

$$\Sigma X_i^2 = \Sigma(x_i - \bar{x})^2$$

$$= \Sigma x_i^2 + n(\bar{x})^2 - 2\bar{x}\Sigma x_i$$

$$= \Sigma n^2 + \frac{n(n+1)^2}{4} - 2\frac{n+1}{2}\cdot\Sigma n$$

$$= \frac{n(n+1)(2n+1)}{4} + \frac{n(n+1)^2}{4} - \frac{n(n+1)^2}{2}$$

$$= \frac{1}{12}(n^3 - n)$$

Similarly $\Sigma Y_i^2 = \frac{1}{12}(n^3 - n)$

Now let $d_i = x_i - y_i$

$\therefore$ $d_i = (x_i - \bar{x}) - (y_i - \bar{y}) = X_i - Y_i$

$\Sigma d_i^2 = \Sigma X_i^2 + \Sigma Y^2 - 2\Sigma X_i Y_i$

$$\Sigma X_i Y_i = \frac{1}{2}(\Sigma X_i^2 + \Sigma Y_i^2 - \Sigma d_i^2)$$

$$= \frac{1}{12}(n^3 - n) - \frac{1}{2}\Sigma d_i^2$$

Hence correlation coefficient between these variables

$$\rho = \frac{\sum X_i Y_i}{\sqrt{\sum X_i^2 \sum Y_i^2}}$$

$$= \frac{\left(\frac{1}{12}\right)(n^3 - n) - \left(\frac{1}{2}\right)\Sigma d_i^2}{[1/12](n^3 - n)}$$

$$= 1 - \frac{6\Sigma d_i^2}{n^3 - n}$$

This is called *rank correlation coefficient.*

Line of Regression

If dots of scatter diagram generally, tend to cluster along a well defined direction which suggests a linear relationship between variables x and y, such a line of best – fit for the given distribution of dots is called *line of regression.*

There are two such lines, one giving the best possible mean values of y for each specified value of x and the other giving the best possible mean values for x for given values of y. The former is called the *line for regression of y on x* and the later is called the *line of regression of x on y.*

First consider the line of regression of y on x.

Let straight line satisfying the general trend of n dots in a scatter diagram be

$$y = a + bx \qquad ...(i)$$

We have to determine the constants a and b so that equation (i) gives for each value of x, the best estimate for the average value of y in accordance with the principle of least squares. Thus normal equations for a and b are

$$\Sigma y = na + b\Sigma x$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 \qquad ...(ii)$$

Equation (ii) gives $\frac{1}{n}\Sigma y = a + b.\frac{1}{n}\Sigma x$

i.e. $$\bar{y} = a + \bar{b}x$$

This shows that $(\bar{x}, \bar{y})$, i.e. means of x and y lie on (i)

Shifting the origin to $(\bar{x}, \bar{y})$, equation (iii) takes the form

$$\Sigma(x - \bar{x})(y - \bar{y}) = a\Sigma(x - \bar{x} + b\Sigma(x - \bar{x})^2$$

But $\Sigma(x - \bar{x}) = 0$

$\therefore$ $$b = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(x-\bar{x})^2} = \frac{\Sigma XY}{\Sigma X^2} = \frac{\Sigma XY}{n\sigma_x^2}$$

$$= r\frac{\sigma_y}{\sigma_x} \qquad \left(\because r = \frac{\Sigma XY}{n\sigma_x\sigma_y}\right)$$

Thus line of best fit becomes

$$(y - \bar{y}) = r\frac{\sigma_y}{\sigma_x}(x - \bar{x})$$

which is equation of the *line of regression of y on x*. Its slope is called *regression co-efficient of y on x*

Interchanging x and y, the line of regression of x on y is

$$(x - \bar{x}) = r\frac{\sigma_x}{\sigma_y}(y - \bar{y})$$

Thus regression co-efficient of y on $x = r\frac{\sigma_y}{\sigma_x}$

and regression co-efficient of x on $y = r\frac{\sigma_x}{\sigma_y}$

Cor. Correlation co-efficient is the geometric mean between the two regression cofficients

$$r\frac{\sigma_y}{\sigma_x} \times r\frac{\sigma_x}{\sigma_y} = r^2$$

Regression co-efficient

It is defined as the hope of lines of regression,

i.e. regression coefficient of line x on $y = r\frac{\sigma_x}{\sigma_y}$

SOLVED EXAMPLES

1. A bag contains 8 white and 6 red balls. Find the probability of drawing two balls of the same colour.

Solution

Two balls out of 14 can be drawn in $^{14}C_2$ ways which is total number of outcomes.

Two white balls out of 8 can be drawn in $^8C_2$ ways.

Thus probability of drawing 2 white balls

$$= \frac{^8C_2}{^{14}C_2} = \frac{28}{91}$$

Similarly 2 red balls out of 6 can be drawn in $^6C_2$ ways.

Thus probability of drawing 2 red balls

$$= \frac{^6C_2}{^{14}C_2} = \frac{15}{91}$$

Hence probability of drawing 2 balls of the same colour (either both white or both red)

$$= \frac{28}{91} + \frac{15}{91} = \frac{43}{91}$$

2. Find probability of drawing an ace or a spade or both from a deck of cards?

Solution.

Probability of drawing an ace from a deck of 52 cards $= \frac{4}{52}$

Similarly probability of drawing a card of spades

$$= \frac{13}{52}$$

and probability of drawing an ace of spades $= \frac{1}{52}$.

Since two events (i.e., a card being an ace and a card being of spades) are not mutually exclusive.

Hence probability of drawing an ace or a spade

$$= \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{4}{13}.$$

3. A pair of dice is tossed twice. Find the probability of scoring 7 points

(*a*) once (*b*) at least once (*c*) twice.

Solution

In a single toss of two dice, the sum 7 can be obtained as (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1) i.e., in 6 ways

Hence probability of getting $7 = \frac{6}{36} = \frac{1}{6}$.

Thus, probability of not getting $7 = 1 - \frac{1}{6} = \frac{5}{6}$.

(*a*) Probability of getting 7 in the first toss and getting 7 in the second toss

$$= \frac{1}{6} \times \frac{5}{6} = \frac{5}{36}.$$

Similarly, probability of not getting 7 in the first toss and getting 7 in the second toss

$$= \frac{5}{6} \times \frac{1}{6} = \frac{5}{36}.$$

Since these are mutually exclusive events, addition law of probability applies.

$$\therefore \text{ Required probability} = \frac{5}{36} + \frac{5}{36} = \frac{5}{18}$$

(*b*) Probability of not getting 7 in either toss

$$= \frac{5}{6} \times \frac{5}{6} = \frac{25}{36}$$

Hence probability of getting 7 at least once

$$= 1 - \frac{25}{36} = \frac{11}{36}.$$

(*c*) Probability of getting 7 twice

$$= \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}.$$

4. There are two groups of subjects; one of which consists of 5 science and 3 engineering subjects, and the other consists of 3 science and 5 engineering subjects. An unbiased die is cast. If the number 3 or number 5 turns up, a subject is selected at randam from the first group, otherwise the subject is selected at randam from the second group. Find probability that an engineering subject is selected ultimately.

Solution

Probability of turning up 3 or 5 $= \frac{2}{6} = \frac{1}{3}$.

Probability of selecting an engineering subject from first group $= \frac{3}{8}$

Hence probability of selecting an engineering subject from first group on turning up 3 or 5

$$= \frac{1}{3} \times \frac{3}{8} = \frac{1}{8}$$

Now probability of not turning 3 or 5

$$= 1 - \frac{1}{3} = \frac{2}{3}$$

Probability of selecting an engineering subject from second group $= \frac{5}{8}$

Hence probability of selecting an engineering subject from second group on turning up 3 or 5

$$= \frac{2}{3} \times \frac{5}{8} = \frac{5}{12}$$

Thus probability of selecting an engineering subject

$$= \frac{1}{8} + \frac{5}{12} = \frac{13}{24}$$

5. A box A contains 2 white and 4 black balls. Another box B contains 4 white and 7 black balls. A ball is transferred from the box A to the box B. Then a ball is drawn from the box B. Find the probability that it is white.

Solution

Probability of drawing a white ball from box B will depend on whether the transferred ball is black or white.

If a black ball is transferred, its probability is $\frac{4}{6}$.

There are now 5 white and 8 black balls in the box B. Then probability of drawing white ball from box B is $\frac{5}{13}$.

Thus probability of drawing a white ball from urn B, if transferred ball is black $= \frac{4}{6} \times \frac{5}{13} = \frac{10}{39}$

Similarly the probability of drawing a white ball from urn B, if transferred ball is white

$$= \frac{2}{6} \times \frac{6}{13} = \frac{2}{13}$$

$$\therefore \text{ Required probability} = \frac{10}{39} + \frac{2}{13} = \frac{16}{39}.$$

6. Two persons A and B toss an unbiased coin alternately on the understanding that the first who gets the head wins. If A starts the game, find their respective chances of winning.

Solution

Probability of getting a head $= \frac{1}{2}$.

Then A can win in 1st, 3rd, 5th, ... throws.

Hence chances of A's winning

$$= \frac{1}{2} + \left(\frac{1}{2}\right)^2 \frac{1}{2} + \left(\frac{1}{2}\right)^4 \frac{1}{2} + \left(\frac{1}{2}\right)^6 \frac{1}{2} + \ldots\ldots$$

$$= \frac{\frac{1}{2}}{1 - \left(\frac{1}{2}\right)^2} = \frac{1}{2} \cdot \frac{4}{3} = \frac{2}{3}\ldots\ldots$$

Hence chance of B's winning $= 1 - \frac{2}{3} = \frac{1}{3}$.

7. (*i*) Is the function defined as follows a density function?

$$f(x) = e^{-x}, x \geq 0 = 0, \quad x < 0$$

(*ii*) If so, determine probability that the variate having this density will fall in the interval (1, 2) ?

(*iii*) Also find cumulative probability function F(2)?

Solution

(*i*) $f(x)$ is clearly ≥ 0 for every x in (1, 2) and

$$\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{0} 0.dx + \int_{0}^{\infty} e^{-x}dx = 1$$

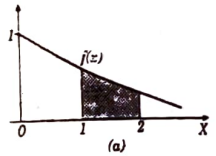
Hence function $f(x)$ satisfies the requirements for a density function.

(*ii*) Required probability = P $(1 \leq x \leq 2)$

$$= \int_1^2 e^{-x}dx = e^{-1} - e^{-2}$$

$$= 0.368 - 0.135 = 0.233.$$

This probability is equal to the shaded area in Fig. (*a*).

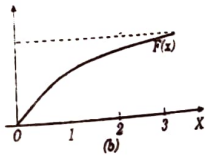


(a)

(*iii*) Cumulative probability function F(2)

$$\int_{-\infty}^{2} f(x)dx = \int_{-\infty}^{0} 0.dx + \int_{0}^{2} e^{-x}dx$$

$$= 1 - e^{-2} = 1 - 0.135 = 0.865$$



(b)

which is shown in Fig.(*b*).

8. X is a continuous random variable with probability density function given by

$$f(x) = kx \ (0 \leq x < 2) = 2k \ (2 \leq x < 4)$$
$$= -kx + 6k \ (4 \leq x < 6)$$

Find k and mean value of X.

Solution.

Since total probability is unity

$$\therefore \quad \int_0^6 f(x)dx = 1$$

i.e., $\int_0^2 kx\,dx + \int_2^4 2k\,dx + \int_4^6 (-kx + 6k)dx = 1$

$$\Rightarrow k\left|\frac{x^2}{2}\right|_0^2 + 2k|x|_2^4 + \left(-k\frac{x^2}{2} + 6kx\right)_4^6 = 1$$

$$\Rightarrow 2k + 4k + (-10k + 12k) = 1 \Rightarrow k = \frac{1}{8}.$$

Mean of X

$$= \int_0^6 xf(x)dx$$

$$= \int_0^2 kx^2dx + \int_2^4 2kx\,dx + \int_4^6 x(-kx + 6k)dx$$

$$= k\left|\frac{x^3}{3}\right|_0^2 + 2k\left|\frac{x^2}{2}\right|_2^4 + \left(-k\left|\frac{x^3}{3}\right|_4^6 + 6k\left|\frac{x^2}{2}\right|_4^6\right)$$

$$= k\left(\frac{8}{3}\right) + k(12) - k\left(\frac{152}{3}\right) + 3k(20) = \frac{1}{8}(24) = 3$$

9. The frequency distribution of a measurable characteristic varying between 0 and 2 is as under $f(x) = x^3, 0 \leq x \leq 1 = (2 - x)^3, 1 \leq x \leq 2$. Calculate standard deviation and also mean deviation about the mean.

Solution

Total frequency, $N = \int_0^1 x^3\,dx + \int_1^2 (2 - x)^3\,dx$

$$= \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$\therefore \ \mu_1'$ (about the origin)

$$= \frac{1}{N}\left[\int_0^1 x.x^3\,dx + \int_1^2 x(2 - x)^3\,dx\right]$$

$$= 2\left\{\left|\frac{x^5}{5}\right|_0^1 + \left|-x.\frac{(2-x)^4}{4}\right|_1^2 - \left|\frac{(2-x)^5}{20}\right|_1^2\right\}$$

$$= 2\left(\frac{1}{5} + \frac{1}{4} + \frac{1}{20}\right) = 1$$

μ_2' (about the origin)

$$= \frac{1}{N}\left[\int_0^1 x^2.x^3dx + \int_1^2 x^2(2 - x)^3\,dx\right]$$

$$= 2\left\{\left|\frac{x^6}{6}\right|_0^1 + \left|-x^2\frac{(2-x)^4}{4}\right|_1^2 + \frac{1}{2}\int_1^2 x(2 - x^4)dx\right\}$$

$$= 2\left\{\frac{1}{6} + \frac{1}{4} + \frac{1}{2}\left[\frac{1}{5} + \frac{1}{30}\right]\right\} = \frac{16}{15}$$

$$\therefore \quad \sigma^2 = \mu_2 = \mu_2' - (\mu_1')^2 = \frac{1}{15}$$

i.e., standard deviation $\sigma = \frac{1}{\sqrt{15}}$

Mean deviation about the mean

$$= \frac{1}{N}\left\{\int_0^1 |x-1|x^3\,dx + \int_1^2 |x-1|(2-x)^3\,dx\right\}$$

$$= 2\left\{\int_0^1 (1-x)x^3\,dx + \int_1^2 (x-1)(2-x)^3\,dx\right\}$$

$$= 2\left\{\left(\frac{1}{4}-\frac{1}{5}\right)+\left(0+\frac{1}{20}\right)\right\} = \frac{1}{5}$$

10. The probability that a pen manufactured by a company will be defective is $\frac{1}{10}$. If 12 such pens are manufactured, find the probability that

(*a*) exactly two will be defective.

(*b*) at least two will be defective.

(*c*) none will be defective

Solution

Probability of a defective pen $= \frac{1}{10} = 0.1$

Hence probability of a non-defective pen

$$1 - 0.1 = 0.9$$

(*a*) Probability that exactly two will be defective

$= {}^{12}C_2\,(0.1)^2\,(0.9)^{10} = 0.2301.$

(*b*) Probability that at least two will be defective

= 1 – (probability that either none or one is non-defective)

$= 1 - [{}^{12}C_0(0.9)^{12} + {}^{12}C_1(0.1)(0.9)^{11}] = 0.3412.$

(*c*) Probability that none will be defective

$= {}^{12}C_{12}\,(0.9)^{12} = 0.2833$

11. In 256 sets of 12 tosses of a coin, in how many cases one can expect 8 heads and 4 tails.

Solution

$$\text{P(head)} = \frac{1}{2} \text{ and P(tail)} = \frac{1}{2}$$

By Binomial distribution, probability of 8 heads and 4 tails in 12 trials,

$$\text{P}(X=8) = {}^{12}C_8\left(\frac{1}{2}\right)^8\left(\frac{1}{2}\right)^4 = \frac{12!}{8!4!}\cdot\frac{1}{2^{12}} = \frac{495}{4096}$$

$\therefore$ Expected number of such cases in 256 sets

$$= 256 \times \text{P}(X=8) = 256\,\frac{495}{4096} = 30.9 \approx 31$$

12. If probability of a bad reaction from a certain injection is 0.001, determine the chance that out of 2,000 individuals more than two will get a bad reaction.

Solution

It follows a Poisson distribution as probability of occurrence is very small.

Mean, $m = np = 2000\,(0.001) = 2$

Probability that more than 2 will get a bad reaction

= 1 – [probability that no one gets a bad reaction + probability that one gets a bad reaction + probability that two get bad reaction]

$$= 1 - \left[e^{-m} + \frac{m^1 e^{-m}}{1!} + \frac{m^2 e^{-m}}{2!}\right]$$

$$= 1 - \left[\frac{1}{e^2} + \frac{2}{e^2} + \frac{2}{e^2}\right] \qquad \ldots[\because m = 2]$$

$$= 1 - \frac{5}{e^2}$$

$$= 0.32 \qquad [\because e = 2.718]$$

EXERCISE

MCQ TYPE QUESTIONS

1. If A, B, C are three events, then P[A ∩ (B ∪ C)] =

(*a*) P(A) + P(B) + P(C) – P(A ∩ B) – P(A ∩ C)

(*b*) P(A) + P(B) + P(C) – P(B) P(C)

(*c*) P(A ∩ B) + P(A ∩ C) – P(A ∩ B ∩ C)

(*d*) P(B ∩ C) + P(A ∩ B) – P(A ∩ B ∩ C)

2. If A and B are two arbitrary events, then P(A ∩ B) cannot be

(*a*) less than P(A) + P(B) – 1

(*b*) greater than P(A) + P(B)

(*c*) equal to P(A) + P(B) – P(A ∪ B)

(*d*) equal to P(A) + P(B) + P(A ∪ B)

3. If M and N are two events, then probability that exactly one of them occurs is not equal to

(*a*) P(M) + P(N) – 2P(M ∩ N)

(*b*) P(M) + P(N) – P(M ∩ N)

(*c*) $\text{P}(\overline{M}) + \text{P}(\overline{N}) - 2\text{P}(\overline{M} \cap \overline{N})$

(*d*) $\text{P}(M \cap \overline{N}) + \text{P}(\overline{M} \cap N)$

4. If A and B are any two events then which one of the following pairs of events are mutually exclusive ?

(*a*) A ∪ B and $A \cap \overline{B}$

(*b*) A ∪ B and $\overline{A} \cap B$

(*c*) A ∪ B and $\overline{A} \cap \overline{B}$

(*d*) A ∪ B and A ∩ B

5. There are three events A, B, C one of which must, and only one can happen, the odds are 8 to 3 against A, 5 to 2 against B. The odds against C are

(*a*) 43 to 34 (*b*) 32 to 23

(*c*) 34 to 13 (*d*) none of these

6. A has one share in a lottery in which there is 1 prize and 2 blanks; B has three shares in a lottery in which there are 3 prizes and 6 blanks; compare the probability of A's success to that of B's success is

(*a*) 7 : 16 (*b*) 16 : 7 (*c*) 6 : 14 (*d*) 14 : 6

7. If A and B are independent, then A and $\overline{B}$ are also

(*a*) independent (*b*) dependent

(*c*) both (*a*) and (*b*) (*d*) none of these

8. Let $S = \left\{1, \frac{1}{2}, \left(\frac{1}{2}\right)^2, \ldots, \left(\frac{1}{2}\right)^n\right\}$.

Let A and B be two events defined as $A = \left\{1, \frac{1}{2}\right\}$ and $B = \left\{\left(\frac{1}{2}\right)^2, \left(\frac{1}{2}\right)^4, \ldots\right\}$.

Then P (A ∩ B) will be

(*a*) 0 (*b*) 1

(*c*) ∞ (*d*) none of these

9. If $E_1, E_2, \ldots, E_n$ are n mutually exclusive events, then the probability of the happening of any one of them is equal to the sum of the probabilities of the happening of separate events.

i.e., $\text{P}(E_1 \cup E_2 \cup E_3 \ldots \ldots \cup E_n)$

$= \text{P}(E_1) + \text{P}(E_2) + \ldots\ldots + \text{P}(E_n)$.

This is

(*a*) Law of total probability

(*b*) Law of probability

(*c*) Both (*a*) and (*b*)

(*d*) None of these

10. What is the probability of correctly choosing an unknown integer between 0 to 9 within three chances ?

(*a*) $\frac{963}{1000}$ (*b*) $\frac{973}{1000}$

(*c*) $\frac{983}{1000}$ (*d*) None of these

11. If $\text{P}(A/C) \geq \text{P}(B/C)$ and $\text{P}(A/\overline{C_1}) \geq \text{P}(B/\overline{C_1})$, then

(*a*) P (A) ≥ P (B) (*b*) P (A) ≤ P (B)

(*c*) P (A) < P (B) (*d*) P (A) > P (B)

12. If A and B are independent and P (C) = 0, then A, B and C are independent.

(*a*) True (*b*) False

13. Six dice are thrown simultaneously. The probability that all will show different faces is

(*a*) $\frac{5!}{6^5}$ (*b*) $\frac{5!}{6^4}$

(*c*) $\frac{5!}{6^3}$ (*d*) none of these

14. Given two events A and B and $\text{P}(A) = \frac{1}{4}$, $\text{P}(B/A) = \frac{1}{2}$ and $\text{P}(A/B) = \frac{1}{4}$.

State that following is true?

(*a*) A is sub-event of B

(*b*) $\text{P}(A/B) = \frac{3}{4}$

(*c*) $\text{P}(A/B) + \text{P}(A/\overline{B}) = 1$

(*d*) None of these

15. If $P(A) = 0$, then $P(A \cap B) = ?$
(a) 0 (b) 1
(a) ∞ (b) none of these

16. Which of the following is correct?
(a) If $P(A) > 0$, $P(B) > 0$ and $P(A/B) = P(B/A)$, then
$P(A) = P(B)$.
(b) If A and B mutually exclusive, then
$P(A/\bar{B}) = P(A)/(1 - P(B))$.
(c) If A and B are independent, then
$P(A \cup B) = 1 - P(\bar{A}) . P(\bar{B})$
(d) All of these

17. If $P(A) = \frac{3}{4}$ and $P(B) = \frac{5}{8}$, then
(a) $P(A \cup B) \geq \frac{3}{4}$ (b) $\frac{3}{8} < P(A \cap B) \leq \frac{5}{8}$
(c) $\frac{1}{8} \leq P(A \cap \bar{B}) \leq \frac{3}{8}$ (d) All of these

18. Three machines M_1, M_2 and M_3 produce identical items. Of their respective output 5%, 4% and 3% of items are faulty. On a certain day, M_1 has produced 25% of the total output, M_2 has produced 30% and M_3 the remainder. An item selected at random is found to be faulty. What are the chances that it was produced by the machine with the highest output?
(a) 0.155 (b) 0.255 (c) 0.355 (d) 0.455

19. In an experiment a coin is tossed 4 times. What is the size of the sample space?
(a) 12 (b) 14 (c) 16 (d) 20

20. For any two events A and B
(a) $P(\bar{A} \cap B) = P(B) - P(A \cap B)$
(b) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
(c) $P(A/B) \leq P(A)$.
(d) All of these

21. Which of the following is correct?
(a) $P(A \cup B/C) = P(A/C) + P(B/C) - P(A \cap B/C)$
(b) $P(A \cap B/C) + P(A \cap B/C) = P(A/C)$
(c) If A and B are independent, then A and $\bar{B}$ are also independent.
(d) All of these

22. Which of the following is correct?
(a) If A, B and C are mutually independent, then $A \cup B$ and C are also independent.
(b) The events $E_1, E_2, \ldots\ldots, E_n$ are mutually exclusive. Let $E = \bigcup_{i=1}^{n} E_i$.
If $P(A/E_i) = P(B/E_i)$, i = 1, 2, ..., n, then
$P(A/E) = P(B/E)$.
(c) Both (a) and (b)
(d) None of these

23. Let p be the probability that a man aged y will get into an accident in a year. What is the probability that a man among n men of all aged y will get into an accident first ?
(a) $\frac{1}{n}\left(1-(1+p)^n\right)$ (b) $\frac{1}{n}\left(1-(1+p)^n\right)$
(c) $n\left(1-(1+p)^n\right)$ (d) None of these

24. What would be the expectation of the number of failures preceding the first success in an infinite series of independent trials with the constant probability of success p ?
(a) $\frac{1}{p}$ (b) $\frac{1}{q}$
(c) $\frac{q}{p}$ (d) None of these

25. A restaurant serves two special dishes A and B to its customers consisting of 60% men and 40% women. 80% of men order dish A and the rest B and 70% women order dish A and the rest B. In what ratio of A to B should the restaurant prepare the dishes?
(a) 19 : 6 (b) 6 : 9 (c) 16 : 7 (d) 7 : 16

26. If A and B are mutually exclusive and $P(A \cup B) = P(A) + P(B)$, then $P(A/A \cup B) =$
(a) $\frac{P(A)}{P(A)+P(B)}$ (b) $\frac{P(A)}{P(A)-P(B)}$
(c) $\frac{P(B)}{P(A)+P(B)}$ (d) None of these

27. If $P(A) = P(\bar{B})$, then
(a) $A = \bar{B}$ (b) $B = 0$
(c) $A \neq B$ (d) None of these

28. If $P(A) = 0$, then
(a) $A = \phi$ (b) $A = 0$
(c) $A = \infty$ (d) none of these

29. The mean of a set of number is $\bar{x}$. If each number is increased by λ, then mean of the new set is
(a) $\bar{x}$ (b) $\bar{x} + \lambda$
(c) $\lambda\bar{x}$ (d) none of these

30. If each observation of raw data whose variance is σ^2, is increased by λ, then variance of the new set is
(a) σ^2 (b) $\lambda^2\sigma^2$
(c) $\lambda + \sigma^2$ (d) $\lambda^2 + \sigma^2$

31. If μ is mean of distribution, then $\sum f_i(\gamma_i - \mu)$ is equal to
(a) M.D (b) stamdard deviation
(c) 0 (d) none of these

32. $b_{xy} \times b_{yx}$ is equal to
(a) $\rho(X, Y)$ (b) cov (x, y)
(c) $[\rho(x, y)]^2$ (d) None of these

33. If cov (X, Y) = 0, then which of the following is negative?
(a) b_{yx} (b) b_{xy}
(c) ρ (X, Y) (d) none of these

34. The mean of a set of observations is $\bar{x}$. If each observation is divided by α, $\alpha \neq 0$ and then is increased by 10, then the mean of new set is
(a) $\frac{\bar{x}}{\alpha}$ (b) $\frac{\bar{x}+10}{\alpha}$
(c) $\frac{\bar{x}+10\alpha}{\alpha}$ (d) $\alpha\bar{x} + 10$

35. S.D. of n observation $a_1, a_2, a_3, \ldots\ldots a_n$ is σ, then If S.D. of the observations $\lambda a_1, \lambda a_2, \lambda a_3, \ldots\ldots \lambda a_n$ is
(a) $\lambda\sigma$ (b) $-\lambda\sigma$
(c) $|\lambda|\sigma$ (d) σ

36. If two lines of regression are $Y = 3x - 5$ and $Y = 2x - 4$, then $\rho(X, Y)$ is equal to
(a) $\sqrt{\frac{2}{3}}$ (b) $\sqrt{\frac{1}{6}}$
(c) $\sqrt{\frac{3}{2}}$ (d) none of these

37. If two lines regression are $3x - y - 5 = 0$ and $2x - y - 4 = 0$, then $\bar{x}$ and $\bar{y}$ are respectively
(a) 1 and –2 (b) –1 and 2
(c) 2 and –1 (d) –2 and –1

38. If regression equation of Y on X is $Y = \lambda x + 4$ and that of X on Y be $4x = Y - 5$, then
(a) $0 < 4\lambda \leq 1$ (b) $0 \leq \lambda \leq 1$
(c) $0 \leq \lambda \leq 4$ (d) none of these

39. If cov (X,Y) = 0, then two lines of the regression are
(a) parallel (b) concident
(c) at right angles (d) none of these

40. If in a regression analysis problem, $b_{yx} = -0.9$ and $b_{xy} = 0.4$, then $\rho(x, y)$ is
(a) 0.6 (b) –0.6
(c) 0 (d) none of these

41. It b_{xy} and b_{yx} be the two regression coefficients, then coefficient of correlation ρ is given by
(a) $\rho = \sqrt{b_{xy}b_{yx}}$
(b) $\rho = -\sqrt{b_{yx}b_{yx}}$
(c) $\rho = (S_{gn}\bar{b}_{yx})\sqrt{b_{xy}b_{yx}}$
(d) none of these

42. It standard deviation for two variables X and Y are 3 and 4 respectively and their covariance is 8, then correlation coefficient between them is
(a) $\frac{2}{3}$ (b) $\frac{8}{3\sqrt{2}}$ (c) $\frac{9}{8\sqrt{2}}$ (d) $\frac{2}{9}$

43. Mean deviation of the data
$a, a+d, a+2d, \ldots a+2nd$ from the mean is equal to
(a) $\frac{n(n+1)|d|}{2n+1}$ (b) $\frac{n(n+1)d}{2n+1}$
(c) $\frac{(n+1)|d|}{2n}$ (d) none of these

44. If two variables X and Y have a perfect correlation (direct or indirect), then they may be connected by a relation of the type
(a) $xy = a^2$ (b) $\frac{a}{x} + \frac{b}{y} = 1$
(c) $\frac{x}{a} + \frac{y}{b} = 1$ (d) none of these

45. If $\rho > 0$ and $m = \frac{b_{yx} + b_{xy}}{2}$, then
(a) $m \geq \rho$ (b) $m = \rho$
(c) $m \leq \rho$ (d) none of these

46. If two lines of regression are $x + 2y = 7$ and $2x + y = 7$, then regression equation of Y and X is
(a) $2x + y = 7$ (b) $x + 2y = 7$
(c) $x + 2y = 0$ (d) $2x + y = 0$

47. Two variables X and Y are connected by the relation $ax + by + c = 0$ where $ab < 0$, then $\rho(x, y)$ is equal to
(a) $\frac{a}{b}$ (b) $\frac{b}{a}$ (c) –1 (d) 1

48. Mean of the numbers 0, 1, 2, 3, ...n with respective weight ${}^nC_0, {}^nC_1, {}^nC_2, \ldots\ldots, {}^nC_n$ is
(a) $\frac{2^n}{n+1}$ (b) $\frac{2^{n+1}}{n+1}$
(c) $\frac{n}{2}$ (d) none of these

NUMERICAL TYPE QUESTIONS

1. A card is taken out of a pack of 52 cards numbered 2 to 53. The probability that the numbers on the card is a prime number less than 20 is _____

2. The probability of getting a number between 1 and 100. Which is divisible by one and itself only is _____

3. If A and B are two elements such tha
$P(A) = \frac{2}{3}$, $P(B) = \frac{3}{4}$, $P(A \cap B) = \frac{1}{2}$, the
$P(A \cup B)$ is _____

4. If A and B are two events such that $P(A \cup B) = 0.65$, $P(A \cap B) = 0.15$, then $P(\overline{A}) + P(\overline{B})$ is _____

5. There are two bags one of which contains 3 black and 4 white balls while other contains 4 black and 3 white balls. A die is cast, if the face 1 or 3 turns up, a ball is taken from first bag and if any other face turns up, a ball is chosen from second bag. The probability of choosing of black ball is_____

6. An urn contains 5 red and 10 black balls. Eight of them are placed in another urn. The chance that the latter then contains 2 red and 6 black balls is_____

7. The chance that a leap year selected at random will contain 53 Sundays is _____

8. A five figure number is formed by the digits 0, 1, 2, 3, 4 without repetition. Then the probability that the number formed is divisible by 4 is____

9. A man alternately tosses a coin and throws a dice, beginning with the coin. Then the probability that he will get a head before he gets a 5 or 6 on dice is_____

10. A box A contains 2 white and 4 black balls. Another box B contains 5 white and 7 black balls. A ball is transferred from the box A to the box B. Then a ball is drawn from the box B. The probability that it is white is _____

11. A problem in mechanics is given to three students A, B and C whose chances of slogging it are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. Then probability that the problem will be solved is _____

12. A and B throw alternately with a pair of dice. A wins if he throws 6 before B throws 7 and B wins if he throws 7 before A throws 6. If A begins, then his chance of winning is _____

13. If mode of a data is 18 and mean is 24, then median is _____

14. Regression equation of Y on X is

$8x - 10y + 66 = 0$ and $\sigma_x = 3$

Hence cov (X, Y) is equal is _____

15. Coefficient of correlation between two variates X and Y is _____

X	1	2	3	4	5
Y	5	4	3	2	1

16. If mean = (3 median − mode) x, then value of x is_____

17. If median = (mode + 2 mean) μ, then μ is equal to_____

18. If two lines of regression are at right angles, then ρ (X, Y) is equal to _____

19. Given : $n = 10$, $\Sigma x = 4$, $\Sigma y = 3$, $\Sigma x^2 = 8$, $\Sigma y^2 = 9$ and $\Sigma xy = 3$, then coefficient of correlation is _____

20. 25% of the items of data are less than 35 and 25% of the items are more than 75. Q.D of the data is_____

21. If X and Y are two independent variables such that $\bar{x} = 5$, $\bar{y} = 10$ $\sigma_x^2 = 4$ $\sigma_y^2 = 9$ and if $u = 3x + 4y$, $v = 3x - y$, then $\rho(u, v)$ is equal to _____

ANSWERS

MCQ Type Questions

1. (c)	2. (c)	3. (b)	4. (c)	5. (a)	6. (a)	7. (a)	8. (a)	9. (a)	10. (b)
11. (a)	12. (a)	13. (b)	14. (d)	15. (a)	16. (a)	17. (d)	18. (c)	19. (c)	20. (d)
21. (a)	22. (a)	23. (a)	24. (c)	25. (a)	26. (a)	27. (a)	28. (a)	29. (b)	30. (b)
31. (c)	32. (c)	33. (d)	34. (c)	35. (c)	36. (a)	37. (a)	38. (c)	39. (c)	40. (d)
41. (c)	42. (a)	43. (a)	44. (c)	45. (a)	46. (b)	47. (d)	48. (c)		

Numerical Type Questions

1. $\frac{2}{13}$	2. $\frac{25}{98}$	3. $\frac{11}{12}$	4. 1.2	5. $\frac{11}{21}$	6. $\frac{140}{429}$	7. $\frac{2}{7}$	8. $\frac{5}{16}$	9. $\frac{3}{4}$
10. $\frac{16}{39}$	11. $\frac{3}{4}$	12. $\frac{30}{61}$	13. 22	14. 7.2	15. −1	16. $\frac{1}{2}$	17. $\frac{1}{3}$	18. 0
19. $\frac{1}{4}$	20. 50	21. 0						

MCQ TYPE QUESTIONS

5. Since odds are 8 to 3 against A

$\therefore \quad P(A) = \frac{3}{11}$; $(P(B) = \frac{2}{7})$

since odds are 5 to 2 aganist B

and $P(A) + P(B) + P(C) = 1$

$\therefore \quad P(C) = 1 - \frac{3}{11} - \frac{2}{7} = \frac{34}{77}$

Hence odds against C are 43 to 34.

6. A can draw a ticket in ${}^3C_1 = 3$ ways.
Number of cases in which A can get a prize is clearly 1.

$\therefore$ Probability of A's success $= \frac{1}{3}$.

Again B can draw a ticket in ${}^9C_3 = \frac{9\times8\times7}{3\times2\times1}$
$= 84$ way.

Number of ways in which B gets all blanks

$= {}^6C_3 = \frac{6\times5\times4}{3\times2\times1} = 20.$

$\therefore$ Number of ways of getting a prize
$= 84 - 20 = 64$

Thus, probability of B's success $= \frac{64}{84} = \frac{16}{21}$

$\therefore$ A's probability of success : B's probability of success

$= \frac{1}{3} : \frac{16}{21} = 7 : 16$

7. $P(A \cap \overline{B}) = P(A) - P(A \cap B)$
$= P(A) - P(A)\,P(B)$
$= P(A)\,(1 - P(B)) = P(A)\,P(\overline{B})$

Hence, A and $\overline{B}$ are independent.

8. $A \cap B = \phi$
$\therefore \quad P(A \cap B) = 0$

10. $S = \{0, 1, 2, ..., 9\}$
Suppose we want a particular integer "3" to be chosen.

Probability of not choosing "3" is $\frac{9}{10}$ (in one chance)
Probability of not choosing 3 in all the three chances

$= \frac{9}{10} \times \frac{9}{10} \times \frac{9}{10}$

Hence probability of choosing it in at least one of the three chances

$= 1 - \frac{27}{1000} = \frac{973}{1000}$

11. $\frac{P(A \cap C)}{P(C)} > \frac{P(B \cap C)}{P(C)}$

$\Rightarrow \quad P(A \cap C) > P(B \cap C)$

and $\frac{P(A \cap \overline{C})}{P(\overline{C})} > \frac{P(B \cap \overline{C})}{P(\overline{C})}$

$\Rightarrow \quad P(A \cap \overline{C}) > P(B \cap \overline{C})$

Adding

$P(A \cap C) + P(A \cap \overline{C}) > P(B \cap C) + P(B \cap \overline{C})$

$P(A) > P(B)$ since $(A \cap C)$ and $A \cap \overline{C}$ are disjoint.

12. $P(C) = 0$
$\Rightarrow \quad C = \phi$
$P(A \cap B \cap C) = P(A \cap B \cap \phi)$
$= P(\phi) = 0$
$P(A) \times P(B) \times P(C) = 0$ ... since $P(C) = 0$
$\therefore \quad P(A \cap B \cap C) = P(A) - P(B) - P(C)$
Hence A, B, C are independent.

13. Number of ways different numbers can occur on six dice = 6!
Total number of ways of occurrence = 6^5

$\therefore$ Required probability, $P = \frac{6!}{6^5} = \frac{5!}{6^4}$

14. $P(A) = \frac{1}{4}$

and $P(B/A) = \frac{P(B \cap A)}{P(A)} = \frac{1}{2}$

$\Rightarrow \quad P(A \cap B) = \frac{1}{8}$, $P(A/B) = \frac{1}{4}$

$\Rightarrow \quad \frac{P(A \cap B)}{P(B)} = \frac{1}{4} \quad \Rightarrow P(B) = \frac{1}{2}$

(a) If A is a sub-event of B, $A \subset B$, then $A \cap B = A$
$\therefore P(A \cap B)$ should be equal to P(A). This is false.

(b) $P(A/B) = \frac{3}{4}$ is false.

(c) $P(A \cap \overline{B}) = P(A) - P(A \cap B) = \frac{1}{4} - \frac{1}{8} = \frac{1}{8}$

$\therefore \quad P(A/\overline{B}) = \frac{P(A \cap \overline{B})}{P(\overline{B})} = \frac{\frac{1}{8}}{1 - \frac{1}{2}} = \frac{1}{4}$

$\therefore \quad P(A/B) + P(A/\overline{B}) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

Hence given statement is false.

15. $P(A) = 0$

$\Rightarrow A = \phi$

$\therefore A \cap B = \phi$

$\therefore P(A \cap B) = P(\phi) = 0$

17. $(A \cup B) \supset A$

(a) $P(A \cup B) \geq P(A) = \frac{3}{4}$

(b) $A \cap B \subset B$

$\therefore P(A \cap B) \leq P(B) = \frac{5}{8}$

Also $P(A \cap B) = 1 - P(A \cap B)^c$

$= 1 - P(A^c \cup B^c)$

$> 1 - [P(A^c) + P(B^c)]$

$= 1 - \left\{\frac{1}{4} + \frac{3}{8}\right\} = \frac{3}{8}$

(c) $A \cap \bar{B} \subset \bar{B}$

$\therefore P(A \cap \bar{B}) \leq P(\bar{B}) = \frac{3}{8}$

Also $A \cap \bar{B} = (\overline{A \cup B})$

$\therefore P(A \cap \bar{B}) = 1 - P(\bar{A} \cup B) \geq 1 - [P(\bar{A}) + P(B)]$

$= 1 - \left\{\frac{1}{4} + \frac{5}{8}\right\} = \frac{1}{8}$

18. Let the event of drawing a faulty item from any of the machines be A, and the event that an item drawn at random was produced by M_i be B_i. To $P(B_i/A)$ proceed as follows :

	M_1	M_2	M_3	Remarks
$P(B_i)$	0.25	0.30	0.45	$\because$ sum = 1
$P(A/B_i)$	0.05	0.04	0.03	
$P(B_i)P(A/B_i)$	0.0125	0.012	0.0135	sum = 0.38
$P(B_i/A)$	$\frac{0.0125}{0.038}$	$\frac{0.0125}{0.038}$	$\frac{0.0135}{0.038}$	By Baye's theorem

The highest output being from M_3, required probability $= \frac{0.0135}{0.038} = 0.355$.

19. Sample space, $S = \{HHHH, HHHT, HHTH, HHTT, \ldots, TTTT\}$

Clearly, $|S| = 2^4 = 16$.

20. (a) $B = (A \cap B) \cup (\bar{A} \cap B)$

$\Rightarrow P(B) = P(A \cap B) + P(\bar{A} \cap B)$ as $A \cap B$ and $\bar{A} \cap B$ are mutually exclusive.

$\Rightarrow P(\bar{A} \cap B) = P(B) - P(A \cap B)$.

(b) $P(A \cup B) = P[A \cup (\bar{A} \cap B)]$

$= P(A) + P(\bar{A} \cap B)$

$= P(A) + P(B) - P(A \cap B)$

(c) $P(A/B) = P(A)\,P(B/A)$

$\leq P(A)$ as $P(B/A) \leq 1$.

21. (a) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\therefore P((A \cap C) \cup (B \cap C)) = P(A \cap C) + P(B \cap C) - P(A \cap C \cap B \cap C)$

and $P((A \cup B) \cap C) = P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)$

$\Rightarrow \frac{P((A \cup B) \cap C)}{P(C)} = \frac{P(A \cap C)}{P(C)} + \frac{P(B \cap C)}{P(C)} - \frac{P(A \cap B \cap C)}{P(C)}$

$\Rightarrow P(A \cup B/C) = P(A/C) + P(B/C) - P(A \cap B/C)$

(b) $P(A \cap \bar{B}/C) + P(A \cap B/C)$

$= \frac{P(A \cap \bar{B} \cap C)}{P(C)} + \frac{P(A \cap B \cap C)}{P(C)}$

$= \frac{P(A \cap \bar{B} \cap C) + P(A \cap B \cap C)}{P(C)}$

$= \frac{P(A \cap \bar{B} \cap C) \cup (A \cap B \cap C)}{P(C)}$

$= \frac{P(A \cap C)}{P(C)} = P(A/C)$

(c) $P(A \cap \bar{B})$

$= P(A) - P(A \cap B) = P(A) - P(A)\,P(B)$

$= P(A)(1 - P(B)) = P(A)\,P(\bar{B})$

Hence A and $\bar{B}$ are independent.

22. (a) $P((A \cup B) \cap C) = P((A \cap C) \cup (B \cap C))$

$= P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)$

$= P(A)\,P(C) + P(B)\,P(C) - P(A)\,P(B)\,P(C)$

$= P(C)(P(A) + P(B) - P(A \cap B))$

$= P(C)\,P(A \cup B)$

(b) $P(A/E) = P(B/E)$

$\frac{P(A \cap E)}{P(E)} = \frac{P(B \cap E)}{P(E)}$

if $P(A \cap E) = P(B \cap E)$

if $P\left(A \cap \bigcup_{i=1}^{n} E_i\right) = P\left(B \cap \bigcup_{i=1}^{n} E_i\right)$

if $P\left(\bigcup_{i=1}^{n} (A \cap E_i)\right) = P\left(\bigcup_{i=1}^{n} (B \cap E_i)\right)$

$\sum_{i=1}^{n} P(A \cap E_i) = \sum_{i=1}^{n} P(B \cap E_i)$

This is true since $P(A \cap E_i) = P(B \cap E_i)$ for each i.

23. Let E_i be the event of a person to get into an accident. Then

$P(E_i) = p \ \forall\ i$

P (at least one man meet with an accident)

$= P(E_1 \cup E_2 \cup \ldots \cup E_n)$

$= 1 - P(\bar{E}_1 \cup \bar{E}_2 \cup \ldots \cup \bar{E}_n)$

$= 1 - P(\bar{E}_1 \cap \bar{E}_2 \cap \ldots \cap \bar{E}_n)$

$= 1 - P(\bar{E}_1)\,P(\bar{E}_2) \ldots P(\bar{E}_n)$

$= 1 - (1-p)(1-p) \ldots (1-p)$

$= 1 - (1-p)^n$

Hence P (at least one man meets with an accident/a person is chosen) $= \frac{1}{n}(1 - (1-p)^n)$.

24. Expectation of X = E(X) = average = mean

$= \frac{1}{n} \sum_{i=1}^{n} x_i$

$= \sum_{i=1}^{n} p_i x_i$

where p_i denote probability of occurrence of x_i.

Let $q = 1 - p$.

Then probabilities of success in 1st, 2nd, 3rd trials are

$p, qp, q^2p, \ldots\ldots$

$\therefore E(X) = (0) \cdot p + (1) \cdot qp + (2) \cdot q^2 p + \ldots$

$= qp(1 + 2q + 3q^2 + \ldots + nq^{n+1} + \ldots)$

$= \frac{qp}{(1-q)^2} = \frac{qp}{p^2} = \frac{q}{p}$

25. Ratio of A and B

$= 60 \times 80 + 40 \times 70 : 60 \times 20 + 40 \times 30$

$= 76 : 24 = \mathbf{19 : 6}$

26. $P(A/A \cup B) = \frac{P(A \cap (A \cup B))}{P(A \cup B)}$

$= \frac{P(A)}{P(A \cup B)} = \frac{P(A)}{P(A) + P(B)}$

27. $P(A) = P(\bar{B})$

$\Rightarrow P(A) = 1 - P(B) \Rightarrow P(A) + P(B) = 1$

$\Rightarrow P(A \cup B) - P(A \cap B) = 1 = P(S)$

Hence $A \cup B = S$ only if $A \cap B = \phi$

i.e. $A = \bar{B}$ only if $A \cap B = \phi$

28. $P(A) = 0$

$\Rightarrow P\left(\frac{A}{S}\right) = 0 \Rightarrow \frac{n(A)}{n(S)} = 0 \Rightarrow n(A) = 0$

$\Rightarrow A = \phi$

29. When each item is increased by λ, the mean is also increased by λ.

30. When each item is increased by λ, the S.D. is multiplied by $|\lambda|$ and variance is multiplied by λ^2.

31. $\Sigma f_i (y_i - \mu) = f_i y_i - M \Sigma f_i = \Sigma f_i y_i - \Sigma f_i Y_i = 0$

32. $b_{xy}\, b_{yx} = \frac{\text{cov}(X,Y)}{\sigma_x^2} \cdot \frac{\text{cov}(X,Y)}{\sigma_y^2}$

$= \left\{\frac{\text{cov}(X,Y)}{\sigma_x \sigma_y}\right\}^2 = |\rho\,X, Y)|^2$

33. Cov (X, Y), b_{xy}, b_{yx} and $\rho(X, Y)$ all are either positive, or 0 or negative simultaneously.

34. When each item is divided by α, *i.e.* it becomes $\frac{\bar{x}}{\alpha}$

When each item is increased by 10, mean also increased by 10.

$\therefore$ New mean $= \frac{\bar{x}}{\alpha} + u = \frac{x + 10\alpha}{\alpha}$

35. Standard Deviation of new data

$= \sqrt{\frac{\Sigma(\lambda a_i)^2}{n} - \left(\frac{\Sigma \lambda a_i}{n}\right)^2} = \sqrt{\lambda^2} \sqrt{\frac{\Sigma a_i^2}{n} - \left(\frac{\Sigma a_i}{n}\right)^2} = |\lambda|\sigma$

36. If we take $y = 3x - 5$ as regression equation of Y on X and $Y = 2x - 4$ as that of X on Y, then

$b_{yx} = 3\, b_{xy} = \frac{1}{2}$

$\Rightarrow b_{yx}\, b_{xy} = \frac{3}{2}$, which is not possible.

Hence, equation $y = 3x - 5$, *i.e.* $x = \frac{1}{3}y + \frac{5}{3}$ is the regression equation of X on Y

and equation, $Y = 2x - 4$ is that Y on X.

Then $b_{yx} = 2$ and $b_{xy} = \frac{1}{3}$

$\Rightarrow b_{xy}\, b_{xy} = \frac{3}{2}$

$\Rightarrow |\rho(X, Y)|^2 = \frac{3}{2}$

$\Rightarrow \rho(XY) = \sqrt{\frac{2}{3}}$

37. Solving equations

$3x - y - 5 = 0$ and $2x - y - 4 = 0$,

we get $x = 1$ and $y = -2$.

38. Here $b_{yx} = \lambda$ and $b_{xy} = \frac{1}{4}$.

Since $0 \leq b_{yx}\, b_{xy} \leq 1$, therefore we must have

$0 \leq \lambda\left(\frac{1}{4}\right) \leq 1 \Rightarrow 0 \leq \lambda \leq 4$

As $b_{xy} \neq 0$, $\therefore b_{yx} \neq 0 \Rightarrow \lambda \neq 0$.

Hence, we must have $0 < \lambda \leq 4$.

39. $\text{Cov}(X, Y) = 0 \Rightarrow b_{yx} = b_{xy} = 0$

So, two lines of regression are $y - \bar{y} = 0$ and $x - \bar{x} = 0$, which are at right angles.

40. Given : $b_{yx} = -0.9$ is negative and $b_{xy} = 0.4$ is positive which is not possible as b_{yx} and b_{xy} are always of same sign. Hence given data is inconsistent.

41. We know $b_{yx}\, b_{xy} = \rho^2 \Rightarrow \rho = \pm\sqrt{b_{yx}\, b_{xy}}$

But b_{yx} and b_{xy} ρ and cov (X, Y) are all of the same sign

$$\therefore \quad \rho = (\text{sgn } b_{yx})\sqrt{b_{yx}\, b_{xy}}$$

42. $$\rho(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_x \sigma_y} = \frac{8}{3 \times 4} = \frac{2}{3}$$

43. Mean, $$\bar{x} = \frac{a + (a+d) + (a+2d) + \ldots + (a+2nd)}{2n+1}$$

$$= \frac{2n+1}{2}\{a + (a + (a+2nd))\} = a + nd$$

Sum of numerical deviations from the mean

$$= \sum_{i=1}^{2n+1} |\{a + (i-1)d\} - \{a + nd\}|$$

$$= |a - a - nd| + |a + d - a - nd| + \ldots$$
$$+ |a + nd - a - nd| + |a + (n+1)\, d - a - nd|$$
$$+ \ldots + |a + 2nd - a - nd|$$
$$= |d|\,|n + (n-1) + (n-2) + \ldots + 1 + 0$$
$$+ 1 + 2 + \ldots + n|$$
$$= |d| \times 2\{1 + 2 + 3 + \ldots + n\} = n(n+1)|d|$$

$$\therefore \quad \text{M.D. (about mean)} = \frac{n(n+1)|d|}{2n+1}$$

44. If two variables X and Y are in perfect correlation (direct or indirect), then there is a linear relationship between the two variables.

45. Since $\rho > 0$, therefore b_{yx} and b_{xy} are also positive

Hence $$\frac{b_{yx} + b_{xy}}{2} \geq \sqrt{b_{yx}\, b_{xy}}$$

$$\Rightarrow \quad \frac{b_{yx} + b_{xy}}{2} \geq \sqrt{\rho^2}$$

$$\Rightarrow \quad \frac{b_{yx} + b_{xy}}{2} \geq \rho \Rightarrow m \geq \rho$$

46. If first time $x + 2y = 7$ is taken as regression equation of Y on X, then we write it as

$$y = -\frac{1}{2}x + \frac{7}{2} \quad \Rightarrow \quad b_{yx} = -\frac{1}{2}$$

Also the second equation (which will be regression equation X on Y) can be written as

$$x = -\frac{1}{2}y + \frac{7}{2} \quad \Rightarrow \quad b_{xy} = -\frac{1}{2}$$

$$\therefore \quad b_{yx}\, b_{xy} = \left(\frac{-1}{2}\right)\left(\frac{-1}{2}\right) = \frac{1}{4} \leq 1$$

Hence regression equation of Y on X is

$$x + 2y = 7$$

47. Since X and Y connected by linear relationship, therefore they are in perfect correlation.

Also $ab < 0$

$$\frac{a}{b} < 0.$$

$$\therefore \quad y = \left(-\frac{a}{b}\right)x - \frac{c}{b} \text{ shows that when } x \text{ increases,}$$

y also increases; so X and Y are in perfect direct correlation

$$\Rightarrow \quad \rho(X, Y) = 1.$$

48. Required mean

$$= \frac{0^n C_0 + 1^n C_1 + 2^n C_2 + \ldots + n^n C_n}{^n C_0 + {}^n C_1 + {}^n C_2 + \ldots + {}^n C_n} = \frac{n2^{n-1}}{2^n}$$

$$= \frac{n}{2} \qquad (\because C_1 + 2C_2 + 3C_3 + \ldots nC_n = n2^{n-1})$$

NUMERICAL TYPE QUESTIONS

5. Let E_1 be the event that a ball is drawn from first bag, E_2 the event that a ball is drawn from second bag and E the event a black ball is chosen, therefore

$$P(E) = P(E_1)\, P\left(\frac{E}{E_1}\right) + P(E_2)\, P\left(\frac{E}{E_2}\right)$$

$$= \frac{2}{6} \times \frac{3}{7} + \frac{4}{6} \times \frac{4}{7} = \frac{11}{21}$$

6. Number of ways in which 8 balls can be drawn out of 15 is $^{15}C_8$.

Number of ways of drawing 2 red balls is $^5C_2$ and corresponding to each of these $^5C_2$ ways of drawing a red ball, there are $^{10}C_6$ ways of drawing 6 black balls.

Hence total number of ways in which 2 red and 6 black balls can be drawn = $^5C_2 \times {}^{10}C_6$.

$$\therefore \quad \text{Required probability} = \frac{^5C_2 \times {}^{10}C_6}{^{15}C_8}$$

$$= \frac{140}{429}$$

7. A leap year consists of 366 days, so that there are 52 full weeks (and hence 52 Sundays) and two extra days. These two days can be

(i) Monday, Tuesday
(ii) Tuesday, Wednesday,
(iii) Wednesday, Thursday
(iv) Thursday, Friday
(v) Friday, Saturday
(vi) Saturday, Sunday
(vii) Sunday, Monday

Of these 7 cases, the last two are favourable

$$\therefore \quad \text{Required probability} = \frac{2}{7}.$$

8. The five digits can be arranged in 5! ways, out of which 4! will begin with zero.

$\therefore$ Total number of 5-figure numbers formed $= 5! - 4! = 96$.

Those numbers formed will be divisible by 4 which will have two extreme right digits divisible by 4,

i.e., numbers ending in 04, 12, 20, 24, 32, 40.

Now, numbers ending in 04 = 3! = 6,
numbers ending in 12 = 3! – 2! = 4,
numbers ending in 20 = 3! = 6,
numbers ending in 24 = 3! – 2! = 4,
numbers ending in 32 = 3! – 2! = 4,
andnumbers ending in 40 = 3! = 6.

[Numbers having 12, 24, 32 in the extreme right are (3! – 2!), since the numbers having zero on the extreme left are to excluded.]

$\therefore$ Total number of favourable ways
$= 6 + 4 + 6 + 4 + 4 + 6 = 30$

Hence, required probability $= \frac{30}{96} = \frac{5}{16}$.

9. $$P(H) = \frac{1}{2},\ P(T) = \frac{1}{2}$$

Let A be the event of getting 5 or 6.

$$\therefore \quad P(A) = \frac{1}{3}$$ and B be the event of getting 1, 2, 3, 4;

$$\therefore \quad P(B) = \frac{2}{3}$$

In order to get a 'Head' before "5 or 6", the following events have to occur:

H, TBH, TBTBH, TBTBTBH, ...

$$\therefore \quad P = \frac{1}{2} + \frac{1}{2} \times \frac{2}{3} \times \frac{1}{2} + \left(\frac{1}{3}\right)^2 \cdot \frac{1}{2} + \left(\frac{1}{3}\right)^3 \cdot \frac{1}{2}$$

$$= \frac{\frac{1}{2}}{1 - \frac{1}{3}} = \frac{3}{4}$$

10. Probability of drawing a white ball from box *B* will depend on whether transferred ball is black or white.

If a black ball is transferred, then its probability is $\frac{4}{6}$.

There are now 5 white and 8 black balls in the box *B*. Then probability of drawing white ball from box *B* $= \frac{5}{13}$

Hence probability of drawing a white ball from urn *B*, if the transferred ball is black

$$= \frac{4}{6} \times \frac{5}{13} = \frac{10}{39}$$

Similarly, probability of drawing a white ball from urn *B*, if transferred ball is white

$$= \frac{2}{6} \times \frac{6}{13} = \frac{2}{13}.$$

$$\therefore \quad \text{Required probability} = \frac{10}{39} + \frac{2}{13} = \frac{16}{39}$$

12. The sum 6 can be obtained as follows : (1, 5), (2, 4), (3, 3), (4, 2), (5, 1), *i.e.* in 5 ways.

Probability of *A*'s throwing 6 with 2 dice $= \frac{5}{36}$

$\therefore$ Probability of *A*'s not throwing 6 $= \frac{31}{36}$

Similarly, probability of *B*'s throwing 7 $= \frac{6}{36}$, i.e. $\frac{1}{6}$

$\therefore$ Probability of *B*'s not throwing 7 $= \frac{5}{6}$

Now A can win, if he throws 6 in the first, third, fifth, seventh etc. throws.

$\therefore$ Chance of *A*'s winning

$$= \frac{5}{36} + \frac{31}{36} \times \frac{5}{6} \times \frac{5}{36} + \frac{31}{36} \times \frac{5}{6} \times \frac{31}{36} \times \frac{5}{6} \times \frac{5}{36} + \ldots\ldots$$

$$= \frac{5}{36} \times \left[1 + \left(\frac{31}{36} \times \frac{5}{6}\right) + \left(\frac{31}{36} \times \frac{5}{6}\right)^2 + \left(\frac{31}{36} \times \frac{5}{6}\right)^3 + \ldots\ldots\right]$$

$$= \frac{5}{36} \times \frac{1}{1 - (31/36) \times (5/6)} = \frac{5}{36} \times \frac{36 \times 6}{61} = \frac{30}{61}$$

13. 18 = 3 Median –2 × 24

14. Given regression limit is

$$8x - 10y + 66 = 0$$

$$\Rightarrow \quad y = \frac{8}{10}x + \frac{66}{10} = \frac{4}{5}x + \frac{33}{5}$$

Hence $$b_{yx} = \frac{4}{5}$$

$$\Rightarrow \quad \frac{\text{cov}(X, Y)}{\sigma_x^2} = \frac{4}{5}$$

$$\Rightarrow \quad \text{cov}(X, Y) = \frac{9 \times 4}{5}$$

15. Since, $x + y = 6$ for all the pairs of observations,

$$b_{xy} = b_{yx} = -1$$

$\therefore$ $$\rho(X, Y) = -1$$

$\Rightarrow$ Mode = 3 Median – 2 Mean

16.

$$\Rightarrow \quad \text{Mean} = \frac{3}{2}\text{Median} - \frac{1}{2}\text{Mode}$$

$$= \frac{1}{2}(3\,\text{Median} - \text{Mode})$$

17. Mode= 3 Median – 2 Mean

$\Rightarrow$ 3 Median= Mode + 2 Mean

$\Rightarrow$ Median= $\frac{1}{3}$ (Mode + 2 Mean)

$\Rightarrow$ $M = \frac{1}{3}$

18. The two lines of regression are at right angles only when

$$b_{yx}\left(\frac{1}{b_{xy}}\right) = -1 \Rightarrow b_{yx} = -b_{xy}$$

But both b_{yx} and b_{xy} are of same sign, therefore we must have

$$b_{yx} = b_{xy} = 0 \Rightarrow \rho(X, Y) = 0$$

19.
$$\rho(X, Y) = \frac{\Sigma xy - \frac{1}{n}\Sigma x \Sigma y}{\sqrt{\Sigma x^2 - \frac{1}{n}(\Sigma x)^2}\sqrt{\Sigma y^2 - \frac{1}{n}(\Sigma y)^2}}$$

$$= \frac{3 - \frac{1}{10}\times 4\times 3}{\sqrt{8 - \frac{1}{10}(4)^2 \times \left\{9 - \frac{1}{10}(3)^2\right\}}} = \frac{1}{4}$$

20. Given : Lower quartile, $Q_1 = 35$

Upper quartile $Q_3 = 75$.

$\therefore$ Quartile deviation $= \frac{Q_3 - Q_1}{2} = \frac{75 - 35}{2} = 20$

21. Here, $\bar{u} = 3\bar{x} + 4\bar{y}$ and $\bar{v} = 3\bar{x} - \bar{y}$

$$u - \bar{u} = 3(x - \bar{x}) + 4(y - \bar{y})$$
$$v - \bar{v} = 3(x - \bar{x}) - (y - \bar{y})$$

$\Rightarrow$ $\text{cov}(u, v) = \frac{1}{n}\Sigma(u - \bar{u})(v - \bar{v})$

$$= \frac{1}{n}\Sigma\{9(x - \bar{x})^2 - 4(y - \bar{y})^2 + 9(x - \bar{x})(y - \bar{y})\}$$

$$= 9\left(\frac{\Sigma(x - \bar{x})^2}{n}\right) - 4\left(\frac{\Sigma(y - \bar{y})^2}{n}\right) + 9\left(\frac{\Sigma(x - \bar{x})(y - \bar{y})}{n}\right)$$

$= 9\,\sigma_x^2 - 4\,\sigma_y^2 + 9$ cov. (x, y)

$= 9 \times 4 - 4 \times 9 + 9(0) = 0$

{since X, Y are independent, therefore cov (X, Y) = 0}

$\Rightarrow$ $\rho(u, v) = \frac{\text{cov}(u, v)}{\sigma_u \sigma_v} = 0$

■■

CHAPTER **5**

Numerical Methods

ALGEBRAIC EQUATIONS

An equation of type $y = f(x)$ is said to be algebraic if it can be expressed in the form

$$f_n y_n + f_{n-1} y_{n-1} + ... + f_1 y_1 + f_0 = 0$$

where f_1 is an ith order polynomial in x.

This equation can be thought of as having a general form

$$f(x, y) = 0$$

This implies that above equation portrays a dependence between the variables x and y.

e.g. $3x + 5y - 21 = 0$ (linear)

$2x + 3xy - 25 = 0$ (non-linear)

$x^3 - xy - 3y^3 = 0$ (non-linear)

These equations have an infinite number of pairs of values of x and y which satisfy them.

POLYNOMIAL EQUATIONS

Polynomial equations are a simple class of algebraic equations that are represented as follows:

$$a_n x^n + a_{n-1} x^{n-1} + ... + a_1 x + a_0 = 0$$

This is called ***n*th degree polynomial** and has n roots. The roots may be

(*i*) real and different

(*ii*) real and repeated

(*iii*) complex numbers

Since complex roots appear in pairs, if n is odd, then polynomial has at least one real root.

e.g. a cubic equation of the type $a_3x^3 + a_2x^2 + a_1x + a_0 = 0$ will have at least one real root and the remaining two may be real or complex roots.

Specific examples of polynomial equations:

(*i*) $5x^5 - x^3 + 3x^2 = 0$

(*ii*) $x^3 - 4x^2 + x + 6 = 0$

(*iii*) $x^2 - 4x + 4 = 0$

TRANSCENDENTAL EQUATIONS

A non-algebraic equation is called *transcendental equation*. These include trigonometric, exponential and logarithmic functions.

e.g. $2\sin x - x = 0$, $e^x \sin x - \frac{1}{2x} = 0$, $\log x^2 - 1 = 0$, $x - e^{1/x} = 0$

A transcendental equation may have finite or infinite number of real roots or may not have real root at all.

METHODS OF SOLVING NON-LINEAR EQUATIONS

(1) Direct Analytical Methods

In certain cases, roots can be found by using direct analytical methods.

e.g. consider a quadratic equation $ax^2 + bx + c = 0$

Solution of this equation is

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$ (x: gives roots of equation)

This equation gives the two roots of quadratic equation. However, there are equations that cannot be solved by analytical methods.

e.g. simple transcendental equation

$$2\sin x - x = 0$$

cannot be solved analytically. Direct methods for solving non-linear equations do not exist except for certain simple cases.

(2) Graphical Methods

These are useful when we are satisfied with approximate solution for a problem. This method involves plotting the given function and determining the points where it crosses the x-axis. These points represent approximate values of the roots of the function.

(3) Trial and Error Methods

This method involves a series of guesses for x, each time evaluating the function to see whether it is close to zero. The value of x that causes the function value closer to zero is one of the approximate roots of the equation.

Although graphical and trial and error methods provide satisfactory approximations for many problem situations, they become cumbersome and time consuming. Moreover, accuracy of the results are inadequate for the requirements of many engineering and scientific problems.

(4) Iterative Methods

It usually begins with an approximate value of the root called *initial guess*, which is then successively corrected iteration by iteration. The process of iteration stops when desired level of accuracy is obtained. Since iterative methods involve a large number of iterations and arithmetic operations to

reach a solution, the use of computers has become inevitable to make the task simple and efficient.

Before initiating an iterative process, we have to determine either an approximate value of root or a *search interval* that contains a root.

One simple method of guessing starting points is to plot the curve of $f(x)$ and to identify a search interval near the root of interest. Graphical representation of a function cannot only provide rough estimates of the roots, but also help in understanding the properties of the function, thereby identifying possible problems in numerical computing.

A plot of $f(x) = x^3 - x - 1$ is shown in the figure.

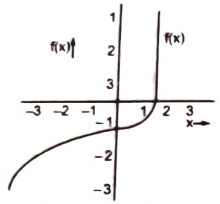


Fig. Plot of $f(x) = x^3 - x - 1$

Although $f(x)$ is a cubic function, it intersects x-axis at only one point. This suggests that remaining two roots are imaginary ones.

In the case of polynomials, many theoretical relation-ships between roots and coefficients are available.

Relation Used for Making Initial Guesses.

For a polynomial represented by

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots a_1 x\ a_0$$

the *largest possible* root is given by $f(x_1) = \dfrac{a_{n-1}}{a_n}$

This value is taken as the initial approximation when no other value is suggested by the knowledge of the problem at hand.

Types of Iterative Methods

There are a number of iterative methods that have been tried and used successfully in various problem situations. All these methods typically generate a sequence of estimates of the solution which is expected to converge to the true solution.

All iterative methods begin their process of solution with one or more guesses at the solution being sought.

Types of iterative methods based on number of guesses they use are

(*i*) Bracketing methods

These are also called *interpolation methods*. It start with two initial guesses that 'bracket' the root and then systematically reduce the width of bracket until the solution is reached.

Types of Braketing methods:

(*a*) Bisection method

(*b*) False position method

These methods are based on the assumption that the function changes sign in the vicinity of a root.

(*ii*) Open end methods

These are also called *extrapolation methods*. It uses a single starting value or two values that do not necessarily bracket the root.

Types of open end methods:

(*a*) Newton-Raphson method

(*b*) Secant method

(*c*) Muller's method

(*d*) Fixed-point method

(*e*) Bairstow's method

Note : Bracketing methods require to find sign changes in the function during every iteration. Open end methods do not require this.

NEWTON-RAPHSON METHOD

If point x_0 is close to the root α, then tangent line to the graph of $f(x)$ at x_0 is a good approximation to $f(x)$ near α. So, root of the tangent line, where line cuts the x-axis ; x_1 is a better approximation to α than x_0 is.

$$\text{Slope of the tangent} = \frac{f(x_0)}{x_0 - x_1} = f'(x_0)$$

$$\therefore \quad x_0 - x_1 = \frac{f(x_0)}{f'(x_0)}$$

$$\Rightarrow \quad x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Repeating process, we obtain a better approximation

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

and, in general, for $n = 0, 1, 2, \dots,$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

If x_0 is sufficiently close to α, $x_n \to \alpha$ as $n \to \infty$.

Order of Convergence

Let $x_0, x_1, x_2, \dots, x_n, \dots$ be a sequence of approximations to a root α produced by a numerical method, where $\lim_{n\to\infty} x_n = \alpha$.

Let $\varepsilon_n = \alpha - x_n$

If $\lim_{n\to\infty} \dfrac{|\varepsilon_{n+1}|}{|\varepsilon_n|^p} = C$, for some p and some *non-zero* constant C, then method has *order of convergence* p, and C is asymptotic error constant (bigger p is, faster the convergence).

Now consider Newton Raphson method, where $f(x)$ is assumed to be twice differentiable, and α is a simple root.

Then $f(\alpha) = f(x_n) + (\alpha - x_n) f'(x_n) + \dfrac{(\alpha - x_n)^2}{2} f''(\xi) = 0$

where ξ is an unknown point between x_n and α;

$$i.e. \quad \alpha - x_n + \frac{f(x_n)}{f'(x_n)} = \frac{-(\alpha - x_n)^2 f''(\xi)}{2f'(x_n)}$$

$$\Rightarrow \quad \alpha - x_{n+1} = \frac{-(\alpha - x_n)^2 f''(\xi)}{2f'(x_n)}$$

$$\Rightarrow \quad \frac{|\varepsilon_{n+1}|}{|\varepsilon_n|^2} = \frac{|f''(\xi)|}{2|f'(x_n)|}$$

$$\Rightarrow \quad \lim_{n\to\infty} \frac{|\varepsilon_{n+1}|}{|\varepsilon_n|^2} = \frac{|f''(\alpha)|}{2|f'(\alpha)|} \text{ if } x_n \to \alpha \text{ as } n \to \infty$$

Thus, in case of a simple root α, Newton's method has second order quadratic—convergence, with asymptotic error constant $\dfrac{|f''(\alpha)|}{2|f'(\alpha)|}$

Note. If α is a root of multiplicity m, then

$$f(x) = (x - \alpha)^m g(x)$$

$$\Rightarrow \quad f_0'(x) = (x - \alpha)^m g'(x) + m(x - \alpha)^{m-1} g(x)$$
$$= (x - \alpha)^{m-1}[(x - \alpha)g'(x) + mg(x)]$$

So, $f'(\alpha) = 0$ unless $m = 1$,

in which case $f'(\alpha) = g(\alpha) \neq 0$.

Hence requirement that α be simple for the method to be quadratically convergent.

Otherwise, if α is not simple,

$$x_{n+1} - \alpha = x_n - \alpha - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \alpha - \frac{(x_n - \alpha)g(x_n)}{(x_n - \alpha)g'(x_n) + mg(x_n)}$$

So, dividing across by $x_n - \alpha$ and taking absolute values of both sides, we have

$$\frac{|x_{n+1} - \alpha|}{|x_n - \alpha|} = \left|1 - \frac{g(x_n)}{(x_n - \alpha)g'(x_n) + mg(x_n)}\right|$$

$$\therefore \lim_{n\to\infty} \frac{|\varepsilon_{n+1}|}{|\varepsilon_n|} = \left|1 - \frac{g(\alpha)}{mg(\alpha)}\right|$$

$$= 1 - \frac{1}{m} = \frac{m-1}{m}; m \geq 2$$

assuming $\lim_{n\to\infty} x_n = \alpha$.

So, order of convergence for multiple roots is only 1—linear and as m, multiplicity of the root gets bigger, convergence slows down since asymptotic error constant $\dfrac{m-1}{m} \to 1$.

However, function $q(x) = \dfrac{f(x)}{f'(x)}$ always has simple roots :

$$q(x) = \frac{(x - \alpha)g(x)}{(x - \alpha)g'(x) + mg(x)}$$

So $q(\alpha) = 0$

$$\therefore q'(x) = \frac{[(x - \alpha)g'(x) + mg(x)][(x - \alpha)g'(x) + g(x)]}{-(x - \alpha)g(x)[(x - \alpha)g''(x) + (m+1)g'(x)]}$$
$$\overline{[(x - \alpha)g'(x) + mg(x)]^2}$$

Hence, $q'(\alpha) = \dfrac{mg(\alpha)^2}{m^2 g(\alpha)^2} = \dfrac{1}{m} \neq 0$

Hence, Newton's method applied to $q(x)$ will be quadratically convergent:

$$x_{n+1} = x_n - \frac{q(x_n)}{q'(x_n)}$$

$$q'(x) = \frac{f'(x)^2 - f(x)f''(x)}{f'(x)^2}$$

$$\Rightarrow \quad \frac{q(x)}{q'(x)} = \frac{f(x)f'(x)}{f'(x)^2 - f(x)f''(x)}$$

Thus, we have *modified Newton method:*

$$x_{n+1} = x_n - \frac{f(x_n)f'(x_n)}{f'(x_n)^2 - f(x_n)f''(x_n)}$$

which will be quadratically convergent to a root of any multiplicity; it is more unwieldy than Newton's method and requires calculation of $f''(x_n)$.

Alternatively

If $f(x) = k(x - \alpha)^m$ for some constant k, then

$$f'(x) = km(x - \alpha)^{m-1}$$

$$\therefore \quad \frac{f(x)}{f'(x)} = \frac{(x - \alpha)}{m}$$

So, $\alpha = x - m\dfrac{f(x)}{f'(x)}$ for any x.

In general case of a multiple root, this becomes an approximation rather than an equality, and we have another modified Newton method:

$$x_{n+1} = x_n - m\frac{f(x_n)}{f'(x_n)}$$

This method does not require calculation of $f''(x_n)$;

but it does require calculation of the multiplicity m from

$$\frac{|\varepsilon_{n+1}|}{|\varepsilon_n|} \approx \frac{m-1}{m}$$

where ε_n and ε_{n+1} are errors in Newton's method.

Example. $f(x) = x - \cos x$ ($\alpha = 0.739085$)

$$f'(x) = 1 + \sin x$$

$$x_{n+1} = x_n - \frac{x_n - \cos x_n}{1 + \sin x_n}$$

$$= \frac{x_n \sin x_n + \cos x_n}{1 + \sin x_n}$$

n	x_n	$\varepsilon_n = \alpha - x_n$	$x_{n+1} - x_n$
0	0	0.739085	1
1	1	–0.260915	–0.249636
2	0.750364	–0.011279	–0.011251
3	0.739113	–0.000028	–0.000028
4	0.739085	0	0
5	0.739085	0	

In this case, $\frac{|\varepsilon_1|}{|\varepsilon_0|^2} = 0.47765$; $\frac{|\varepsilon_2|}{|\varepsilon_1|^2} = 0.165681$;

$$\frac{|\varepsilon_3|}{|\varepsilon_2|^2} = 0.220098$$

Note : $x_{n+1} - x_n$ is a good approximation for $\alpha - x_n$ as x_n approaches α.

Example : $f(x) = x^2 - 4x + 4 = 0$;

$\alpha = 2$ $m = 2$

$f'(x) = 2x - 4$

From Newton's method :

$$x_{n+1} = x_n - \frac{x_n^2 - 4x_n + 4}{2x_n - 4} = \frac{x_n^2 - 4}{2x_n - 4} = \frac{x_n + 2}{2}$$

n	x_n	$\varepsilon_n = \alpha - x_n$	$x_{n+1} - x_n$
0	1	1	0.5
1	1.5	0.5	0.25
2	1.75	0.25	0.125
3	1.875	0.125	0.0625
4	1.9375	0.0625	

Here,

$$\frac{|\varepsilon_{n+1}|}{|\varepsilon_n|} = \frac{|\alpha - x_{n+1}|}{|\alpha - x_n|} = \frac{1}{2} = \frac{|x_{n+1} - x_n|}{|x_n - x_{n+1}|} \text{ for all } n.$$

Thus convergence is only linear :

$$\frac{m-1}{m} = \frac{1}{2}$$

$$\Rightarrow \quad 2m - 2 = m$$

$$\Rightarrow \quad m = 2$$

So an alternative method should be used.

Using first modified method :

$$f(x) = x^2 - 4x + 4$$

$$\Rightarrow \quad f'(x) = 2(x-2)$$

$$\Rightarrow \quad f''(x) = 2$$

$$\therefore \quad x_{n+1} = x_n - \frac{2(x_n - 2)^3}{4(x_n - 2)^2 - 2(x_n - 2)^2}$$

$$= x_n - (x_n - 2) = 2$$

Using second method :

$$x_{n+1} = x_n - 2\frac{f(x_n)}{f'(x_n)}$$

$$\Rightarrow \quad x_{n+1} = x_n - \frac{2(x_n - 2)^3}{2(x_n - 2)} = 2$$

Example. $f(x) = x^4 - 2x^3 + 2x - 1 = 0$; & $\alpha = 1$ $f'(x) = 4x^3 - 6x^2 + 2$

Using Newton's method,

$$x_{n+1} = x_n - \frac{x_n^4 - 2x_n^3 + 2x_n - 1}{4x_n^3 - 6x_n^2 + 2} = \frac{3x_n^4 - 4x_n^3 + 1}{4x_n^3 - 6x_n^2 + 2}$$

n	x_n	$\varepsilon_n = \alpha - x_n$	$x_{n+1} - x_n$	$\frac{\lvert\alpha - x_{n+1}\rvert}{\lvert\alpha - x_n\rvert}$	$\frac{\lvert x_{n+2} - x_{n+1}\rvert}{\lvert x_{n+1} - x_n\rvert}$
0	0	1	0.5	0.5	0.375
1	0.5	0.5	0.1875	0.625	0.59227
2	0.6875	0.3125	0.11102	0.64474	0.62836
3	0.79852	0.20148	0.06976	0.65376	0.64464
4	0.86828	0.13172	0.04497	0.65859	0.65288
5	0.91325	0.08675	0.02936	0.66156	
6	0.94261	0.05739			

Here $\frac{m-1}{m} \approx \frac{2}{3} \Rightarrow 3m - 3 = 2m \Rightarrow m = 3$

Since root is multiplicity 3, we use modified method.

$$x_{n+1} = x_n - 3\frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{3(x_n^4 - 2x_n^3 + 2x_n - 1)}{4x_n^3 - 6x_n^2 + 2}$$

$$= \frac{x_n^4 - 4x_n^3 + 3}{4x_n^3 - 6x_n^2 + 2}$$

Using this method, we have

n	x_n
0	0
1	1.5
2	1.03125
3	1.00016
4	0.99993
5	0.99961
6	1.00006
7	1.00000

So, there is a root of multiplicity 3 at $x = 1$.

In practice, Newton's method would be terminated if $\frac{|x_{n+2} - x_{n+1}|}{|x_{n+1} - x_n|}$ approached a non-zero limit after a preset number of steps and whatever the most recent value of x_n was would be used as an initial guess for the modified method.

Limitations of Newton-Raphson Method

This method fails in the following situations :

(*i*) Division by zero may occur if $f'(x_i)$ is zero or very close to zero.

(*ii*) If initial guess is too far away from the required root, the process may converge to some other root.

(*iii*) A particular value in the iteration sequence may repeat, resulting in an infinite loop. This occurs when to the curve $f(x)$ at $x = x_{i+1}$ cuts the x-axis again at $x = x_i$.

Example. Find by Newton Raphson's method, real root of equation $xe^x - 2 = 0$ correct to three decimal places.

Solution :

Let $f(x) = xe^x - 2$ ...(*i*)

Then $f(0) = -2$ and $f(1) = e - 2 = 0.7183$

So a root of (*i*) lies between 0 and 1 and nearer to 1.

Let us take $x_0 = 1$.

Also, $f'(x) = xe^x + e^x$

and $f'(1) = e + e = 5.4366$

By Newton's rule,

first approximation, $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$

$$= 1 - \frac{0.7183}{5.4366} = 0.8679$$

$\therefore$ $f(x_1) = 0.0672$,

and $f'(x_1) = 4.4491$

second approximation,

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 0.8679 - \frac{0.0672}{4.4491} = 0.8528$$

Thus, required root is 0.0853 correct to 3 decimal places.

SECANT METHOD

This method may be regarded as an approximation to Newton's method where, instead of $f'(x_n)$, the quotient $\frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}$ is used; *i.e.* instead of tangent to the graph of $f(x)$ at x_n, we use *secant joining* the points $(x_{n-1}, f(x_{n-1}))$ and $(x_n, f(x_n))$.

Equating two expressions for the slope of the secant, we have

$$\frac{f(x_n)}{x_n - x_{n-1}} = \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}$$

$$\Rightarrow \quad x_{n+1} = x_n - \frac{(x_n - x_{n-1}) f(x_n)}{f(x_n) - f(x_{n-1})}$$

Convergence rate of the secant method is 1.618, *i.e. almost quadratic*, if α is a simple root.

Advantage

$f'(x)$ need not be evaluated; so the number of function evaluations is half that of Newton's method. However, its convergence rate is slightly less than Newton and it requires two initial guesses x_0 and x_1.

In common with Newton's method, its convergence rate deteriorates for multiple roots, and if m can be estimated, the modification

$$x_{n+1} = x_n - m\frac{(x_n - x_{n-1}) f(x_n)}{f(x_n) - f(x_{n-1})} \text{ should be used.}$$

Also in common with Newton's method, convergence is not guaranteed at all unless x_0 and in case of secant method x_1 also is a sufficiently good initial guess. For this reason a starting method such as Bisection method or *Regula Falsi* is used where applicable for a few iterations to obtain a good initial guess for one of the more accurate methods.

SOLUTION OF NON-LINEAR SIMULTANEOUS EQUATIONS

1. Newton-Raphson Method

Consider equations, $f(x, y) = 0$ and $g(x, y) = 0$

If an initial approximation (x_0, y_0) to a solution has been found by graphical method or otherwise, then a better approximation (x_1, y_1) can be obtained as follows :

Let $x_1 = x_0 + h$ and $y_1 = y_0 + k$

$\therefore \quad f(x_0 + h, y_0 + k) = 0; \quad g(x_0 + h, y_0 + k) = 0$

Expanding each of the functions by Taylor's series to first degree terms, we get approximately

$$f_0 + h\frac{\partial f}{\partial x_0} + k\frac{\partial f}{\partial y_0} = 0$$

$$g_0 + h\frac{\partial g}{\partial x_0} + k\frac{\partial g}{\partial y_0} = 0$$

where, $f_0 = f(x_0, y_0)$, $\dfrac{\partial f}{\partial x_0} = \left(\dfrac{\partial f}{\partial x}\right)_{x_0, y_0}$ etc.

Solving above equations for h and k, we get a new approximation to the root as

$$x_1 = x_0 + h, y_1 = y_0 + k$$

This process is repeated till we get the values to the desire accuracy.

Example. Solve the system of non-linear equations

$$x^2 + y = 11$$
$$y^2 + x = 7$$

Solution

An initial approximation to the solution is obtained from a rough graph of above equations, as

$$x_0 = 3.5 \text{ and } y_0 = -1.8$$

We have, $f = x^2 + y - 11$

and $g = y^2 + x - 7$

$\therefore \quad \dfrac{\partial f}{\partial x} = 2x, \dfrac{\partial f}{\partial y} = 1$

and $\dfrac{\partial g}{\partial x} = 1, \dfrac{\partial g}{\partial y} = 2y$

Newton-Raphson's equations above will be

$$7h + k = 0.55$$

and $h - 3.6k = 0.26$

Solving these, we get

$$h = 0.0855$$

Repeating the above process, replacing (x_0, y_0) by (x_1, y_1), we get

$$x_2 = 3.5844,$$
$$y_2 = -1.8482$$

2. Bisection Method

It is one of the simplest and most reliable of iterative methods for the solution of nonlinear equations. This method, also called *binary chopping or half-interval method*, relies on the fact that if $f(x)$ is real and continuous in the interval $a < x < b$, and $f(a)$ and $f(b)$ are of opposite signs, i.e.

$$f(a)f(b) < 0$$

then there is at least one real root in the interval between a and b (there may be more than one root in the interval).

Let $x_1 = a$ and $x_2 = b$.

Let us also define another point x_0 to be the midpoint between a and b, i.e.

$$x_0 = \frac{x_1 + x_2}{2}$$

Now, there exists following three conditions:

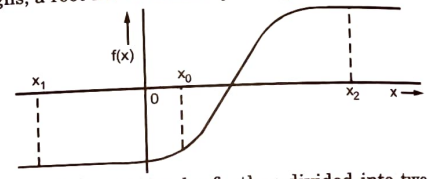
(i) If $f(x_0) = 0$, we have a root at x_0.

(ii) If $f(x_0)\, f(x_1) < 0$, there is a root between x_0 and x_1.

(iii) If $f(x_0)\, f(x_2) < 0$, there is a root between x_0 and x_2.

It follows that by testing sign of the function at midpoint, we can deduce which part of the interval contains the root.

Figure shows that, since $f(x_0)$ and $f(x_2)$ are of opposite signs, a root lies between x_0 and x_2.



This subinterval can be further divided into two halves to locate a new sub-interval containing the root. This process can be repeated until interval containing the root is as small as needed.

Example. Find a root of the equation,

$x^2 - 4x - 10 = 0$ using bisection method.

Solution

First step is to guess two initial values that would bracket a root.

Using equation

$$\left|x^*_{\max}\right| = \sqrt{\left(\frac{a_n - 1}{a_n}\right)^2 - 2\left(\frac{a_n - 2}{a_n}\right)}$$

maximum absolute of the solution can be decided.

Thus $x_{\max} = \sqrt{\left(\dfrac{-4}{1}\right)^2 - 2\left(\dfrac{-10}{1}\right)} = 6$

Therefore, both the roots in the interval $(-6, 6)$. The table below gives values of $f(x)$ between –6 and 6 and shows that there is a root in the interval $(-2, -1)$ and another in $(5, 6)$.

x	–6	–5	–4	–3	–2	–1	0	1	2	3	4	5	6
f(x)	50	35	22	11	2	–4	–10	–13	–14	–13	–10	–5	2

Now, take $x_1 = -2$ and $x_2 = -1$

Then $x_0 = \dfrac{-2 - 1}{2} = -1.5$

$f(-2) = 2$ and $f(-1.5) = -1.75$

Since $f(-2)\, f(-1.5) < 0$

the root must be in the interval $(-2, -1.5)$.

The next step begins.

$$x_1 = -1, x_2 = -1.5 \text{ and } x_0 = -1.75$$

$$f(-1.75) = 0.0625$$

Since $f(-1.75)$ and $f(-1.5)$ are of opposite sign, the root lies in the interval $(-1.75, -1.5)$, another iteration begins.

$$x_1 = -1.75, x_2 = -1.5 \text{ and } x_0 = -1.625$$

$$f(-1.625) = -0.859$$

Now, root lies in the interval $(-1.75, -1.625)$

$$\therefore \quad x_0 = -1.6875$$

$$f(-1.6875) = -0.40$$

Again $x_0 = -\dfrac{1.75 + 1.6875}{2} = -1.72$

$$f(-1.72) = -0.1616$$

Again, $x_0 = -\dfrac{1.75 + 1.72}{2} = -1.735$

$$f(-1.735) = -0.05$$

Again, $x_0 = 1.7425$

$$f(-1.7425) = +0.0063$$

The root lies between – 1.735 and – 1.7425.

Approximate root is – 1.7416.

3. Regula Falsi Method

This is a variation of the bisection method, where m_{k+1}, instead of being midpoint of $I_k = (a_k, b_k)$ is the point of intersection with the x-axis of the secant joining $(a_k, f(a_k))$ and $(b_k, f(b_k))$.

$$m_{k+1} = \frac{a_k f(b_k) - b_k f(a_k)}{f(b_k) - f(a_k)}$$

Like bisection method, it only work for odd roots, and has linear convergence rate.

Advantages

(*i*) It always converges to odd roots.

(*ii*) Error bounds are easily obtained.

(*iii*) Convergence is unaffected by the multiplicity of the root.

Disadvantages

(*i*) Slow to converge.

(*ii*) Skips even roots.

NUMERICAL INTERGRATION

The process of evaluating a definite integral from a set of tabulated values of the integrand $f(x)$ is called *numerical integration*. This process when applied to a function of a single variable is called *quadrature*.

1. Newton-Cote's quadrature formula

Let $I = \int_a^b f(x)dx$

where $f(x)$ takes the values $y_0, y_1, y_2, \ldots, y_n$.

For $x = x_0, x_1, x_2 \ldots x_n$ divide the interval (a, b) into n sub-intervals of width h, so that

$x_0 = a, x_1 = x_0 + h, x_2 = x_0 + 2h, \ldots, x_n = x_0 + nh = b$.

Then $I = \int_{x_0}^{x_0 + nh} f(x)dx$

Putting $x = x_0 + rh$

$\Rightarrow \quad dx = hdr$

$\therefore \quad I = h\int_0^n f(x_0 + rh)dr$

By Newton's forward interpolation formula

$$I = h\int_0^n \left[y_0 + r\Delta y_0 + \frac{r(r-1)}{2!}\Delta^2 y_0 + \frac{r(r-1)(r-2)}{3!}\Delta^3 y_0 + \ldots\right]dr$$

Integrating term by term, we get

$$\int_{x_0}^{x_0 + nh} f(x)dx = nh\left[y_0 + \frac{n}{2}\Delta y_0 + \frac{n(2n-3)}{12}\Delta^2 y + \frac{n(n-2)^2}{24}\Delta^3 y + \ldots\right] \ldots (i)$$

2. Trapezoidal rule

For definite integrals such as

$$\int_0^1 \sqrt{1 - x^3}\,dx \quad \text{or} \quad \int_0^1 e^{-x^2}dx$$

fundamental theorem of calculus cannot be used to evaluate them since there are no elementary functions that are antiderivatives of $\sqrt{1 - x^3}$ or e^{-x^2}. *Approximation methods are used for such integrals.*

Trapezoidal rule is a numerical method that approximates value of a definite integral.

Consider the definite integral $\int_a^b f(x)dx$.

Assume that $f(x)$ is continuous on $[a, b]$ and divide $[a, b]$ into n subintervals of equal length

$$\Delta x = \frac{b - a}{n}$$

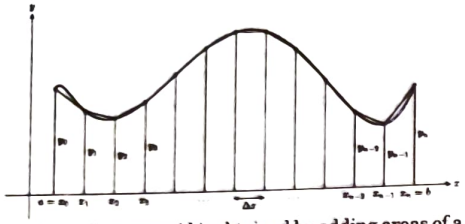
using $n + 1$ points $x_0 = a, x_1 = a + \Delta x, x_2 = a + 2\Delta x, \ldots,$ $x_n = a + n\Delta x = b$.

Compute value of $f(x)$ at these points.

$y_0 = f(x_0), y_1 = f(x_1), y_2 = f(x_2), \ldots, y_n = f(x_n)$

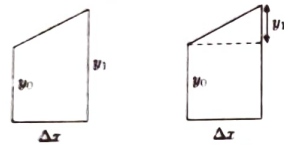
Approximate integral by using n trapezoids formed

by using straight line segments between points (x_{i-1}, y_{i-1}) and (x_i, y_i) for $1 \le i \le n$ as shown in the figure below.



Area of a trapezoid is obtained by adding areas of a rectangle and triangle.

$$A = y_0\,\Delta x + \frac{1}{2}(y_1 - y_0)\Delta x = \frac{(y_0 + y_1)\Delta x}{2}$$



Adding area of the n trapezoids, the approximation obtained is

$$\int_a^b f(x)\,dx \approx \frac{(y_0 + y_1)\Delta x}{2} + \frac{(y_1 + y_2)\Delta x}{2} + \frac{(y_2 + y_3)\Delta x}{2} + \cdots + \frac{(y_{n-1} + y_n)\Delta x}{2}$$

which simplifies to the trapezoidal rule formula.

$$\int_a^b f(x)dx \approx \frac{\Delta x}{2}\left(y_0 + 2y_1 + 2y_2 + \ldots + 2y_{n-1} + y_n\right)$$

Example 1. Use trapezoidal rule with $n = 8$ to estimate $\int_1^5 \sqrt{1 + x^2}\,dx$.

Solution For $n = 8$, $\Delta x = \dfrac{5-1}{8} = 0.5$.

Compute the values of $y_0, y_1, y_2, \ldots, y_8$.

x	1	1.5	2	2.5	3	3.5	4	4.5	5
$y = \sqrt{1+x^2}$	$\sqrt{2}$	$\sqrt{3.25}$	$\sqrt{5}$	$\sqrt{7.25}$	$\sqrt{10}$	$\sqrt{13.25}$	$\sqrt{17}$	$\sqrt{21.25}$	$\sqrt{26}$

Therefore,

$$\int_1^5 \sqrt{1+x^2}\,dx \approx \frac{0.5}{2}\left(\sqrt{2} + 2\sqrt{3.25} + 2\sqrt{5} + 2\sqrt{7.25} + 2\sqrt{10} + 2\sqrt{13.25} + 2\sqrt{17} + 2\sqrt{21.25} + \sqrt{26}\right)$$

$$\approx 12.76$$

Example 2. Following points were found empirically.

x	2.1	2.4	2.7	3.0	3.3	3.6
y	3.2	2.7	2.9	3.5	4.1	5.2

Use the trapezoidal rule to estimate $\int_{2.1}^{3.6} y\,dx$.

Solution

By inspection, $\Delta x = 0.3$

$$\therefore \int_{2.1}^{3.6} y\,dx \approx \frac{0.3}{2}(3.2 + 2(2.7) + 2(2.9) + 2(3.5) + 2(4.1) + 5.2)$$

$$\approx 5.22$$

3. Simpson's one-third rule

Putting $n = 2$ in equation (i) above and taking the curve through (z_0, z_0) (z_1, y_1) and (z_2, y_2) as a parabola, *i.e.* a polynomial of second order so that differences of order higher than second vanish, we get

$$\int_{x_0}^{x_0+2h} f(x)dx = 2h\left(y_0 + \Delta y_0 + \frac{1}{6}\Delta^2 y_0\right)$$

$$= \frac{h}{3}(y_0 + 4y_1 + y_2)$$

Similarly, $\int_{x_0+2h}^{x_0+4h} f(x)dx = \dfrac{h}{3}(y_2 + 4y_3 + y_4)$

$$\int_{x_0+(n-2)h}^{x_0+nh} f(x)dx = \frac{h}{3}(y_{n-2} + 4y_{n-1} + y_n)$$

n being even.

Adding all these integrals, when n is even,

$$\int_{x_0}^{x_0+nh} f(x)dx = \frac{h}{3}\left[(y_0 + y_n) + 4(y_1 + y_3 + \ldots + y_{n-1}) + 2(y_2 + y_4 + \ldots + y_{n-2})\right]$$

4. Simpson's three-eighth rule

Putting $n = 3$ in equation (i) above and taking the curve through (x_i, y_i) $i = 0, 1, 2, 3$ as a polynomial of third order so that differences above the third order vanish, we get

$$\int_{x_0}^{x_0+3h} f(x)dx = 3h\left(y_0 + \frac{3}{2}\Delta y_0 + \frac{3}{4}\Delta^2 y_0 + \frac{1}{8}\Delta^3 y_0\right)$$

$$= \frac{3h}{8}(y_0 + 3y_1 + 3y_2 + y_3)$$

Similarly, $\int_{x_0+3h}^{x_0+6h} f(x)dx = \dfrac{3h}{8}(y_3 + 3y_4 + 3y_5 + y_6)$

Adding all such expressions from x_0 to $x_0 + nh$, where n is a multiple of 3, we get

$$\int_{x_0}^{x_0+nh} f(x)dx = \frac{3h}{8}\left[(y_0 + y_n) + 3(y_1 + y_2 + y_4 + y_5 + \ldots + y_{n-1}) + 2(y_3 + y_6 + y_{n-3})\right]$$

Example. Evaluate $\int_0^6 \dfrac{dx}{1+x^2}$ by using Trapezoidal rule, Simpson's $\dfrac{1}{3}$ rule, and Simpson's $\dfrac{3}{8}$ rule,

Solution

Divide the interval (0, 6) into six parts each of width $h = 1$.

The values of $f(x) = \dfrac{1}{1+x^2}$ are given below :

x	0	1	2	3	4	5	6
$f(x)$	1	0.5	0.2	0.1	0.0588	0.0385	0.027
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

(i) By Trapezoidal rule,

$$\int_0^6 \frac{dx}{1+x^2} = \frac{h}{2}[(y_0 + y_6) + 2(y_1 + y_2 + y_3 + y_4 + y_5)]$$

$$= \frac{1}{2}[(1+.027) + 2(.5+.2+.1+.0588+.0385)]$$

$$= \mathbf{1.4108}$$

(ii) By Simpson's $\dfrac{1}{3}$ rule

$$\int_0^6 \frac{dx}{1+x^2} = \frac{h}{3}[(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

$$= \frac{1}{3}[(1+.027) + 4(0.5+0.1+0.0385) + 2(0.2+0.0588)]$$

$$= \mathbf{1.3662}$$

(iii) By Simpson's $\dfrac{3}{8}$ rule,

$$\int_0^6 \frac{dx}{1+x^2} = \frac{3h}{8}[(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2y_3]$$

$$= \frac{1}{8}[(1 + 0.027) + 3(0.5 + 0.2 + 0.0588 + 0.0385 + 2(0.1)]$$

$$= \mathbf{1.3571}$$

NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS

1. Runge-Kutta Method

This methods do not require calculations of higher order derivatives. These methods agree with Taylor's series solution upto the terms in h^r, where r differs from method to method and is called *order of that method*. Euler's method, Modified Euler's method and Runge's method are Runge-Kutta methods of the first, second and third order respectively.

The fourth-order Runge-Kutta method is most commonly used and is often referred to as 'Runge-Kutta method only'.

Working rule for finding increment k of y corresponding to an increment h of x by Runge-Kutta method from, $\dfrac{dy}{dx} = f(x, y), y(x_0) = y_0$, is as follows :

Calculate successively,

$$k_1 = hf(x_0, y_0)$$

$$k_2 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1\right)$$

$$k_3 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2\right)$$

$$k_4 = hf(x_0 + h, y_0 + k_3)$$

and finally compute,

$$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

which gives required approximate value

$y_1 = y_0 + k$.

(Note : k is weighted mean of k_1, k_2, k_3 and k_4)

Example. Apply Runge-Kutta fourth order method, to find an approximate value of y when $h = 0.2$.

Given: $\dfrac{dy}{dx} = x + y$ and $y = 1$, when $x = 0$.

Solution

$x_0 = 0,\ y_0 = 1,\ h = 0.2,\ f(x_0, y_0) = 1$

$\therefore\ k_1 = hf(x_0, y_0) = 0.2 \times 1 = 0.2$

$$k_2 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1\right)$$

$$= 0.2 \times f(0.1, 1.1) = 0.2400$$

$$k_3 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2\right)$$

$$= 0.2 \times f(0.1, 1.12) = 0.2440$$

and $k_4 = hf(x_0 + h, y_0 + k_3)$

$$= 0.2 \times f(0.2, 1.244) = 0.2888$$

$$\therefore\ k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$= \frac{1}{6}(0.2000 + 0.4880 + 0.2888) = \mathbf{0.2428}$$

Hence, required approximate value of

$Y_1 = Y_0 + K$

$= 1 + 0.2428 = 1.2428$

2. Taylor's Series Method

Given initial value problem

$$y' = \frac{dy}{dx} = f(x, y), \text{ with } y(x_0) = y_0 \quad \ldots(i)$$

Then $y'' = \dfrac{d^2y}{dx^2} = f_x + f_y y' = f_x + f_y f$

$$y''' = \frac{d^3y}{dx^3} = f_{xx} + 2f_{xy}f + f_{yy}f^2 + f_x f_y + f_y^2 f \ldots(ii)$$

By Taylor's theorem, series about a point $x = x_0$

$$y = y_0 + (x - x_0)(y')_0 + \frac{(x - x_0)^2}{2!}(y'')_0 + \frac{(x - x_0)^3}{3!}(y''')_0 + \ldots \ldots(iii)$$

From equation (iii) we can find the value y_1 of y for $x = x_1$ and y', y'', y''',... can be found at $x = x_1$ with equations (i) and (ii) and so on.

3. Euler's Method

Given initial value problem

$$y' = \frac{dy}{dx} = f(x, y), \text{ with } y(x_0) = y_0$$

By Taylor series, for $h \to 0$,

$$y(x + h) = y(x) + hy'(x) = y(x) + hf(x, y)$$

This gives, $y_{n+1} = y_n + hf(x_n, y_n)$

where $h = \dfrac{x_n - x_0}{n}$ (i.e., $x_n = x_0 + nh$)

EXERCISE

MCQ TYPE QUESTIONS

1. The order of error is the Simpson's rule for numerical integration with a step size h is
   (a) h (b) h^2
   (c) h^3 (d) h^4

2. Four arbitrary points (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4) are given in the x, y-plane. Using the method of least squares, if regressing y upon x gives the fitted line $y = ax + b$; and regressing x upon y gives the fitted line $x = cy + d$, then
   (a) two fitted lines must coincide
   (b) two fitted lines need not coincide
   (c) it is possible that $ac = 0$
   (d) a must be $1/c$

3. The accuracy of Simpson's rule quadrature for a step size h is
   (a) $O(h^2)$ (b) $O(h^3)$
   (c) $O(h^4)$ (d) $O(h^5)$

4. The Newton's-Raphson iterative formula for finding $f(x) = x^2 - 1$, is
   (a) $x_{i+1} = \frac{x_i^2 - 1}{2x_i}$ (b) $x_{i+1} = \frac{x_i^2 + 1}{2x_i}$
   (c) $x_{i+1} = \frac{2x_i^2 + 1}{2x_i}$ (d) $x_{i+1} = \frac{2x_i}{2x_i^2 + 1}$

5. The convergence of which of the following method is sensitive to starting value?
   (a) False position
   (b) Gauss seidal method
   (c) Newton-Raphson method
   (d) All of these

6. Newton-Raphson method of solution of numerical equation is not preferred when
   (a) graph of A(B) is vertical
   (b) graph of $x(y)$ is not parallel
   (c) graph of $f(x)$ is nearly horizontal where it crosses the x-axis.
   (d) none of these

7. In which of the following methods, proper choice of initial value is very important ?
   (a) Bisection method
   (b) False position
   (c) Newton-Raphson
   (d) Bairsto method

8. Newton-Raphson method is applicable to the solution of
   (a) both algebraic and trans-cendental equations
   (b) both algebraic and trans-cendental and also used when the roots are complex
   (c) algebraic equations only
   (d) transcendental equations only

9. The root of $x^3 - 2x - 5 = 0$ correct to three decimal places by using Newton-Raphson method is
   (a) 2.0946 (b) 1.0404
   (c) 1.7321 (d) 0.7011

10. Using Newton-Raphson method, a root correct to three decimal places of the equation $e^x = 1 + 2x$ is
    (a) 1.256 (b) 1.255
    (c) 1.286 (d) none of these

11. A real root of equation $x^3 + x^2 - 1 = 0$ by iteration (method of successive approximation) method is
    (a) 0.7548765 (b) 0.7548756
    (c) 0.7548776 (d) 0.7548764

12. A real root of equation $\cos x = 3x - 1$ correct to seven decimal places by the method of successive approximation is
    (a) 0.6071016 (b) 0.6071015
    (c) 0.60711516 (d) none of these

13. From certain experiment following data has been obtained:

x	1	3	4
y	4	12	19

Using Langrange's inverse formula, the value of x for which y = 7, will be
(a) 2.124 (b) 1.857
(c) 2.429 (d) 1.946

14. The Lagrange interpolating polynomial of degree 2 approximating the function $y = \ln x$ defined by the following table of values. The value of ln 2.7 will be

x	2	2.5	3.0
y = ln x	0.69315	0.91629	1.09861

(a) 0.9941162 (b) 0.9941164
(c) 0.9941165 (d) 0.9941166

15. The function $y = \sin x$ is tabulated below

x	0	$\pi/4$	$\pi/2$
y = sin x	0	0.70711	1.0

Using Lagrange's interpolation formula, the value of sin $(\pi/6)$ will be
(a) 0.52743 (b) 0.52742
(c) 0.51743 (d) none of these

NUMERICAL TYPE QUESTIONS

1. Solving $x^2 - 2 = 0$ by Newton Raphson technique, when initial guess $x_0 = 1.0$, then subsequent estimate of x (*i.e.* x_1) will be _____
2. Following are the values of a function
   $y(x) : y(-1) = 5, y(0), y(1) = 8$
   $\frac{dy}{dx}$ at $x = 0$ as per Newton's central difference scheme is _____
3. Using Newton-Raphson method, a root correct to three decimal places of the equation $x^3 - 3x - 5 = 0$ is _____
4. Using Newton-Raphson method, a root correct to three decimal places of the equation $\sin x = 1 - x$ is _____
5. A root of the equation $x^3 - x - 11 = 0$ correct to four decimals using bisection method, is _____
6. Using Bisection method, the negative of $x^3 - x + 11 = 0$ is
7. A real root of the equation $x^3 - 2x - 5 = 0$ by the method of false position correct to three decimal places is _____
8. Using Bisection method, the negative root of $x^3 - 4x + 9 = 0$ correct to three decimal places is _____
9. A real root of the equation $x - \cos x = 0$ by the method of false position correct to four decimal places is
10. The double (repeated) root of $4x^3 - 8x^2 - 3x + 9 = 0$ by Newton-raphson method will be
11. Using trapezoidal rule, and the table given below, $\int_4^{5.2} \ln x \, dx$ will be

x :	4	4.2	4.4	4.6	4.8	5	5.2
ln x:	1.39	1.44	1.48	1.53	1.57	1.61	1.65

12. Using trapezoidal rule, taking 10 equal interval $\int_0^{\pi} \sin x \, dx$ will be _____
13. If $y_1 = 4$, $y_3 = 12$, $y_4 = 19$ and $y_x = 7$, then x will be _____

ANSWERS

MCQ Type Questions

1. (b)	2. (d)	3. (d)	4. (b)	5. (c)	6. (c)	7. (c)	8. (a)	9. (a)	10. (a)
11. (c)	12. (a)	13. (b)	14. (b)	15. (c)					

Numerical Type Questions

1. 1.5	2. 1.5	3. 2.279	4. 0.511	5. 2.3736	6. – 2.3736
7. 2.0944	8. – 2.506	9. 0.7391	10. 1.5	11. 1.8277	12. 1.984
13. 1.86					

EXPLANATIONS

MCQ TYPE QUESTIONS

2. $y = ax + b$ ....(i)
   $x = cy + d$
   $\Rightarrow \quad x - d = cy$
   $\Rightarrow \quad y = \frac{1}{c}x - d$
   From equatons (i),
   $a = \frac{1}{c}$ and $b = -d$

14. $L_0(x) = \frac{(x - 2.5)(x - 3.0)}{(-0.5)(-1.0)}$
    $= 2x^2 - 11x + 15$
    Similarly,
    $L_1(x) = -(4x^2 - 20x + 24)$
    $L_2(x) = 2x^2 - 9x + 10$
    $\therefore \quad L_2(x) = (2x^2 - 11x + 15)(0.69315)$
    $- (4x^2 - 20x + 24)(0.091629)$
    $+ (2x^2 - 9x + 10)(1.09861)$
    $= -0.08164x^2 + 0.81366x - 0.60761$
    which is the required quadratic polynomial.

Putting $x = 2.7$ in the above polynomial, we get

$\ln 2.7 \approx L_2(2.7)$

$= -0.08164(2.7)^3 + 0.81366(2.7) - 0.60761$

$= 0.9941164$

15. $\sin\frac{\pi}{6} \approx \frac{\left(\frac{\pi}{6}-0\right)\left(\frac{\pi}{6}-\frac{\pi}{2}\right)}{\left(\frac{\pi}{4}-0\right)\left(\frac{\pi}{4}-\frac{\pi}{2}\right)}(0.70711) + \frac{\left(\frac{\pi}{6}-0\right)\left(\frac{\pi}{6}-\frac{\pi}{4}\right)}{\left(\frac{\pi}{2}-0\right)\left(\frac{\pi}{2}-\frac{\pi}{4}\right)}(1)$

$= \frac{8}{9}(0.70711) - \frac{1}{9}$

$= \mathbf{0.51743}$

NUMERICAL TYPE QUESTIONS

1. $x^2 - 2 = 0$

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$

$= 1 + \frac{1}{2} = 1.5$

$f(x_0) = 1^2 - 2 = -1$

$f(x_0) = 2x$

$f(x_0) = 2x_0 = 2(1) = 2$

2. $\left(\frac{dy}{dx}\right)_{at\ x=0} = \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{y(1) - y(-1)}{1 - (-1)}$

$= \frac{8-5}{2} = 1.5$

13. $x = \frac{(-5)(-12)}{(-8)(-15)}(1) + \frac{(3)(-12)}{(8)(-7)}(3) + \frac{(3)(-5)}{(15)(7)}(4)$

$= \frac{1}{2} + \frac{27}{14} - \frac{4}{7} = \mathbf{1.86}$

■■

Mock Tests and Solved Paper

INSTRUCTIONS

1. Total of 65 questions carrying 100 marks, out of which 10 questions carrying a total of 15 marks are in General Aptitude (GA)
2. The Engineering Mathematics will carry around **15% of the total marks,** the General Aptitude section will carry **15% of the total marks** and the **remaining 70% of the total marks.**
3. **Types of Questions**
   (*a*) **Multiple Choice Questions (MCQ)** carrying 1 or 2 marks each in all papers and sections. These questions are objective in nature, and each will have a choice of four options, out of which the candidate has to mark the correct answer(s).
   (*b*) **Numerical Answer Questions** of 1 or 2 marks each in all papers and sections. For these questions the answer is a real number, to be entered by the candidate using the virtual keypad. No choices will be shown for these type of questions.
4. For **1-mark** multiple-choice questions, **1/3 marks** will be deducted for a wrong answer. Likewise, for **2-marks** multiple-choice questions, **2/3** marks will be deducted for a wrong answer. There is no negative marking for numerical answer type questions.

GENERAL APTITUDE (GA)

(Q.1 – 5) : Carry One Mark Each

1. If length of base of a triangle is increased by 20%, then by what percentage its height must be decreased so that the area remains constant?

(*a*) 20% (*b*) 16.67%

(*c*) 15% (*d*) 25%

2. The given sentence below has two blanks; each blank indicates that something has been omitted. Beneath the sentence has four lettered words. Choose the word for each blank that best fits the meaning of the sentence as a whole:

It's true that those who _____ from alcohol tend to be from lower socioeconomic classes, since drinking can be_____.

(*a*) pleasure.....stimulating

(*b*) regale.....blissful

(*c*) abstain.....expensive

(*d*) repel.....carcinogenic

3. In the given question, a related pair of words is followed by four lettered pairs of words. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

YAK : FUR

(*a*) horse : hoof

(*b*) snake : hiss

(*c*) crocodile : tears

(*d*) seal : blubber

4. Following question presents a sentence, part of which is underlined. Beneath the sentence you will find four ways of phrasing the underlined part. The first of these repeats the original; the other three are different. If you think the original is best, choose the first answer; otherwise choose one of the others:

For those concerned about the probable effects of climate change, forests contain twice as much carbon as the atmosphere, in plant-matter and the soils they cover, and <u>when they are razed and their soils disturbed most is emitted.</u>

(*a*) and when they are razed and their soils disturbed most is emitted

(*b*) but when it is razed and its soils disturbed most is emitted

(*c*) and when they raze and their soils are disturbed most are emitted

(*d*) and when they are razed and their soils are disturbed most are emitted

5. Typically, the teacher is perceived as an individual who possesses a great deal of knowledge and delivers it to students in a classroom in a school. However, simply possessing knowledge and delivering in a school or even a college or university does not make someone a teacher. If the person is a true teacher, he has to command respect from society as well as students, who look up to someone who not merely has theoretical knowledge but can effectively transform it to those needy and implements himself.

1.2

Which of the following conclusions can be drawn from the above passage?

(a) The society seldom holds high opinion of professionals who are not inclined to teach others.

(b) Only those should teach others who have enough knowledge to impart and are brave enough to implement the theory.

(c) True teaching calls for more than precept and practice

(d) Practice makes man perfect but it is teaching that makes one respectful.

(Q.6 –10) : Carry Two Marks Each

6. A certain sum of amount becomes triple in 7 years at CI (interest compounded annually). If the sum is ₹ 1000, then the interest for 21 years (in ₹.) is____

7. If x, y and z are positive integers which satisfy the equations

$x^2 + xy + yz + zx = 672$;

$y^2 + xy + yz + zx = 720$ and

$z^2 + xy + yz + zx = 840$,

then the value of $x + y + z$ will be_____

8. Read the following statements (I, II) in each questions and give your response as A, B, C and D respectively when answer can be found using I only, using II only, using both the statements and by using either of the statements.

A farmer has a certain number of hens and goats. Find the number of hens.

I. Total number of heads is 50

II. Total number of legs is equal to 4 times, 15 less than the total number of heads.

9. In the given table, numbers arranged in rows-columns. Each row and column is satisfying a common property. Find $p + q + r$.

(a) 130

(b) 129

(c) 131

(d) 128

44	53	33
p	43	q
54	34	r

10. *Here a related pair of words is followed by four lettered pairs of words. Click on the lettered pair that best expresses a relationship similar to that expressed in the original pair.*

MUSHROOM : FUNGUS

(a) circus: entertainment

(b) biology : zoo

(c) trapeze: qualidrilateral

(d) fiber: wheat

TECHNICAL SECTION

(Q.1 – 25) : Carry One Mark Each

1. Heparin is composed of the following repeating units

(a) D-glucuronic acid and N-acetyl glucosamine

(b) D-glucuronic acid N-acetyl galactosamine 4-sulfate

(c) D-glucuronate 2 sulfate and N-sulfo-glucosamine 6-sulfate

(d) L-iduronic acid and N-acetyl glucosamine 4-sulfate.

2. Which amino acid residue is most likely to be found in the interior of a water soluble globular protein?

(a) Ser (b) Arg

(c) Val (d) Asp

3. For which of the following property does the A-form double helix DNA not differ from that of B-form double helix?

(a) Appearance of the major and minor grooves

(b) Thinkness of the helix

(c) Tilt of the bases

(d) Polarity ofthe two complementary strands

4. Which of the following hormone passes through the cell membrane, binds to its intracellular receptors and activates it?

(a) Estrogen (b) Non-epinephrine

(c) Acetylcholine (d) Glucagons

5. Which of the following statements is/are correct?

(a) T-helper lymphocytes secrete a factor, which stimulates formation of other types of lymphocytes

(b) T-suppressor lymphocytes cause lowering of the intensity of the immune response

(c) T-cytotoxic lymphocytes recognize nonself antigens on cells and lyses them

(d) All of these

6. Iron-containing compounds that acts as hydrogen acceptors in the respiratory chain are

(a) flavoproteins (b) dehydrogenases

(c) oxidases (d) cytochromes

7. Adipose tissue cannot resynthesize triacyl glycerols from glycerol released during lipolysis, because

(a) glycolysis generates DHAP which is reduced to glycerol 3-phosphate

(b) adipose tissue lacks glycerol kinase

(c) both (a) and (b)

(d) none of these

8. What is the activated reactant of the following biosynthesis Phosphatidyl serine from serine?

(a) CDP-ethanolamine

(b) Acyl co A

(c) CDP-diacylglycerol

(d) UDP-glucose

9. Which of the following increases free surface of the cell to enhance absorption?

(a) Microfilaments (b) Microtubules

(c) Microbodies (d) Microvilli

10. Which of the following is not an enzyme linked receptor?

(a) Receptor gnanyl cyclases

(b) Receptor tyrosine kinases

(c) Receptor tyrosine phosphatases

(d) None of these

11. Which of the following statements is true?

(a) A temperate phage produces a specific repressor that blocks transcription of phage gene

(b) In the usual type of lysogeny, the prophage remains free in the cytoplasm as a plasmid

(c) Non-enveloped viruses are called rare viruses

(d) Early proteins are coded by late mRNA

12. Which of the following is not a feature of eulkaryotic gene expression?

(a) Polycistonie mRNAs are very are

(b) RNA synthesis and protein synthesis are coupled as in prokaryotes

(c) Many genes are interrupted by non-coding DNA sequence

(d) mRNA is often extensively modified before translation

13. Aminopterine is used during the production of hydridoma cells because it

(a) blocks the De novo pathway

(b) prevents the growth of B cells

(c) prevents the growth of myeloma cells

(d) blocks the synthesis of Ig by B cell

14. In animal cell cultures, the addition of serum to media is essential for providing

(a) amino acid for protein synthesis

(b) nucleotide for DNA synthesis

(c) growth factors

(d) all of these

15. Which of the following compounds is a precursor in the biosynthesis of both cholesterol and β-hydroxybutyrate (a- ketone body)?

(a) Farnesyl pyrophosphate

(b) β-hydroxy, β-methyl Iglutaryl-coenzymeA (HMGCoA)

(c) Methyl malonyl CoA

(d) All of these

16. Cell line used in Hepatitis B virus vaccine is

(a) chick embryo

(b) human diploid cells

(c) human leucocytes

(d) human plasma derived

17. The velocity of the reaction $A + B \rightarrow P + Q$ at 35°C is twice as great as the velocity at 25°C. The activation energy is _____ cal/mole.

18. Biosensors which make use of ion-selective electrodes in order to transduce the biological reaction into an electrical signal are

(a) potentiometric biosensors

(b) amperometric biosensors

(c) conductometric biosensors

(d) whole cell biosensors

19. Which of the following reagents is used for harvesting anchorage dependent animal cells from culture vessels?

(a) Trypsin/Collagenase

(b) Trypsin/Collagen

(c) Collagen/Fibronectin

(d) DMSO

20. Ehlers-Danlor syndrome type V is caused due to deficiency of

(a) lysozyme (b) pyruvate dehydrogenase

(c) collagen (d) myosin

21. Sakaguci test is used for the detection of

(a) tyrosine (b) cysteine

(c) histidine (d) arginine

22. Which of the following is expressed as a sequence tag?

(a) Protein coding sequence of a gene

(b) Characteristic gene expression in cell

(c) A partial sequence of a colon randomly seleccted from DNA library

(d) None of these

23. TAP tags are useful for

(a) protein resolution (b) genome sequencing

(c) peptide sequencing (d) proteome exploration

24. Who demonstrated that DNA replicates in a semi-conservative manner?

(a) McLeod and McCarty

(b) Meselson and Stahl

(c) Jacob and Monad

(d) Watson and Crick

25. Genetic elements which possess the dual capacity to exist either as chromosomal or an extrachromosomal entity are called

(a) Autosomes (b) Oxysomes

(c) Mesosomes (d) Episomes

Q. 26 - 55 : Carry Two Marks Each

26. If $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and $P^{-1}AP = \text{diag}(\lambda_1, \lambda_2)$,

where λ_1 and λ_2 are Eigen values of A, then matrix P is

(a) $\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

27. Five letters were to be placed in envelopes for posting. Unfortunately the letters were dropped before being put in their envelopes and they were placed in at random. Probability that at least one is in the correct envelope is _____

28. Using Simpson's rule with n = 3, the value of $\int_0^3 \sqrt{x^2}\,dx$ is _____

29. Suitable integrating factor to solve the non exact equation $y + (x + x^3y^2)\frac{dy}{dx} = 0$ is

(a) $\frac{1}{2xy + x^4y^2}$ (b) e^x

(c) e^{3x^2y} (d) None

30. Mean of 100 items is 49. It was discovered that three items which should have been 60, 70 and 80 were wrongly read as 40, 20 and 50. The correct mean is _____

31. Which of the following are selectable marker genes?

(a) Lux (b) Hpt IV

(c) npt II (d) both (b) and (c)

32. The conditions which are most suitable for biogas production is

(a) 30-40° temperature and pH 6-8

(b) 50° C temperature and pH 3-4

(c) 30-40° temperature and pH 3-4

(d) 40-50° temperature and pH 6-8

33. Depth filtration is preferred over absolute filtration because of

(a) improved filtration efficiency

(b) effective physical separation of micro-organisms from media

(c) efficient with media containing suspended solids

(d) none of these

34. Which micro-organism's genome was first sequenced?

(a) Haemophilus influenzae

(b) Mycobacterium tuberculosis

(c) Pseudomonas sp.

(d) Bacillus subtilis

35. Alkaptoneuria is caused due to the absence of

(a) tyrosinase

(b) arginase

(c) carbamoyl phosphate

(d) homogentisate 1, 2-dioxygenase

36. Which of the following plant hormones is antagonistic in their effects?

(a) Gibberellin (b) Abscissic acid

(c) Both (a) and (b) (d) Ethylene

37. Half of the patients with allergic diseases have increased serum

(a) IgE levels (b) IgD levels

(c) IgG levels (d) IgM levels

38. Match the products in ListI with its appropriate functions in List II.

List I	List II
A. Transferrin	1. Blood clot
B. Anti-fibrin 99	2. Animal cell separ-ation
C. Callagenase	3. Binding and trans port of iorn
D. Erythropoietin	4. Anaemia

Codes :

	A	B	C	D
(a)	3	1	2	4
(b)	3	2	1	4
(c)	2	3	1	4
(d)	3	1	4	2

39. The main functions of laffles in a bioreactor is

(a) to reduce aeration rate

(b) to increase aeration

(c) to prevent a vortex

(d) both (b) and (c)

40. One microgram of a pure enzyme (MW = 92,000) catalyzed a reaction at a rate of 0.50 μ moles/min under optimum condition.

The specific activity of the enzyme in terms of units/mole is _____

41. Which of the following co-enzyme acts in conversion of homocysteine to meth-ionine in some bacteria?

(a) Hydroxocobalamin

(b) Co-enzyme A

(c) Methylcobalamin

(d) None of these

42. Mercaptoethanol is used as a stabilizer to prevent

(a) precipitation of enzymes

(b) oxidation of sulphydryl groups and thus loss of enzyme activity

(c) denaturation of enzyme by decreasing the temperature

(d) none of these

43. Cell viability test can be determined by which of the following approaches?

P. Fluorescein diacetate (FDA)

Q. 0.3 M mannitol

R. 2, 3, 4-tri phynyitetrazolium chloride (TTC)

S. Gelling agent

(a) P, R (b) P, S

(c) R, Q (d) R, S

44. Which of the following enzyme is responsible for class switch recombination during immuno-globulin generation?

(a) Telomerase

(b) RAG

(c) Activation-induced deaminase

(d) MRN complex

45. Statins reduce cholesterol levels by inhibiting which of the following enzyme?

(a) Mevalonate kinase

(b) HMG-CoA reductase

(c) HMG-CoA synthetase

(d) NADPH oxidase

46. FASTA algorithm was published by

(a) Pearson and Lipman

(b) F. Sanger

(c) Joseph Sambrook

(d) Altschuf et. al

47. Aromatic amino acids exhibit an absorption peak in

(a) 290-349 nm region

(b) 240-310 nm region

(c) 300-350 nm region

(d) 360-410 nm region

48. Match List I with one appropriate from List II and select the correct answer from the codes given below the lists :

List I

A. B-adrenergic receptor

B. T cell receptor

C. 5-hydroxytryptamine

D. Colchicine

List II

1. Neurotransmitter
2. Mitotic spindle
3. Adenylate cyclase
4. CD3 complex

Codes:

	A	B	C	D
(a)	3	4	1	2
(b)	2	4	3	1
(c)	4	1	2	3
(d)	3	2	1	4

49. Match List I with appropriate from List II and select the correct answer from the codes given below the lists :

List I

A. Lesch-Nyhan syndrome

B. Tay-sach disease

C. 5-hydroxytryptamine

D. Immunodeficiency disease

List II

1. Hexosaminidase - A
2. Hypoxanthine-guanine phosphoribosyl transferase
3. Adenosine deaminase
4. Chloride channel

1.6

Codes:

	A	B	C	D
(a)	3	1	4	2
(b)	2	4	1	3
(c)	2	1	4	3
(d)	4	2	3	1

50. The structural membrane proteins are characterized by

(a) don't transport small molecule

(b) exist only by beta sheets

(c) shear pre static carbohydrate groups on the cytoplasmic side

(d) have a hydrophobic transmembrane domain

51. Which of the following statements is false with respect to the protein kinase C(PKC)?

(a) PKC is a ca^{2+} dependent kinase

(b) PKC can be activated by diacylglycerol

(c) Activated PKC can phosphorylate the specific tryptophan residue on the target protein

(d) Activation of PKC can lead to the activation of MAP kinase

52. Decreasing order of permeability through a lipid bi-layer of the molecules/ions is

(a) $O_2 > Na^+ >$ Isoleucine > Tyrosine

(b) $O_2 >$ Isoleucine > tyrosine > N_a^+

(c) Isoleucine > Tyrosine > $O_2 > N_a^+$

(d) Isoleucine > Tyrosine > $N_a^+ > O_2$

53. Match List I (complexes) with List II (its alternative names) and select the correct answer from the codes given below the lists :

List I	List II
A. Complex I	1. Reduced COQ : Cytochrome reductase
B. Complex II	2. ATPase
C. Complex III	3. Succinate : COQ reductase
D. Complex V	4. NADH : COQ reductase

Codes:

	A	B	C	D
(a)	4	3	1	2
(b)	3	2	4	1
(c)	2	4	3	1
(d)	2	4	1	3

54. Injection of Dinitrophenol (DNP) into a rat causes an immediate increase in its body temperature because it

(a) acts as an inhibitor of mitochondrial ATPase

(b) blocks electron transport chain

(c) inhibits succinic dehydrogenase

(d) uncouples electron transport from oxidative phosphorylation

55. BLAST algorithm was published by

(a) Pearson & Lipman

(b) F. Sanger

(c) Joseph Sambrook

(d) Eugene Myers, Stephen Altschul

ANSWERS

General Aptitude (GA)

1. (b) **2.** (c) **3.** (d) **4.** (a) **5.** (c) **6.** 26000 **7.** 41 **8.** (b) **9.** (b) **10.** (a)

Technical Section

1. (c) **2.** (c) **3.** (d) **4.** (a) **5.** (d) **6.** (d) **7.** (b) **8.** (c) **9.** (d) **10.** (c)

11. (a) **12.** (b) **13.** (a) **14.** (c) **15.** (b) **16.** (d) **17.** 12,627 **18.** (a) **19.** (a) **20.** (c)

21. (d) **22.** (c) **23.** (d) **24.** (b) **25.** (d) **26.** (c) **27.** $\frac{19}{30}$ **28.** 6.235 **29.** (d) **30.** 50

31. (d) **32.** (a) **33.** (c) **34.** (a) **35.** (d) **36.** (c) **37.** (a) **38.** (b) **39.** (d)

40. 4.6×10^{10} **41.** (c) **42.** (b) **43.** (a) **44.** (c) **45.** (b) **46.** (a) **47.** (b) **48.** (a)

49. (c) **50.** (d) **51.** (c) **52.** (b) **53.** (a) **54.** (d) **55.** (d)

MOCK TEST – 2

INSTRUCTIONS

1. Total of 65 questions carrying 100 marks, out of which 10 questions carrying a total of 15 marks are in General Aptitude (GA)
2. The Engineering Mathematics will carry around **15% of the total marks,** the General Aptitude section will carry **15% of the total marks** and the **remaining 70% of the total marks.**
3. **Types of Questions**

   (a) **Multiple Choice Questions (MCQ)** carrying 1 or 2 marks each in all papers and sections. These questions are objective in nature, and each will have a choice of four options, out of which the candidate has to mark the correct answer(s).

   (b) **Numerical Answer Questions** of 1 or 2 marks each in all papers and sections. For these questions the answer is a real number, to be entered by the candidate using the virtual keypad. No choices will be shown for these type of questions.
4. For **1-mark** multiple-choice questions, **1/3 marks will be deducted for a wrong answer. Likewise, for 2-marks** multiple-choice questions, **2/3 marks will be deducted for a wrong answer. There is no negative** marking for numerical answer type questions.

GENERAL APTITUDE (GA)

(Q.1 – 5) : Carry One Mark Each

1. Mr. Keshav earned a profit of 15% by selling a fan. If he had to sell at ₹ 640 less, he may incur a loss of 25%. The initial selling price of the fan (in ₹) is _____

2. Following question presents a sentence, which is underlined. Beneath the sentence you will find four ways of phrasing the underlined part. The first of these repeats the original; the other three are different. If you think the original is best, choose the first answer; otherwise choose one of the others :

The plan should also open the door to private financing, which the firm has not had access for two years.

(a) The plan should also open the door to private financing, which the firm has not had access for two years

(b) The plan should also open the door to private financing, for which the firm has not had access since two years

(c) The plan also should open the door to private financing, which the firm has not had access for two years

(d) The plan should also open the door to private financing, to which the firm has not had access for two years

3. Given sentence below has two blanks; each blank indicates that something has been omitted. Beneath the sentence are four lettered words. Choose the word for each blank that best fits the meaning of the sentence as a whole.

Within a complex web of ideologies, most of today's _____ radicals are linked by self-described Patriot beliefs, which emphasize resistance to _____ by force of arms and reject the idea that elections can fix what ails the country.

(a) innovative - - - - - - - - democracy

(b) armed - - - - - - - - tyranny

(c) jingoistic - - - - - - - - nepotism

(d) timid - - - - - - - - oppression

4. Study the information given in the following problem and answer the question.

Statements:

Some engineers are teachers.

Some teachers are post graduates.

Conclusions:

I. Some engineers are post graduates.

II. Some post graduates are engineers.

III. All post graduates are teachers.

IV. All teachers are engineers.

(a) Only I and III follow

(b) Only, II and IV follow

(c) All follow

(d) None of these

2.2

5. In the given question, a related pair of words is followed by four lettered pairs of words. Choose the best expresses a relationship similar to that expressed in the original pair.

DALMATIAN : DOG ::

(a) shark : walrus (b) stable : horse

(c) oriole : bird (d) antler : ant

(Q.6 –10) : Carry Two Marks Each

6. In the following question part of sentence is underlined. Four alternative meanings of the highlighted part of the sentence are given below the sentence. Mark as your answer that alternative meaning which you think is correct.

Jefferson paid an arm and a leg for his car but he really enjoys driving it.

(a) unduly high price

(b) no attention

(c) something against one's wish

(d) very low price

7. A total amount of ₹ 50000 invested in two banks, the rate of interest is 15% and 18% per annum, under simple interest for 8 years and 10 years respectively. If the interest be equal by the end of the period, then find the total interest (in ₹) is _____

8. $p_1, p_2, p_3, p_4, \ldots p_{10}$ are positive integers which are not divisible by 5. the remainder when

$$p_1^2 + p_2^2 + p_3^2 + p_4^2 + \ldots\ldots\ldots + p_{10}^2$$

is divided by 5 is

(a) 0 or 2 (b) 1 or 3

(c) 0 or 2 or 4 (d) 0, 1, 2, 3 or 4

9. Read the following statements (I, II) in each questions and give your response as a, b and c respectively when answer can be found using I only, using II only, using both statements. And give d, when answer cannot be found even by using both.

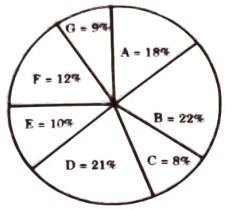
Find the value of P.

I. $P^2 < P$ II. $P^2 > P$

10. Study the following pie-chart and answer the question that follows

The pie-chart shows the budget estimated by a family for their monthly expense for different heads.

Budget estimated by a family for different heads, A, B, C, D, E, F, G is shown in the pie diagram below.



Total monthly salary of the family = ₹ 140000

Family spent ₹ 1400 less on E than budgeted. The new expenditure of E is equal to the expenditure of which of the following heads?

(a) A (b) G

(c) B (d) F

TECHNICAL SECTION

(Q.1 – 25) : Carry One Mark Each

1. Initiation of protein synthesis from most eukaryotic mRNA requires

(a) 3′ poly-A tail. (b) 5′ poly-A tail.

(c) 3′ cap (d) 5′ cap

2. Major groove of DNA is lined by

(a) N3 of purine and N1 of pyrimidine

(b) N7 of purine and O2 of pyrimidine

(c) N7 of purine and C4 of pyrimidine

(d) none of these

3. Which of the following statement does not describe phosphoglycerides ?

(a) They are both amphipathic and amphoteric

(b) They arise from phosphatidic acid

(c) They are found in cell membranes

(d) They are a major store of metabolic energy

4. Which of the following biochemical reactions is most commonly utilized by living cells to propagate intracellular signals ?

(a) Acylation

(b) Phosphorylation

(c) Methylation

(d) Decarboxylation

5. The class of immunoglobulin that can get transported across epithelial cells is

(a) IgG (b) IgE

(c) IgA (d) IgM

6. Which among the following is transferred from chloroplastof mesophyll cells to the chloroplast of bundle sheath cells?

(a) Oxaloacetic acid (b) ATP and ADP

(c) Aspartic acid (d) None of these

7. Which technique is used to identify the protein binding regions of DNA?

(a) Finer printing (b) Southern blotting

(c) Foot printing (d) Western blotting

8. The hydrolysis of fats is accelerated by the presence of which of the following ions/cations?

(a) Mg^{++} (b) Ca^{++}

(c) Na^{+} (d) Fe^{++}

9. Most sensitive immunodiagnostic test in terms of amount of antibody detection is

(a) Gel precipitaion test

(b) Radio-immune assay

(c) Complement of fixation test

(d) Haemagglutination test

10. Equation for exponential phase in case of S-curve is given by

(a) $\frac{dx}{dt} = x$ (b) $\frac{dx}{dt} = \mu x$

(c) Both (a) and(b) (d) None of these

11. Which of the following ions bind strongly to valinomycin?

(a) K^{+} (b) Na^{+}

(c) Al^{3+} (d) Ca^{++}

12. When organic acid is oxidized in respiration, the value of R.Q becomes

(a) less than one (b) zero

(c) one (d) more than one

13. Aspirin, used as a common antipyretic and anti-inflammatory agent, inhibits synthesis of

(a) arachidonic acid (b) prostaglandins

(c) glucocorticoids (d) histamine

14. When a protein is denatured, the freedom of rotation about bounds in both polypeptide backbone and the side chain is

(a) increased

(b) decreased

(c) increased in the backbone only

(d) increased in the side chain only

15. Which of the following is a hydrophilic hormone with cell-surface receptors?

(a) Thyroxin (b) Epinephrine

(c) Aspirin (d) Progesterone

2.3

16. Two proteins have same molecular weight but differ in their amino acid composition. They can be separated by

(a) reverse phase chromatography

(b) gel filtration

(c) ion-exchange

(d) hydrophobic chromatography

17. The active form of methionine is

(a) S-adenosylmethionine

(b) heated methioxine

(c) nucleated methionine

(d) none of these

18. Dinitrophenol would be most likely to inhibit cell function by disrupting

(a) TCA cycle

(b) glycolysis

(c) hepatic gluconeogenesis

(d) oxidative phosphorylation

19. Cellulose cell wall can better be stained by

(a) Fehling's solution (b) Metachite green

(c) Millions reagent (d) None of these

20. What is the phenotypic ratio in F2 generation in case of complementary gene?

(a) 9 : 3 : 1 (b) 12 : 3 : 1

(c) 9 : 7 (d) 15 : 1

21. The travel of gene expression and gene activation can be measured by using

(a) Reporter gene (b) Marker gene

(c) Promoter element (d) Gene sequence

22. Which step translation does not consume high-energy phosphate bond?

(a) Translocation

(b) Peptidyl transferees reaction

(c) Amino acid activation

(d) Aminoacyl RNA binding to aside

23. When a molecule of pyruvic acid is subjected to anaerobic oxidation and forms lactic acid, then there will be

(a) gain of 2 ATP molecules

(b) gain of 4 ATP molecules

(c) loss of 3 ATP molecules

(d) loss of 6 ATP molecules

24. EMS, MMS, dES belongs to _____ under chemical mutagens.

(a) acridine dyes (b) alkylating agents

(c) methylating agents (d) epoxides

25. According to Howarth repre-sentation of the sugars, OH group at the reducing centre is present below the plane of the ring, then sugar is said to be in

(a) β-form (b) α-form
(c) both (a) and (b) (d) none of these

Q. (26 - 55) : Carry Two Marks Each

26. If sum of Eign values of the matrix $\begin{pmatrix} 6 & -2 & 2 \\ -2 & x & -1 \\ 2 & -1 & x \end{pmatrix}$ is given by the equation $x^2 - 4x + 15$, then x is _____

27. The value of $\int_c \frac{\cos \pi z}{(z-1)(z-2)} dz$ where C = circle; $|z| = \frac{3}{2}$, will be _____

28. One call out of 15 calls between 2 pm and 4 pm of a week is busy. Find the probability 2 calls will be busy, out of 6 randomly selected telephone numbers dialed.

(a) 0.0505 (b) 0.909
(c) 0.48 (d) 0.5

29. The value of integral $\int_0^\infty \frac{dx}{(1+x^2)^3}$ is _____

30. Percentage error in the area of an ellipse, if there is an error of 1% each in measuring the major and minor axes, will be _____ %

31. Human genome comprises of how many genes and base pairs?

(a) ~37,544 & 3.3×10^8
(b) ~20,500 & 3.3×10^9
(c) ~19,300 & 3.3×10^9
(d) ~19,500 & 3.9×10^8

32. Prints software is used for

(a) detection of genes from genome sequence
(b) identification of functional domains/motifs of proteins
(c) detection of tRNA genes
(d) prediction of function of a new gene

33. Match List I (Method of separation) with List II (principle of separation) and select the correct answer from the codes given below the lists.

List I	List II
A. Filtration	1. Specific gravity
B. Ultrafiltration	2. Electric charge
C. Iron-exchangers	3. Molecular size
D. Centrifugation	4. Particle size

Codes:

	A	B	C	D
(a)	3	4	2	1
(b)	4	3	2	1
(c)	4	1	2	3
(d)	3	1	4	2

34. Match List I (precipitating agents) with List II (example) and select the correct answer from the codes given below the lists.

List I	List II
A. Ionic polymers	1. Ammonium sulfate
B. Non-ionic polymers	2. Ethanol
C. Neutral salt	3. Polyethyleneimine
D. Organic solvents	4. Polyethylene glycol

Codes:

	A	B	C	D
(a)	4	1	3	2
(b)	4	1	2	3
(c)	3	4	1	2
(d)	4	2	1	3

35. One molecule of palmitate acid with 16 C-atoms, on complete oxidation will produce _____ number of ATP molecules .

36. What is $\Delta G°$ for the reaction in which cytochrome C is oxidized by oxygen from the ferrous to the ferric state with $E° = +0.56V$ and F = 23,061 cal/v per equivalent ?

(a) –26, 540 cal/mole (b) –12, 914 cal/mole
(c) –25, 828 cal/mole (d) –20, 860 cal/mole

37. Match List I with List II and select the correct answer from the codes given below the lists.

List I	List II
A. Luciferase	1. Fluorescence detection
B. Nopaline syntheses	2. Bioluminescence
C. β-glucuronidase	3. Electrophoresis
D. Nopaline synthase	4. Radioactive detection

Codes:

	A	B	C	D
(a)	2	4	1	3
(b)	3	1	2	4
(c)	3	4	2	1
(d)	1	4	3	2

38. If rings of monosaccharide are 5-membered and 6-membered, then they are respectively called

(a) pyranose and furanose sugar
(b) structural and storage polysac-charide
(c) furanose and pyranose sugar
(d) storage and structural

39. Isotopes used in proving semi-conservative mode of replication of DNA are

(a) ^{14}N, ^{14}C (b) ^{14}C, ^{31}P
(c) ^{14}N, ^{15}C (d) ^{10}C, ^{13}P

40. Protein reacts with concentrated HNO_3 gives yellow substitution products, which turn orange upon addition of alkali. The yellow product is

(a) nitrobenzene (b) trinitrophenol
(c) aminobenzaldehyde (d) none of these

41. Fixation of one CO_2 molecule in C_3 cycle of photosynthesis requires, how many NADPH & ATP?

(a) 2NADPH & 2ATP (b) 3NADPH & 2ATP
(c) 2NADPH & 3ATP (d) 1NADPH & 1ATP

42. Mutation results in loss of function

(a) due to synthesis of incomplete polypeptide chain
(b) sudden changes in sequence of bases in genes
(c) amino acid codon mutes to form non-sense codon
(d) all of these

43. Purines and pyrimidines are cleaved by

(a) dimethyl sulfate
(b) hydrazine
(c) piperazine
(d) dimethyl sulfate and hydrazine respectively

44. Match the items in List I with their functions in List II and select the correct answer from the codes given below the lists.

List I	List II
A. Virulence genes	1. Hairy root induction
B. Opines	2. T-DNA transfer and integration
C. Rol genes	3. Food and energy source
D. Aux and cyt genes	4. Tumor formation

Codes:

	A	B	C	D
(a)	2	4	1	3
(b)	2	3	4	1
(c)	4	2	1	3
(d)	1	3	4	2

45. If an enzyme solution is centrifuged and the high concentration of ammonium sulfate is added, then what is observed immediately?

(a) Solution colour changes to blue
(b) Enzyme particles dissolve completely
(c) Crystallization of enzymes occur
(d) O.D. of the solution decreaes

46. Which of the following enzymes can be used in labelling of DNA 5?

(P) Alkaline phosphatase
(Q) DNA ligase
(R) Terminal transferees
(S) Polynucleotide kinase

(a) P, R (b) P, S
(c) R, Q (d) R, S

47. In the bioreactor, baffles are incorporated to

(a) prevent vortex and to improve aeration efficiency
(b) maintain uniform suspension of cells
(c) minimize the size of air bubble for greater aeration
(d) maintain uniform nutrient medium

48. Match List I (items) with List II (inhibitors) and select the correct answer from the codes given below the lists :

List I	List II
A. DNA polymerase α	1. Phenyl methyl sulfonyl fluoride (PMSF)
B. RNA polymerase II	2. Aphidicolin
C. Serine protease	3. α-amanitin
D. P-amino benzoic acid	4. Sulfanilamide

Codes:

	A	B	C	D
(a)	2	3	1	4
(b)	4	2	3	1
(c)	3	2	4	1
(d)	4	3	2	1

49. Protoplast fusion will be successful in plant cells, if

(a) fusion agents other than polyethylene glycol should be used.
(b) cell walls of two strains of cells should not be damaged.
(c) DNA between two cells should be compatible.
(d) none of these

50. Order of the predicted rate of diffusion across a lipid bi-layer is (highest to lowest)

(a) Toluene, Phenylalanine, galactose, chloride ion
(b) Galactose, phenylalanine, toluene, chloride ion
(c) Toluene, galactose, phenylalanine, chloride ion
(d) Chloride ion, toluene, galactose, phenylalanine

51. The direct effect of cyclic AMP in the protein kinase 'A' pathway is to

(*a*) activate adenylate cyclase

(*b*) dissociate regulatory submits form prtein kinase 'A'

(*c*) phosphorylated protein kinase 'A'

(*d*) release hormones from target tissue

52. For which of the following pairs of substances does the inner mitochondrial membrane contain specific transport proteins?

(*a*) Ca^{+2} and Mg^{+2}

(*b*) AMP & ADP

(*c*) Malate & Aspartate

(*d*) Citrate& Oxaloacetate

53. One microgram of a pure enzyme (MW = 92,000) catalyzed a reaction at a rate of 0.50 μ moles/min under optimum condition. The turn over number per min is_____

54. Which of the following cross-linking agents may be used for the immobilization of enzymes?

(*a*) Glutaraldehyde (*b*) Cyanogen bromide

(*c*) Thionyl chloride (*d*) All of these

55. Match List I (method) with List II (their performing the jobs) and select the correct answer from the codes given below the lists:

List I	List II
A. Prosite	1. Searching family data base
B. Bem launcher	2. Finding alignments
C. Boxshade	3. Displaying alignments
D. PSI-BLAST	4. Searching for multiple alignments

Codes:

	A	B	C	D
(*a*)	2	1	3	4
(*b*)	2	1	4	3
(*c*)	3	1	4	2
(*d*)	4	2	3	1

ANSWERS

General Aptitude (GA)

1. 1840	**2.** (*d*)	**3.** (*b*)	**4.** (*d*)	**5.** (*c*)	**6.** (*a*)	**7.** 72000	**8.** (*d*)	**9.** (*d*)	**10.** (*b*)

Technical Section

1. (*d*)	**2.** (*c*)	**3.** (*d*)	**4.** (*b*)	**5.** (*c*)	**6.** (*d*)	**7.** (*c*)	**8.** (*b*)	**9.** (*b*)	**10.** (*b*)
11. (*a*)	**12.** (*d*)	**13.** (*b*)	**14.** (*a*)	**15.** (*b*)	**16.** (*a*)	**17.** (*a*)	**18.** (*d*)	**19.** (*d*)	**20.** (*c*)
21. (*b*)	**22.** (*b*)	**23.** (*d*)	**24.** (*b*)	**25.** (*b*)	**26.** 3	**27.** 0	**28.** (*a*)	**29.** $\frac{3\pi}{16}$	**30.** 4
31. (*b*)	**32.** (*b*)	**33.** (*b*)	**34.** (*c*)	**35.** 129	**36.** (*c*)	**37.** (*a*)	**38.** (*c*)	**39.** (*c*)	**40.** (*b*)
41. (*c*)	**42.** (*d*)	**43.** (*d*)	**44.** (*a*)	**45.** (*c*)	**46.** (*b*)	**47.** (*a*)	**48.** (*a*)	**49.** (*c*)	**50.** (*c*)
51. (*c*)	**52.** (*c*)	**53.** 4.6×10^4	**54.** (*d*)	**55.** (*a*)					

MOCK TEST – 3

INSTRUCTIONS

1. Total of 65 questions carrying 100 marks, out of which 10 questions carrying a total of 15 marks are in General Aptitude (GA)
2. The Engineering Mathematics will carry around **15% of the total marks**, the General Aptitude section will carry **15% of the total marks** and the **remaining 70% of the total marks.**
3. **Types of Questions**
   - (*a*) **Multiple Choice Questions (MCQ)** carrying 1 or 2 marks each in all papers and sections. These questions are objective in nature, and each will have a choice of four options, out of which the candidate has to mark the correct answer(s).
   - (*b*) **Numerical Answer Questions** of 1 or 2 marks each in all papers and sections. For these questions the answer is a real number, to be entered by the candidate using the virtual keypad. **No choices will be shown for these type of questions.**
4. For **1-mark** multiple-choice questions, **1/3 marks** will be deducted for a **wrong answer. Likewise, for 2-marks** multiple-choice questions, **2/3** marks will be deducted for a wrong answer. **There is no negative** marking for numerical answer type questions.

GENERAL APTITUDE (GA)

(Q.1 – 5) : Carry One Mark Each

1. In the following question, four pairs of analogous words are given. Three pairs can be classified as one group due to some underlying similarity. Detect the similarity and find the pair which does not belong to the group.

(*a*) Scold : jay (*b*) Smack : jellyfish

(*c*) Lounge : lizard (*d*) Bevy : heroin

2. *The following question presents a sentence, all or part of which is underlined. Beneath the sentence you will find four ways of phrasing the underlined part. The first of these repeats the original; the other three are different. If original is best, choose the first answer; otherwise choose one of the others.*

Under New York State rules, the only ways for Mr. Lazio to leave the Conservative line were death, leave the state or nomination for a judgeship.

(*a*) were death, leave the state or nomination for a judgeship

(*b*) were to die or to leave the state or being nominated as a judge

(*c*) were to die, to leave the state or to be nominated for a judgeship

(*d*) were dying, leaving the state or nominating for judgeship

3. *Given sentence below has two blanks; each blank indicates that something has been omitted. Beneath the sentence are four lettered words. Choose the word for each blank that best fits the meaning of the sentence as a whole.*

The location of and enrollment at some of the flourishing after-school programs in mathematics that have sprung up in the past two decades suggest they are a response to the _____ mathematically literate parents perceive in their public school's mathematics coursework and____.

(*a*) growth.....wisdom

(*b*) deficiencies.....pedagogy

(*c*) complexities.....puzzle

(*d*) affluence.....creativity

4. *Given passage consists of six sentences. First and sixth sentences are given. The middle four sentences in each have been removed and jumbled up. These are labeled as P, Q, R and S. Find out the proper order for the four sentences.*

I. Doing well on the A.P. calculus exam

P. and research shows that the best American students

Q. of calculus eual to the best high school

R. have a conceptual understanding

S. requires solid but not exceptional algebra skills

II. calculus students in Japan.

(*a*) SRQP (*b*) SPRQ

(*c*) PQSR (*d*) SPQR

5. Which of the following is not the synonym of the capitalized word, "PLUMB" ?

(*a*) Ascend (*b*) Fathom

(*c*) Measure (*d*) Investigate

3.2

(Q.6 –10) : Carry Two Marks Each

6. If sum of two numbers is 720 and their HCF is 24, then highest possible LCM of the numbers is _____

7. Given that $14x + 9y = 167$.

If $14x$ is increased by 27% and $9y$ is increased by 39%, then 167 is increased by 33. If $14x$ is decreased by 27% and $9y$ is decreased by 39%, then 167 is decreased by z. Find z.

(a) 27 (b) 39

(c) 43 (d) 33

8. A fruit vendor had some apples and gave half of the apples and one extra apple to his first customer. He had given half of the remaining apples and 1 apple extra to his second customer. Then he had given half of the further remaining apples and 1 apple extra to his third customer. At the stage, the fruit vendor left with any apples. Sum of the digits of the number of apples with the vendor initially is _____

9. A, B, C and D worked together and shared the money in the ratio of 3 : 4 : 5 : 6 instead of $\frac{1}{3} : \frac{1}{4} : \frac{1}{5} : \frac{1}{6}$.

How many members made profit due to this mistake?

(a) 0 (b) 1

(c) 2 (d) 3

10. Mr. Kranthi borrowed Rs.8000 from a moneylender for 3 years at CI compounding annually. If the interest for third year is Rs. 2304, then rate of interest per annum will be _____

TECHNICAL SECTION

(Q.1 – 25) : Carry One Mark Each

1. Which of the following is desirable characteristic for a cloning plasmid?

(a) A site at which replication can be initiated

(b) One or more unique restriction endonuclease sites

(c) One or more antibiotic-resistance or drug resistance genes

(d) All of these

2. The tryptophan biosynthetic operon is an example of a

(a) positively controlled repressible operon

(b) negatively controlled inducible operon

(c) positively controlled inducible operon

(d) negatively controlled repressible operon

3. Which pair of amino acids will have the highest absorbance at 280 nm? (Assume equimolar concentrations)

(a) Thr and His (b) Phe and Pro

(c) Trp and Tyr (d) Phe and His

4. z-DNA helix

(a) has fewer base pairs per turn than the B-DNA

(b) is favoured by an alternating GC sequence

(c) tends to be found at the 3′ end of genes

(d) is most common conformation of DNA

5. Which of the following is involved in cell signaling process?

(a) Cytochrome P450 (b) Protein kinase C

(c) Zinc fingers (d) All of these

6. The method of promoting competence in transformation is by adding

(a) calcium chloride

(b) magnesium sulphate

(c) magnesium chloride

(d) calcium sulphate

7. Cause for conformational change in biomolecules is

(a) pH (b) ionic strength

(c) surface groups (d) all of these

8. Tissue culture plastic adhesivity was precisely varied by applying different concentrations of

(a) SDS

(b) sodium silicates

(c) poly (2-Hydroxyethyl Methacrylate)

(d) GABA

9. Micro RNA has

(a) enzymatic function

(b) inhibitor role

(c) regulatory role

(d) promoter role

10. Change in the frequency of a gene variant in a population due to random sampling is called

(a) genetic drift (b) gene flow

(c) natural selection (d) mutation

11. Biofilm formation is a

(a) chemical process

(b) physical process

(c) natural process

(d) both (a) and (c)

12. Xylanase synthesized in solid fermen-tation is maximally active at _____ °C

13. Cell-mediated immunity is directed primarily at microbes that survive in

(a) vacuole (b) mesosome

(c) phagocyte (d) episome

14. Avidity is influenced by

(a) valence of the antibody

(b) specificity of receptors

(c) valence of the antigen

(d) both (a) and (c)

15. Alloimmunity can be regulated by

(a) neonatal B cells (b) natural killer cells

(c) neonatal T cells (d) macrophages

16. The vascular tissue, which is used to transfer the hormones from root to foliage, is

(a) phloem (b) sieve cells

(c) sieve tubes (d) xylem

17. The antifungal added to media for the manipulations to be carried out on culture cells is

(a) penicillin (b) amphotericin B

(c) streptomycin (d) tetracycline

18. The processes which cannot be performed by prokaryotes is

(a) glycosylation (b) amidation

(c) phosphorylation (d) all of these

19. The largest portion of the category of cell culture products on the market is

(a) media (b) sera

(c) reagent (d) all of these

20. Natural hyper-accumulators in plants that can tolerate high levels of toxic metals is

(a) metallophyte (b) pesticides

(c) chelators (d) explosives

21. The agent that provides a suitable chemical environment for optimum activity of PCR is

(a) divalent cations (b) DNA template

(c) taq polymerse (d) buffer solution

22. The differences between the various FISH techniques are usually due to

(a) variations in the sequence

(b) different combinations of FISH

(c) labelling of the probes

(d) all of these

3.3

23. A database of annotated protein models calculated by homology modelling is

(a) OCA

(b) SWISS-MODEL Respository

(c) OPM

(d) MSD

24. A sequencing technology, which is developed, based on reversible dye-terminators is

(a) solexa sequencing

(b) high-throughput sequencing

(c) 454 pyrosequencing

(d) solid requencing

25. Which of the following statement/s is true about hydroxylamine?

(a) Hydroxylamine (NH_2OH) reacts with purine bases.

(b) Mutations induced by hydroxylamine can be reversed with hydroxylamine.

(c) Hydroxylamine effect is strong on cytosine while on uracil the effect is less.

(d) All of these

Q. (26 - 55) : Carry Two Marks Each

26. $\lim_{x \to \infty} \frac{x^3 - \cos x}{x^2 + 1}$ equals

(a) ∞ (b) 0

(c) 2 (d) does not exist

27. Taylor series expansion of the function, $f(x) = \frac{x}{1+x}$ around $x = 0$ is

(a) $x + x^2 + x^3 + x^4$ .....

(b) $1 + x + x^2 + x^3 + x^4$+.....

(c) $2x + 4x^2 + 8x^3 + 16x^4$.....

(d) $x - x^2 + x^3 - x^4 +$ .....

28. Probability of guessing atleast 6 out of 10 answers in an examination, if each question consists of four options will be _____

29. The function $z = (x - 1)^2 + 2y^2$ has a

(a) minimum point

(b) maximum point

(c) neither minimum nor maximum

(d) none of these

30. Magnitude of the gradient of the function $f = xyz^3$ at (1, 0, 2) is _____

31. Which of the following is the study of fluid deformation?

(a) Rheology (b) Hydrology

(c) Meteorology (d) Fluidol

32. Match List I (primary lymphoid organs) with List II (their respective site of maturation) and select the correct answer from the codes given below the lists :

List I

A. Mammals

B. Rabbit

C. Sheep

D. Birds

List II

1. Bursa of Fabricius
2. Bone marrow & Thymus
3. Peyer's Patches and Appendix
4. Peyer's Patches and Gut

Codes :

	A	B	C	D
(a)	2	3	4	1
(b)	2	1	4	3
(c)	2	3	1	4
(d)	2	4	3	1

33. Lack of agglutination at high concentration of antibodies is

(a) Titre (b) Prozone effect

(c) Rhesus factor (d) Epitope

34. Auxins are the compounds, which positively influence on

(a) cell enlargement

(b) root initiation

(c) bud formation

(d) all of these

35. Most common example of unipotent cell is

(a) stem cells

(b) endothelial cells

(c) skin cells

(d) smooth muscle cells

36. The use of antibodies as enzymes is described as

(a) abzyme (b) transferase

(c) lysozyme (d) transposase

37. The complete degradation of an organic chemical to carbon dioxide, water and potentially other inorganic compounds including salts is

(a) Cometabolism (b) Biostimulation

(c) Mineralization (d) Bioremediation

38. Immobilization method in which the enzymes are trapped in insoluble beads or micro-spheres is a/an

(a) adsorption (b) entrapment

(c) cross linkage (d) none of these

39. Which of the following is the RAPD marker?

(a) Dimer (b) Decamer

(c) Tetramer (d) Pentamer

40. In humans, mini-satellites are found at

(a) subtelomeric region

(b) centromeric region

(c) telomeric regions

(d) non-telomeric region

41. A term coined to describe behaviour resulting from the combination of evolutionary and ecological processes is

(a) phylodynamics

(b) microbial Phylogenetics

(c) phylogeography

(d) phylocode

42. Micro-array technology evolved from

(a) northern blotting technique

(b) chromatographic technique

(c) southern blotting technique

(d) electrophoresis

43. An enzyme was assayed at an initial substrate concentration of 2×10^{-5}M. In 6 min half of the substrate had been used. Then the value of V_{max} is _____ μmoles $\times$litre$^{-1}\times$min^{-1} (k_m for substrate is 5×10^{-8}M)

44. Which of the following statements belongs to transduction?

(a) DNA can be absorbed and recycled for spare parts.

(b) The bacterial DNA can match up with a homologous DNA in the recipient cell and exchange it. The recipient cell now has DNA from both itself and the other bacterial cell.

(c) DNA can insert itself into the genome of the recipient cell as if still acting like a virus resulting in a double copy of the bacterial genes.

(d) All of these

45. Match List I (viruses) with List II (their respective genetic material) and select the correct answer from the codes given below the lists :

List I	List II
A. Adenovirus	1. ssRNA
B. Parvovirus	2. dsRNA
C. Reovirus	3. ssDNA
D. Togavirus	4. ds DNA

Codes :

	A	B	C	D
(a)	4	3	2	1
(b)	4	1	3	2
(c)	3	1	4	2
(d)	4	3	1	2

46. Which of the following statements are true about receptors?

(a) The immunosuppressant drugs tacrolimus and cyclosporine inhibit calcineurin thus reducing the threat of transplant rejection by B-cells.

(b) TNF-α is made by macrophages.

(c) Notch proteins are single-pass transmemebrane glycoprotein.

(d) Both (b) and (c)

47. Which of the following statements are true about Cyclins?

(a) Cyclins have catalytic activity.

(b) CDKs are active in the absence of a partner cyclin.

(c) Cyclins form the regulatory subunits and CDKs the catalytic subunits of an activated heterodimer.

(d) All of these

48. Folate derivatives are required for the synthesis of which of the following deoxynucleotides?

(a) Adenylate and guanylate

(b) Cytidylate and thymidylate

(c) Adenylate, guanylate, thymidylate

(d) Adenylate, guanylate and cytidylate

49. Match List I with List II and select the correct answer from the codes given below the lists :

List I

A. β-ketoacyl-ACP synthase

B. β-Ketoacyl-ACP reductase

C. Enoyl p ACP reductase

D. Acyl carrier protein (ACP)

List II

1. Reduces double bond forming saturated acyl-ACP Hexosamini dase-A
2. Reduces β-keto group to β-hydroxyl group
3. Condenses acyl and malonyl group
4. Carries acyl group in thioester linkages

Codes:

	A	B	C	D
(a)	2	1	3	4
(b)	3	2	1	4
(c)	4	1	2	3
(d)	3	1	2	4

50. Viral encoded Ras oncogene transforms normal mammalian cells into cancer cells. Viral Ras protein differs from its normal counterpart by

(a) diminished GTP are activity

(b) increased GTP are activity

(c) diminished ATP are activity

(d) increased ATP are activity

51. Which of the following reagents is used in detection of amino acid cysteine?

(a) α-Naphthol (b) Glyoxylic acid

(c) HNO_3 (d) Sodium nitroprusside

52. Concentration of low density lipoproteins (LDL) in plasma is

(a) 120-150 mg% (b) 200 mg%

(c) 350% (d) 150-400 mg%

53. Anti metabolites act by

(a) competitive inhibition when they combine irreversibly with an enzyme

(b) competitive inhibition when they combine reversibly with an enzyme

(c) non competitive inhibition by combining with the prosthetic group

(d) by disrupting the normal growth of a cell

54. During glyoxylate cycle, 4 carbon oxaloacetate is generated from two carbon acetate. The correct sequence in which intermediate compounds appear from citrate is

(a) citrate, Succinate, fumarate, glyoxylate

(b) citrate, fumarate, malate, glyoxylate

(c) citrate, isocitrate, glyoxylate, malate

(d) Citrate, isocitrate, malate, fumarate

55. PAM 250 scoring matrix means

(a) 250 mutations per 10 amino acids are assumed

(b) 50 mutations per 100 amino acids are assumed

(c) 250 mutations per 100 amino acids are assumed

(d) none of these

1. (*d*)	**2.** (*c*)	**3.** (*b*)	**4.** (*b*)	**5.** (*a*)	**6.** 5304	**7.** (*d*)	**8.** 5	**9.** (c)	**10.** 20

Technical Section

1. (*d*)	**2.** (*d*)	**3.** (c)	**4.** (*b*)	**5.** (*b*)	**6.** (*a*)	**7.** (*d*)	**8.** (c)	**9.** (c)	**10.** (*a*)
11. (c)	**12.** 50	**13.** (c)	**14.** (*d*)	**15.** (*a*)	**16.** (*d*)	**17.** (*b*)	**18.** (*d*)	**19.** (*d*)	**20.** (*a*)
21. (*d*)	**22.** (*d*)	**23.** (*b*)	**24.** (*a*)	**25.** (*c*)	**26.** (*a*)	**27.** (*d*)	**28.** 0.09	**29.** (*a*)	**30.** 8
31. (*a*)	**32.** (*a*)	**33.** (*b*)	**34.** (*d*)	**35.** (*c*)	**36.** (*a*)	**37.** (*c*)	**38.** (*b*)	**39.** (*b*)	**40.** (*a*)
41. (*a*)	**42.** (c)	**43.** 575	**44.** (*d*)	**45.** (*a*)	**46.** (*d*)	**47.** (c)	**48.** (c)	**49.** (*b*)	**50.** (c)
51. (*a*)	**52.** (*b*)	**53.** (*d*)	**54.** (*c*)	**55.** (*c*)					

GENERAL APTITUDE

1. "When she fell down the _____, she received many _____ but little help." The words that best fill the blanks in the above sentence are
   (a) stairs, stares (b) stairs, stairs
   (c) stares, stairs (d) stares, stares

2. "In spite of being warned repeatedly, he failed to correct his _____ behaviour." The word that best fills the blank in the above sentence is
   (a) rational (b) reasonable
   (c) errant (d) good

3. For $0 \le x \le 2\pi$, $\sin x$ and $\cos x$ are both decreasing functions in the interval _____.
   (a) $\left(0, \frac{\pi}{2}\right)$ (b) $\left(\frac{\pi}{2}, \pi\right)$
   (c) $\left(\pi, \frac{3\pi}{2}\right)$ (d) $\left(\frac{3\pi}{2}, 2\pi\right)$

4. The area of an equilateral triangle is $\sqrt{3}$. What is the perimeter of the triangle?
   (a) 2 (b) 4
   (c) 6 (d) 8

5. Arrange the following three-dimensional objects in the descending order of their volumes:
   (i) A cuboid with dimensions 10 cm, 8 cm and 6 cm
   (ii) A cube of side 8 cm
   (iii) A cylinder with base radius 7 cm and height 7 cm
   (iv) A sphere of radius 7 cm
   (a) (i), (ii), (iii), (iv) (b) (ii), (i), (iv), (iii)
   (c) (iii), (ii), (i), (iv) (d) (iv), (iii), (ii), (i)

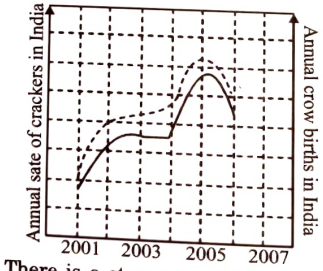
6. An automobile travels from city A to city B and returns to city A by the same route. The speed of the vehicle during the onward and return journeys were constant at 60 km/h and 90 km/h, respectively. What is the average speed in km/h for the entire journey?
   (a) 72 (b) 73
   (c) 74 (d) 75

7. A set of 4 parallel lines intersect with another set of 5 parallel lines. How many parallelograms are formed?
   (a) 20
   (b) 48
   (c) 60
   (d) 72

8. To pass a test, a candidate needs to answer at least 2 out of 3 questions correctly. A total of 6,30,000 candidates appeared for the test. Question A was correctly answered by 3,30,000 candidates. Question B was answered correctly by 2,50,000 candidates. Question C was answered correctly by 2,60,000 candidates. Both questions A and B were answered correctly by 1,00,000 candidates. Both questions B and C were answered correctly by 90,000 candidates. Both questions A and C were answered correctly by 80,000 candidates. If the number of students answering all questions correctly is the same as the number answering none, how many candidates failed to clear the test?
   (a) 30,000 (b) 2,70,000
   (c) 3,90,000 (d) 4,20,000

9. If $x^2 + x - 1 = 0$ what is the value of $x^4 + \frac{1}{x^4}$?
   (a) 1 (b) 5
   (c) 7 (d) 9

10. In a detailed study of annual crow births in India, it was found that there was relatively no growth during the period 2002 to 2004 and a sudden spike from 2004 to 2005. In another unrelated study, it was found that the revenue from cracker sales in India which remained fairly flat from 2002 to 2004, saw a sudden spike in 2005 before declining again in 2006. The solid line in the graph below refers to annual sale of crackers and the dashed line refers to the annual crow births in India. Choose the most appropriate inference from the above data.



   (a) There is a strong correlation between crow birth and cracker sales.
   (b) Cracker usage increases crow birth rate.
   (c) If cracker sale declines, crow birth will decline.
   (d) Increased birth rate of crows will cause an increase in the sale of crackers.

BIOTECHNOLOGY

1. Consider an unfair coin. The probability of getting heads is 0.6. If you toss this coin twice, what is the probability that the first or the second toss is heads?
   (a) 0.56 (b) 0.64
   (c) 0.84 (d) 0.96

2. If serum is removed from the growth medium of human embryonic kidney cell line (HEK), then the cells will
   (a) proliferate faster
   (b) proliferate normally
   (c) undergo cell cycle arrest
   (d) undergo immediate apoptosis

3. The repeat sequence of telomere in humans is
   (a) 5′-TATAAT-3′ (b) 5′-TTAGGG-3′
   (c) 5′-GGGCCC-3′ (d) 5′-AAAAAA-3′

4. If a segment a of sense stand of DNA is 5′-ATGGACCAGA-3′, then the resulting RNA sequence after transcription is
   (a) 5′-AGACCAGGTA-3′
   (b) 5′-UCUGGUCCAU-3′
   (c) 5′-UACCUGGUCU-3′
   (d) 5′-AUGGACCAGA-3′

5. Which one of the following is an example of a neurotoxin?
   (a) Cholera toxin (c) Streptolysin-O
   (c) Botulinum toxin (d) Diphtheria toxin

6. Which of the following components constitute a molecular mechanics force field?
   P. Bond stretching
   Q. Bond angle bending
   R. Torsional bond rotation
   S. Non-bonded interactions
   (a) P and Q only (b) P, Q and R only
   (c) P, Q and S only (d) P, Q, R and S

7. Which one of the following BLAST search programs is used to identify homologous of an genomic DNA query in a protein sequence database?
   (a) blastp (b) blastn
   (c) blastx (d) tblastn

8. A mixture contains three similarly sized peptides P, Q and R. The peptide P is positively charged, Q is weakly negative and R is strongly negative. If this mixture is passed through an ion-exchange chromatography column containing an anionic resin, their order of elution will be
   (a) P, Q, R
   (b) R, Q, P
   (c) Q, R, P
   (d) P, Q and R elute together

9. Which one of the following is **INCORRECT** about protein structures?
   (a) A protein fold is stabilized by favorable non-covalent interactions
   (b) All parts of a fold can be classified as helices, strands or turns
   (c) Two non-covalent atoms cannot be closer than the sum of their van der Waals radii
   (d) The peptide bond is nearly planar

10. Which one of the following metabolic processes in mammalian cells does **NOT** occur in the mitochondria?
    (a) Citric acid cycle
    (b) Oxidative phosphorylation
    (c) Fatty acid β-oxidation
    (d) Glycolysis

11. Which one of the following is **NOT** A principal component of innate immunity?
    (a) Mucosal epithelia
    (b) Dendritic cells
    (c) Complement system
    (d) Memory B-cells

12. Which of the following technique(s) can be used to study conformational changes in myoglobin?
    P. Mass spectrometry
    Q. Fluorescence spectroscopy
    R. Circular dichroism spectroscopy
    S. Light microscopy
    (a) P only (b) P and S only
    (c) Q and R only (d) S only

13. Which one of the following bioreactor configurations is the basis for a trickling biological filter?
    (a) Stirred tank (b) Packed bed
    (c) Air lift (d) Fluidized bed

14. Cell type A secretes molecule X into the culture medium. Cell type B in the same culture responds to the molecule X by expressing protein Y. Which one of the following modes of signaling represents the interaction between A and B?
    (a) Autocrine (b) Juxtacrine
    (c) Paracrine (d) Intracrine

15. Which one of the following statements is true for actin?
(a) Actin filament is structurally polarized and the two ends are not identical
(b) *De novo* actin polymerization is a single-step process
(c) The pointed end of the actin filaments is the fast growing end
(d) Actin forms spindle fibres during mitosis

16. Standard error is
(a) the probability of a type I error in a statistical test
(b) the error in estimating a sample standard deviation
(c) the standard deviation of a variable that follows standard normal distribution
(d) the standard deviation of distribution of sample means

17. Which one of the following techniques is used to monitor RNA transcripts, both temporally and spatially?
(a) Northern blotting
(b) *In situ* hybridization
(c) Southern blotting
(d) Western blotting

18. Identify the character based method(s) used for the construction of a phylogenetic tree.
P. Maximum parsimony
Q. Neighbor joining
R. Maximum likelihood
S. Bootstrapping
(a) Q only (b) P and R only
(c) Q and S only (d) S only

19. Which one of the following is the solution for $\cos^2 x + 2\cos x\ 1 = 0$, for values of x in the range of $0 + < x < 360°$?
(a) 45° (b) 90°
(c) 180° (d) 270°

20. Which one of the following plant secondary metabolites is a natural insecticide?
(a) Digitoxin (b) Pyrethrin
(c) Salicyclic acid (d) Avenacin A-1

21. The determinant of the matrix $\begin{pmatrix} 4 & -6 \\ -3 & 2 \end{pmatrix}$ is _____.

22. The variable z has a standard normal distribution. If $P(0 \le z \le 1) = 0.34$, then $P(z^2 > 1)$ is equal to (up to two decimal places) _____.

23. The absorbance of a solution of tryptophan measured at 280 nm in a cuvette of 2.0 cm path length is 0.56 at pH 7. The molar extinction coefficient (ε) for tryptophan at 280 nm is 5600 $M^{-1}\ cm^{-1}$ at pH 7. The concentration of tryptophan (in M) in the solution is _____.

24. A single stem cell undergoes 10 asymmetric cell divisions. The number of stem cells at the end is _____.

25. Genomic DNA isolated from a bacterium was digested with a restriction enzyme that recognizes a 6-base pair (bp) sequence. Assuming random distribution of bases, the average length (in bp) of the fragments generated is _____.

26. In leguminous plants, both the rhizobium genes and the plant genes influence nodulation and nitrogen fixation. Which one of the following functions is NOT encoded by the host plant genes?
(a) Production of inducers that modify rhizobial cell wall
(b) Production of flavonoid inducers
(c) Establishment of contact between bacteria and legume
(d) Root hair curling

27. Which of the following cytokines are endogenous pyrogens?
P. Tumor necrosis factor-α
Q. Interleukin-I
R. Transforming growth factor-β
S. Interleukin-10
(a) P and Q only (b) P and R only
(c) R and S only (d) Q and S only

28. Match the classes of RNA molecules in Group I with their functions in Group II.

Group I	Group II
P. snoRNA	1. Protects germline from transposable elements
Q. piRNA	2. Blocks translation of selected mRNA
R. miRNA	3. Template for telomere elongation
S. snRNA	4. Modification and processing of rRNA
	5. Splicing of RNA transcripts

(a) P-3, Q-5, R-2, S-4
(b) P-1, Q-3, R-2, S-5
(c) P-1, Q-4, R-5, S-2
(d) P-4, Q-1, R-2, S-5

29. Determine the correctness or otherwise of the following Assertion [a] and the Reason [r]
Assertion : *Ab initio* gene finding algorithms that predict protein coding genes in eukaryotic genomes are not completely accurate.
Reason : Eukaryotic slice sites are difficult to predict.
(a) Both [a] and [r] are false
(b) [a] is true but [r] is false
(c) Both [a] and [r] are true and [r] is the correct reason for [a]
(d) Both [a] and [r] are true but [r] is not the correct reason for [a]

30. Which one of the following amino acids is catalyzed by activated macrophages to produce reactive nitrogen species?
(a) Arginine (b) Asparagine
(c) Cysteine (d) Histidine

31. Determine the correctness or otherwise of the following Assertion [a] and the Reason [r]
Assertion : The association constant in water for the G-C base pair is three times lower than that for the A-T base pair.
Reason : There are three hydrogen bonds in the G-C base pair and two in the A-T base pair.
(a) Both [a] and [r] are true and [r] is the correct reason for [a]
(b) [a] is false but [r] is true
(c) Both [a] and [r] are false
(d) Both [a] and [r] are true and [r] is not the correct reason for [a]

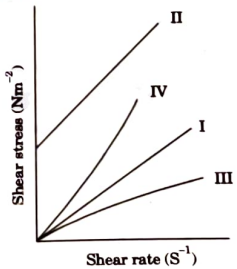
32. Which one of the combinations of the following statements is true about antibody structure?
P. Limited proteolysis of rabbit IgG with the enzyme pepsin generates two antigen-binding regions (Fab) and an Fc fragment
Q. Limited proteolysis of rabbit IgG with the enzyme papain generates a single bivalent antigen-binding region $F(ab')_2$ and peptide fragments
R. The Fc fragment of IgG can self-associate and crystallize into a lattice
S. The $F(ab')_2$ fragment of IgG is composed of both light and heavy chains
(a) P and Q only
(b) P and R only
(c) R and S only
(d) Q and S only

33. Which one of the following statements is true with regard to processing and presentation of protein antigens?
(a) In the class II MHC pathway, protein antigens in the cytosol are processed by proteasomes
(b) In the class I MHC pathway, extracellular protein antigens are endocytosed into vesicles and processed
(c) In the class I MHC pathway, transporter associated antigen processing (TAP) protein is required for translocating processed peptides generated in the cytosol
(d) Invariant chain in endoplasmic reticulum is involved in transport of peptides and loading of class I MHC

34. Which of the following are true about bacterial superoxide dismutase?
P. Present in obligate aerobes
Q. Present in facultative aerobes
R. Present in aerotolerant anaerobes
S. Absent in obligate aerobes
(a) P and Q only (b) P, Q and R only
(c) P and R only (d) Q and S only

35. Which of the following are true with regard to anaerobic respiration in bacteria?
P. The final electron acceptor is an inorganic substance other than molecular oxygen
Q. The number of ATP molecules produced per glucose molecule is more than that produced in aerobic respiration
R. The number of ATP molecules produced per glucose molecule is less than that produced in aerobic respiration
S. Only substrate level phosphorylation is used to generate ATP
(a) P and S only (b) Q and S only
(c) P and R only (d) P, Q and S only

36. Shear stress versus shear rate behaviour of four different types of fluids (I, II, III and IV) are shown in the figure below.



Which one of the following options is correct?

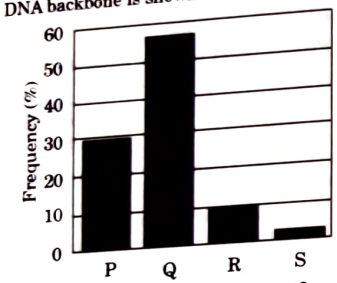
(*a*) I-Newtonian, II-Bingham plastic, III-Dilatant, IV-Pseudoplastic

(*b*) I-Pseudoplastic, II-Dilatant, III-Newtonian, IV-Bingham plastic

(*c*) I-Newtonian, II-Pseudoplastic, III-Bingham plastic, IV-Dilatant

(*d*) I-Newtonian, II-Bingham plastic, III-Pseudoplastic, IV-Dilatant

37. An analysis of DNA-protein interactions was carried out using all DNA-protein complexes in the protein data bank (PDB). The frequency distribution of four amino acid residues, represented as P, Q, R and S, occurring in non-covalent interactions between the protein and DNA backbone is shown below.



Which one of the following is correct?

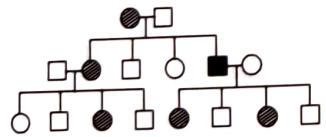
(*a*) P-Lys, Q-Arg, R-Gln, S-Glu

(*b*) P-Gln, Q-Glu R-Lys, S-Arg

(*c*) P-Asn, Q-Asp, R-Arg, S-Lys

(*d*) P-His, Q-Glu, R-Gln, S-Lys

38. A pedigree of an inheritable disease is shown below.



Unaffected Female Carrier Female Affected Female

Unaffected Male Carrier Male Affected Male

What type of inheritance does the disease follow?

(*a*) Autosomal dominant

(*b*) X-linked dominant

(*c*) X-linked recessive

(*d*) Autosomal recessive

39. Match the industrial products mentioned in Group I with their producer organisms in Group II

Group I	Group II
P. Citric acid	1. Trichoderma viride
Q. Cellulase	2. Clostridium acetobutylicum
R. Vitamin B_{12}	3. Aspergillus niger
S. Butanol	4. Propionibacterium freudenreichii

(*a*) P-4, Q-3, R-1, S-2

(*b*) P-3, Q-1, R-2, S-4

(*c*) P-2, Q-1, R-4, S-3

(*d*) P-3, Q-1, R-4, S-2

40. 5′ capping of mRNA transcripts in eukaryotes involves the following events:

P. Addition of GMP on the 5′ end

Q. Removal of γ-phosphate of the triphosphate on first base at the 5′ end

R. 5′-5′ linkage between GMP and the first base at 5′ end

S. Addition of methyl group to N7 position of guanine

Which one of the following is the correct sequence of events?

(*a*) P, Q, R, S (*b*) P, R, Q, S

(*c*) Q, P, R, S (*d*) Q, P, S, R

41. Calculate the following integral (up to two decimal places)

$$\int_0^1 (x+3)(x+1)\,dx = ____$$

42. The probability distribution for a discrete random variable X is given below.

X	1	2	3	4
$P(X)$	0.3	0.4	0.2	0.1

The expectation value of X is (up to one decimal place)____

43. If $1 + r + r^2 + r^3 + \ldots\infty = 1.5$, then $1 + 2r + 3r^2 + 4r^3 + \ldots\infty =$ (up to two decimal places) ____

44. Moist heat sterilization of spores at 121°C followes first order kinetics as per the expression:

$$\frac{dN}{dt} = -k_d N$$

where, N is the number of viable spores, t is the time, k_d is the rate constant and $\frac{dN}{dt}$ is the rate of change of viable spores.

If k_d value is 1.0 min^{-1}, the time (in minutes) required to reduce the number of viable spores from an initial value of 10^{10} to a final of 1 is (up to two decimal places) ____

45. An aqueous solution containing 6.8 mg/L of an antibiotic is extracted with amyl acetate. If the partition coefficient of the antibiotic is 170 and the ratio of water to solvent is 85, then the extraction factor is ____

46. A microbial strain is cultured in a 100 L stirred fermenter for secondary metabolite production. If the specific rate of oxygen uptake is 0.4 h^{-1} and the oxygen solubility in the broth is 8 mg/L, then the volumetric mass transfer coefficient ($K_L a$) (in s^{-1}) of oxygen required to achieve a maximum cell concentration of 12 g/L is (up to two decimal places) ____

47. In a chemostat, the feed flow rate and culture volume are 100 ml/h and 1.0 L, respectively. With glucose as substrate, the values of $_{max}$ and K_s are 0.2h^{-1} and 1 g/L, respectively. For a glucose concentration of to g/L in the feed, the effluent substrate concentration (in g/L) is ____

48. Mammalian cells in active growth phase were seeded at a density of 1×10^5 cell/ml. After 72 hours, 1×10^6 cells/ml were obtained. The population doubling time of the cells in hours is (up to two decimal places) ____

49. Yeast converts glucose to ethanol and carbon dioxide by glycolysis as per the following reaction:

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$$

Assuming complete conversion, the amount of ethanol produced (in g) from 200 g of glucose is (up to two decimal places)____

50. At the end of a batch culture, glucose solution is added at a flow rate of 200 ml/h. If the culture volume after 2 h of glucose addition is 1000 ml, the initial culture volume (in ml) is ____

51. Consider the following alignment of two DNA sequences:

AGTAAC

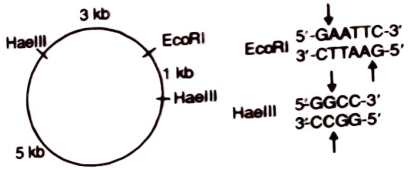
AA--AC

Assuming an affine gap scoring scheme of an identity matrix for substitution, a gap initiation penalty of 1 and a gap extension penalty of 0.1, the score of the alignment is (up to one decimal place) ____

52. First order deactivation rate constants for soluble and immobilized amyloglucosidase enzyme are 0.03 min^{-1} and 0.005 min^{-1}, respectively. The ratio of half-life of the immobilized enzyme to that of the soluble enzyme is (rounded off to the nearest integer)____

53. Consider a simple uni-substrate enzyme that follows Michaelis-Menten kinetics. When the enzyme catalyzed reaction was carried out in the presence of 19 nM concentration of an inhibitor, there was no change in the maximal velocity. However, the slope of the Lineweaver-Burk plot increased 3-fold. The dissociation constant for the enzyme-inhibitor complex (in nM) is ____

54. The product of complete digestion of the plasmid shown below with *Eco*RI and *Hae*III was purified and used as a template in a reaction containing Klenow fragment of DNA polymerase, dNTPs and [α-^{32}P]-dATP in a suitable reaction buffer. The product thus obtained was purified and subjected to gel electrophoresis followed by autoradiography.



The number of bands that will appear on the X-ray film is____

55. A rod shaped bacterium has a length of 2 m, diameter of 1 m and density the same as that of water. If proteins constitute 15% of the cell mass and the average protein has a mass of 50 kDa, the number of proteins in the cell is ____

(1 Da = 1.6×10^{-24}g)

ANSWERS

General Aptitude (GA)

1. (a)	2. (c)	3. (b)	4. (c)	5. (d)	6. (a)	7. (c)	8. (d)	9. (c)	10. (a)

Technical Section

1. (c)	2. (c)	3. (b)	4. (d)	5. (c)
6. (d)	7. (c)	8. (b)	9. (b)	10. (d)
11. (d)	12. (c)	13. (b)	14. (c)	15. (a)
16. (d)	17. (b)	18. (b)	19. (c)	20. (b)
21. (– 10)	22. (0.32)	23. (50)	24. (1)	25. (4096)
26. (d)	27. (a)	28. (d)	29. (c)	30. (a)
31. (b)	32. (c)	33. (c)	34. (b)	35. (c)
36. (d)	37. (a)	38. (c)	39. (d)	40. (c)
41. (5.33)	42. (2.1)	43. (2.25)	44. (23.04)	45. (2)
46. (0.167)	47. (1)	48. (21.67)	49. (102.22)	50. (600)
51. (1.8)	52. (6)	53. (5)	54. (2)	55. (2945000)

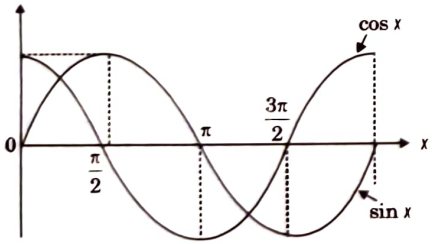
EXPLANATIONS

GENERAL APTITUDE

1. Stares - To look at someone for a long time

   Stairs - A construction designed to bridge a large vertical distance by dividing it into smaller vertical distances, called steps

2. Errant means misbehaving, exhibiting inappropriate behaviour / offending conduct.

3. 

   From the curve it is clear that sin x and cos x both are decreasing in the interval $\left(\frac{\pi}{2}, \pi\right)$

4. Area of equilateral triangle $= \frac{\sqrt{3}}{4}a^2$

   $\therefore \quad \sqrt{3} = \frac{\sqrt{3}}{4}a^2$

   $\therefore \quad a^2 = 4$

   $\therefore \quad a = 2$

   Now perimeter of equilateral triangle $= 3a = 3 \times 2 = 6$

5. (i) Volume of cuboid $= (8 \times 10 \times 6)\ cm^3 = 480\ cm^3$

   (ii) Volume of cube $= (8 \times 8 \times 8)\ cm^3 = 512\ cm^3$

   (iii) Volume of cylinder $= \pi r^2 h$

   $$\left(\frac{22}{7} \times 7 \times 7 \times 7\right) cm^3 = 1078\ cm^3$$

   (iv) Volume of Sphere $= \frac{4}{3}\pi r^3 = \left(\frac{4}{3}\pi \times (7)^2\right) cm^3$

   $= 1436.75\ cm^3$

   Hence, the descending order of their volumes is (iv), (iii), (ii), (i).

6. Average speed $= \frac{2(S_1 \times S_2)}{S_1 + S_2}$

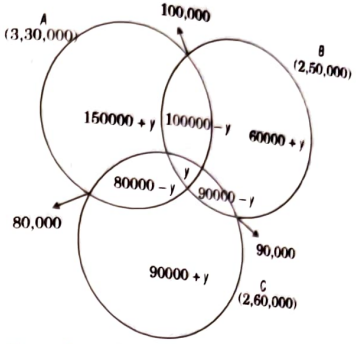
   $= \frac{2(60 \times 90)}{(60 + 90)}$ km/h

   $= \frac{2 \times 60 \times 90}{150} = 72$ km/h

7. Number of parallelogram $= {}^4C_2 \times {}^5C_2$

   $= \frac{4!}{2! \times 2!} \times \frac{5!}{3! \times 2!}$

   $= 6 \times \frac{5 \times 4}{2} = 60$

8. 

   From above diagram,

   $\therefore \quad 6,30,000 = 2y + 1,50,000 + 100000 + 80000 + 60000 + 90000 + 90000$

   $\therefore \quad 63000 = 2y + 57000$

   $\therefore \quad 6,30,000 - 5,70,000 = 2y$

   $\therefore \quad 60000 = 2y$

   $\therefore \quad y = 30000$

   Students who failed to clear the test

   $= 150000 + 60000 + 90000 + 4y$

   $= 300000 + 4 \times 30000 \qquad (\because y = 30000)$

   $= 420000$

   Students who failed to clear the test = 420000

9. $x^2 - 1 = -x$

   $x - \frac{1}{x} = -1$

   $x^2 + \frac{1}{x^2} = 0.2$

   Now, $x^4 + \frac{1}{x^4} = \left(x^2 + \frac{1}{x^2}\right)^2 - 2$

   $= (3)^2 - 2$

   $= 9 - 2$

   $= 7$

BIOTECHNOLOGY

1. Probability of getting head (p) = 0.6

   $\Rightarrow$ Probability of getting tail (q) = 11 – p = 0.4

   Desired outcomes = {HT, TH, HH}

   $= 0.6 \times 0.4 + 0.4 \times 0.6 \times 0.6 \times 0.6$

   $= 0.24 + 0.24 + 0.36$

   $= 0.84$

   $\therefore$ Option (c) is correct.

2. During G1, cells synthesize all of the molecules needed for a new round of cell division. The serum in growth media contains everything needed for this, including growth factors, which are important signals for cell division. Therefore, starving cells of serum can prevent them from dividing.

   Human embryonic kidney cells 293, also often referred to as **HEK 293** or less precisely as **HEK cells**, are a specific cell line originally derived from human embryonic kidney cells grown in tissue culture.

   Cell Cycle Arresting Compounds rely on differing mechanisms of action to prevent the normal progression of the cell cycle. Some of these compounds interfere with CDK/cyclin complexes leaving cells stuck at the G_2/M phase border, while others affect CaMKII phosphorylation, inducing arrest at the G_1 phase. Other mechanisms of action include interference with RNA function and inhibition of protein synthesis. Many of these compounds ultimately induce apoptosis as a result of their interruption of the cell cycle.

   The cell cycle checkpoints play an important role in the control system by sensing defects that occur during essential processes such as DNA replication or chromosome segregation, and inducing a cell cycle arrest in response until the defects are repaired.

3. **Telomeres** are sections of DNA found at the ends of each of our chromosomes. They consist of the same **sequence** of bases **repeated** over and over. In **humans** the **telomere sequence** is TTAGGG. This **sequence** is usually **repeated** about 3,000 times and can reach up to 15,000 base pairs in length.

4. Wherever a gene exists on a DNA molecule, one **strand** is the **coding strand** (or **sense strand** or non-**template strand**), and the other is the noncoding **strand** (also called the **antisense strand**, anticoding **strand**, **template strand**, or transcribed **strand**). Sequence of the transcribed strand is complimentary to the sequence of sense strand. Hence the sequence of mRNA will also be same as that of sense strand (Ts will be replaced with Us).

5. **Botulinum toxin** (BTX) is a neurotoxic protein produced by the bacterium *Clostridiumb* **botulinum** and related species. It prevents the release of the neurotransmitter acetylcholine from axon endings at the neuromuscular junction and thus causes flaccid paralysis. Infection with the bacterium causes the disease botulism.

protein complex secreted by the bacterium *Vibrio cholerae*. CTX is responsible for the massive, watery diarrhea characteristic of cholera infection.

Streptolysin O is a toxin secreted by *Streptococcus pyogenes* and is a prototype member of poreforming bacterial cytolysins. It is used permeabilize cell membranes to permit cellular uptake of large or charged molecules. It is used to study macromolecule delivery. It is a potential anticancer agent and is used to study suicide cancer gene therapy.

Diphtheria toxin is an exotoxin (extracellular protein) secreted by *Corynebacterium diphtheriae*, the pathogenic bacterium that causes diphtheria, inhibits protein synthesis and kills susceptible cells. The gene that encodes DT (tox) is present in some corynephages, and DT is only produced by *C.* diphtheriae isolates that harbor tox$^+$ phages.

6. Molecular mechanics describes the potential energy of a molecule in terms of a set of potential energy equations that are reminiscent of classical mechanics in physics. The potential energy equations to calculate the energy, the "atom types" that define different atoms in a molecule, and the parameters/constants used in the equations are known as a force-field.

   The total potential energy of a molecular system is the sum of individual potential components. Simple molecular mechanics force fields include bond stretching, angle bending, torsion, and van der Waals interactions in their make-up. The sums extend over all bonds, bond angles, torsion angles, and non-bonded interactions in the molecule. Modern force fields add other terms for greater accuracy.

7. BLAST is actually a family of programs (all included in the blastall executable). These include:

   Nucleotide-nucleotide BLAST (blastn)

   This program, given a DNA query, returns the most similar DNA sequences from the DNA database that the user specifies.

   Protein-protein BLAST (blastp)

   This program, given a protein query, returns the most similar protein sequences from the protein database that the user specifies.

   Position-Specific Iterative BLAST (PSI-BLAST) (blastpgp)

   Nucleotide 6-frame translation-protein (blastx)

   This program compares the six-frame conceptual translation products of a nucleotide query sequence (both strands) against a protein sequence database.

   Nucleotide 6-frame translation-nucleotide 6-frame translation (tblastx)

   This program is the slowest of the BLAST family. It translates the query nucleotide sequence in all six possible frames and compares it against the six-frame translations of a nucleotide sequence database. The purpose of tblastx is to find very distant relationships between nucleotide sequences.

   Protein-nucleotide 6-frame translation (tblastn)

   This program compares a protein query against the all six reading frames of a nucleotide sequence database.

   Large numbers of query sequences (megablast)

   When comparing large numbers of input sequences via the command-line BLAST, "megablast" is much faster than running BLAST multiple times. It concatenates many input sequences together to form a large sequence before searching the BLAST database, then post-analyzes the search results to glean individual alignments and statistical values.

8. Anion exchange resins exchange the negative ions readily, so when various peptides with different charges pass through it, then they will be eluted in order from strong negative charge to positive charge. Therefore the order of elution will be R (Strongly negative), followed by Q (Weakly negative) and the last P (Positively charged) peptides.

9. When the van der Waals attraction between two atoms exactly balances the repulsion between their two electron clouds, the atoms are said to be in *van der Waals contact*. Each type of atom has a van der Waals radius at which it is in van der Waals contact with other atoms. The van der Waals radius of an H atom is 0.1 nm, and the radii of O, N, C, and S atoms are between 0.14 and 0.18 nm. Two covalently bonded atoms are closer together than two atoms that are merely in van der Waals contact. For a van der Waals interaction, the internuclear distance is approximately the sum of the corresponding radii for the two participating atoms. Thus the distance between a C atom and an H atom in van der Waals contact is 0.27 nm, and between two C atoms is 0.34 nm. In general, the van der Waals radius of an atom is about twice as long as its covalent radius. For example, a C—H covalent bond is about 0.107 nm long and a C—C covalent bond is about 0.154 nm long.

10. Glycolysis occurs in most organisms in the cytosol of the cell. The most common type of glycolysis is the *Embden–Meyerhof–Parnas (EMP pathway)*.

    In mammalian cells, both peroxisomes and mitochondria contain a beta-oxidative pathway. Beta-oxidation is a key pathway for the breakthrough of fatty acids. In yeasts and plants, fatty acid oxidation occurs uniquely in peroxisomes.

    Conversion of pyruvate to acetyl CoA, the citric acid cycle takes place in the matrix of the mitochondria. Oxidative phosphorylation is a process occurring in the intermembrane space of the mitochondria that results in the formation of ATP from the flow of electrons to oxygen.

11. The main components of the innate immune system are (1) physical epithelial barriers, (2) phagocytic leukocytes, (3) dendritic cells, (4) a special type of lymphocyte called a natural killer (NK) cell, and (5) circulating plasma proteins. Memory B-cells is NOT a principal component of innate immunity

12. Electrospray ionization mass spectrometry (ESI-MS) was employed to monitor the heme release and the conformational changes of myoglobin (Mb) under different solvent conditions, and to observe ligand bindings of Mb. ESI-MS, complemented by circular dichroism and fluorescence spectroscopy, was used to study the mechanism of acid- and organic solvent-induced denaturation by probing the changes in the secondary and the tertiary structure of Mb. The light microscope can be used to provide information about the activity of cells and to look at very small structures such as nanostructures.

13. A trickling filter, is a fixed-bed, biological reactor that operates under (mostly) aerobic conditions. Pre-settled wastewater is continuously 'trickled' or sprayed over the filter. As the water migrates through the pores of the filter, organics are aerobically degraded by the biofilm covering the filter material.

14. Paracrine signaling is a form of cell-to-cell communication in which a cell produces a signal to induce changes in nearby cells, altering the behavior of those cells.

    Autocrine : mode of hormone action to which hormons bind to receptors on to the cell and affects the cell that produces it.

    Paracrine : describes hormone action where hormones are released from cells and bind to receptor on nearby cells and affects their function. Hence the answer is Paracrine.

15. In the presence of physiological concentrations of monovalent and divalent cations, actin monomers polymerize into filaments. The actin filament is structurally polarized and the two ends are not identical.

16. The standard error (SE) of a parameter is the standard deviation of its sampling distribution or an estimate of the standard deviation. If the parameter or the statistic is the mean, it is called the standard error of the mean (SEM).

    The standard error of the sample mean depends on both the standard deviation and the sample size, by the simple relation SE = SD/√(sample size). The standard error falls as the sample size increases, as the extent of chance variation is reduced—this idea underlies the sample size calculation for a controlled trial, for example. By contrast the standard deviation will not tend to change as we increase the size of our sample.

17. In situ hybridization (ISH) is a powerful technique for localizing specific nucleic acid targets within fixed tissues and cells, allowing you to obtain temporal and spatial information about gene expression and genetic loci.

18. Construction of the phylogenetic tree : (1) Distance methods (2) Character methods: (a) Maximum parsimony and (b) Maximum likelihood.

    Distance based methods (eg. Neighbour Joining (NJ)): find a tree such that branch lengths of paths between sequences (species) fit a matrix of pairwise distances between sequences.

    Maximum Parsimony : find a phylogenetic tree that explains the data, with as few evolutionary changes as possible.

    Maximum likelihood : find a tree that maximizes the probability of the genetic data given the tree.

19. $\cos^2 x + 2\cos x + 1 = 0$

Let $\cos x = y$

$\therefore\ y^2 + 2y + 1 = 0$

$\therefore\ (y+1)^2 = 0$

$\therefore\ \boxed{y = -1}$

$\therefore\ \cos x = -1$

$\therefore\ x = \cos^{-1}(-1)$

$\therefore\ x = \pi - \cos^{-1}(1)$

$= \pi - 0$

$\therefore\ \boxed{x = \pi}$

$\therefore\ x = 180°$

$\therefore$ Option (*c*) is correct.

20. **The pyrethrins are a class of organic compounds (mixture of six chemicals) normally derived from Chrysanthemum cinerariifolium flowers that have potent (toxic) insecticidal activity by targeting the nervous systems of insects. Pyrethrins are commonly used to control mosquitoes, fleas, flies, moths, ants, and many other pests.**

21. **The given matrix is** $A = \begin{bmatrix} 4 & -6 \\ -3 & 2 \end{bmatrix}$

$\therefore\ A = 4(2) - (-3)(-6)$

$= 8 - 18$

$= -10$

23. $A = \varepsilon cl$, $l = 2cm$, $A = 0.56$, $\varepsilon = 5600\ m^{-1}cm^{-1}$

$\therefore$ concentration of tryptophan $= A/(\varepsilon l)$

$$= \frac{0.56}{2\times 5600} = 5 \times 10^{-5}\ M$$

$= 50 \times 10^{-6}\ M = 50\ \mu m$

24. **When stem cells undergoes the asymmetric cell divisions then these gives rise to one cell of the same type while another differentiated into a specialized cell types such as neurons etc.**

Thus when one stem cells undergoes 10 asymmetric cell divisions only one cell will have same property of stem cell while the rest cells will have any specified cell type like neurons.

25. DNA have 4 bases = C, G, A, T

Now, restriction enzyme recognizes 6 bp sequence.

So, Avg. length (bp) of fragment generated

$= 4^6 = 4096$ (bp)

26. **A critical step in establishing a successful nitrogen-fixing symbiosis between rhizobia and legume plants is the entrapment of the bacteria between root hair cell walls, usually in characteristic 180° to 360° curls, shepherd's crooks, which are formed by the host's root hairs. Purified bacterial signal molecules, the nodulation factors (NFs), which are lipochitooligosaccharides, induce root hair deformation in the appropriate host legume and have been proposed to be a key player in eliciting root hair curling.**

27. **In essence, all endogenous pyrogens are cytokines, molecules that are a part of the immune system. They are produced by activated immune cells and cause the increase in the thermoregulatory set point in the hypothalamus. Major endogenous pyrogens are interleukin 1 (α and β) and interleukin 6 (IL-6). Minor endogenous pyrogens include interleukin-8, tumor necrosis factor-β, macrophage inflammatory protein-α and macrophage inflammatory protein-β as well as interferon-α, interferon-β, and interferon-γ. Tumor necrosis factor-α also acts as a pyrogen. It is mediated by interleukin 1 (IL-1) release.**

These cytokine factors are released into general circulation, where they migrate to the circumventricular organs of the brain due to easier absorption caused by the blood–brain barrier's reduced filtration action there. The cytokine factors then bind with endothelial receptors on vessel walls, or interact with local microglial cells. When these cytokine factors bind, the arachidonic acid pathway is then activated.

28. **Small nucleolar RNAs (snoRNAs) are a class of small RNA molecules that primarily guide chemical modifications of other RNAs, mainly ribosomal RNAs, transfer RNAs and small nuclear RNAs.**

In animals, small RNA molecules termed PIWI-interacting RNAs (piRNAs) silence transposable elements (TEs), protecting the germline from genomic instability and mutation. piRNAs have been detected in the soma in a few animals, but these are believed to be specific adaptations of individual species.

MicroRNAs (miRNA) regulate the expression of protein-coding genes in animals and plants. They function by binding to mRNA transcripts with complementary sequences and inhibit their expression. The level of sequence complementarity between the microRNA and mRNA transcript varies between animal and plant systems. Owing to this subtle difference, it was initially believed that animal and plant microRNAs act in different ways. Recent developments revealed that, although differences still remain in the two kingdoms, the differences are smaller than fi rst thought. It is now clear that both animal and plant microRNAs mediate both translational repression of intact mRNAs and also cause mRNA degradation.

A small nuclear RNA (snRNA) is one of many small RNA species confined to the nucleus; several of the snRNAs are involved in splicing or other RNA processing reactions. Many snRNAs of eukaryotic cells are involved in the maturation of primary transcripts of messenger RNAs (mRNAs). Certain snRNAs are involved in the formation and function of the spliceosome, a large ribonucleoprotein structure in which pre-mRNA splicing takes place.

29. ***Ab initio* gene finding in eukaryotes, especially complex organisms like humans, is considerably more challenging for several reasons. First, the promoter and other regulatory signals in these genomes are more complex and less well-understood than in prokaryotes, making them more difficult to reliably recognize. Two classic examples of signals identified by eukaryotic gene finders are CpG islands and binding sites for a poly(A) tail.**

Second, splicing mechanisms employed by eukaryotic cells mean that a particular protein-coding sequence in the genome is divided into several parts (exons), separated by non-coding sequences (introns). (Splice sites are themselves another signal that eukaryotic gene finders are often designed to identify.) A typical protein-coding gene in humans might be divided into a dozen exons, each less than two hundred base pairs in length, and some as short as twenty to thirty. It is therefore much more difficult to detect periodicities and other known content properties of protein-coding DNA in eukaryotes.

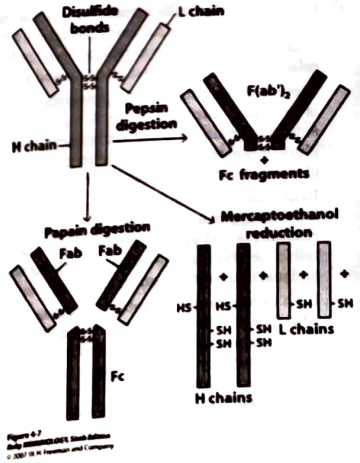
30. **L-arginine-dependent production of reactive nitrogen intermediates (RNIs: nitric oxide, nitrite, and nitrate) by mammalian macrophages has been proposed to occur via an L-arginine oxidative deimination pathway and is known to be responsible for certain antineoplastic and antimicrobial effect or functions.**

31. **Assertion** : The association constant in water for the G-C base pair is three times lower than that for the A-T base pair.

Reason : There are three hydrogen bonds in the G-C base pair and two in the A-T base pair.

Assertion is false. Reason is correct. The "reason statement" is itself an explanation why the assertion is false.

32. **Porter fragmented rabbit IgG with the proteolytic enzyme papain in the presence of the reducing agent cysteine. Papain hydrolyzed peptide bonds in IgG and produced three fragments (I, II, and III). The three fragments had similar molecular weights (50 kD) but different charges. Two of the three fragments were identical and retained the ability to bind antigen. These two fragments were called the Fab fragments (for fragments of antigen-binding). Because the intact IgG is bivalent and the two Fab fragments each could bind antigen, Porter concluded that the Fab fragments must be univalent. The third fragment produced by papain digestion did not bind with antigen and crystallized during cold storage. Porter called this piece the Fc fragment (for fragment crystallizable). Thus the ratio of Fab to Fc is 2:1. Edelman confirmed Porter's results by cleaving and electrophoresing human IgG into two antigenically different fractions equivalent to the two fragments from rabbit IgG (Figure 4-2). In similar studies, Alfred Nisonoff used pepsin, which hydrolyzes different sites on the IgG molecule than does papain. IgG treated with pepsin yielded one large fragment with a molecular weight (100 kD) double that of one Fab fragment, and many small fragments. Nisonoff called the large fragment F(ab9)2. This fragment also could bind antigen, but unlike the Fab fragment, it led to a visible serologic reaction. It had both of the antigen-binding sites of IgG (the chains remained linked) and could be treated further with reducing agents to yield two Fab-like fragments called Fab9. Collectively, the two enzymes cleave at about the same region of the IgG molecule. Papain splits the molecule on one side, and pepsin on the other side, of the bond that holds Fab fragments together.**



33. Endogenous antigens are processed by cytosolic pathway. In the cytoplasm antigen is first processed by proteasomes and then smaller peptides are transported inside endoplasmic reticulum by TAP protein.

Protein antigens in the cytosol are processed by proteasomes in the class I MHC pathway.

In the class II MHC pathway, extracellular protein antigens are endocytosed into vesicles and processed.

Invariant chain in endoplasmic reticulum is involved in transport of peptides and loading of class II MHC pathway.

34. Distribution of superoxide dismutase, catalase and peroxidase in prokaryotes with different O_2 tolerances.

Group	Superoxide dismutase	Catalase	Peroxidase
Obligate aerobes and most facultative anaerobes (e.g. Enterics)	Yes	Yes	No
Most aerotolerant anaerobes (e.g. Streptococci)	Yes	No	Yes
Obligate anaerobes (e.g. Clostridia, Methanogens, Bacteroides)	No	No	No

35. For aerobic, the cells involved include those in most organisms and body cells; however, anaerobic may occur in muscle cells and red blood cells, and is some types of bacteria and yeast.

Lactic acid is produced during Anaerobic, and none is produced during aerobic. A high amount of energy is produced in aerobic respiration with 38 glucose molecules, and low energy with only 2 glucose molecules during anaerobic.

In addition, the reactants for aerobic respiration is both oxygen and glucose, yet for anaerobic the reactant is just glucose. The reaction site in the cell for aerobic is in the cytoplasm or mitochondria, and just in the cytoplasm for anaerobic respiration.

Both aerobic and anaerobic respiration involves a first stage called glycolysis. The additional stage in anaerobic is fermentation, but in aerobic there is the Krebs cycle and electron transfer chain. Finally, combustion is complete in aerobic respiration, and incomplete during anaerobic respiration.

The following are true with regard to anaerobic respiration in bacteria:

(P) The final electron acceptor is an inorganic substance other than molecular oxygen

(R) The number of ATP molecules produced per glucose molecule is less than that produced in aerobic respiration

36. Curve II is bingham plastic follows the τ

$$= \tau_0 + k\frac{du}{dy}$$

Now, for rest type i.e. I, III, IV they follows the

$$\tau = k\left(\frac{du}{dy}\right)^n$$

n = flow behaviour index

Now if n = 1 then fluid is newtonian which is curve I.

If n < 1 then fluid is pseudoplastic which is curve III.

If n > 1 then fluid is Dilatant which is curve IV.

Now, curve for newtonian (I) will be standard line passing through origin.

Curve for pseudoeplastic (III) will be concave downward passing through origin.

Curve for Dilatant (IV) will be concave upward through passing through origin.

37. DNA has net negative charge due to the presence of phosphate groups. Lys, Arg and Gln are basic amino acids whereas Glu is acidic amino acid. Arg (Q) attracts more phosphate and forms extensive hydrogen bonding with its five H-bond donors present in its structure that stabilizes the interactions in DNA-protein complexes reflecting the highest percentage frequency in the graph. Followed by Arg the next higher frequency is shown for Lys (P) due to its basic nature and hydrogen bonding capacity which is followed by the frequency of Gln (R). Glu (s) due to negative charge has least frequency.

38. It can't be autosomal dominant as carriers are not possible in such cases. It can't be X-linked dominant. If that was the case, all female carriers actually would have been affected. It can't be autosomal recessive as affected children in such cases are born when both the parents are carriers. Answer is X-Linked Recessive as most of the following traits of X-linked recessive inheritance are fulfilled by the given pedigree chart:

1. More males than females are affected
2. Affected sons are usually born to unaffected mothers, thus the trait skips generations
3. Approximately 1/2 of carrier mothers' sons are affected
4. It is never passed from father to son
5. All daughters of affected fathers are carriers

39.

Product	Producing Organism
Citric acid	*Aspergillus niger*
Cellulase	*Trichoderma viride*
Vitamin B12	*Propionibacterium freudenreichii*
Butanol	*Clostridium acetobutylicum*

40. The starting point for capping with 7-methylguanylate is the unaltered 5′ end of an RNA molecule, which terminates at a triphosphate group. This features a final nucleotide followed by three phosphate groups attached to the 5′ carbon. The capping process is initiated before the completion of transcription, as the nascent pre-mRNA is being synthesized.

1. One of the terminal phosphate groups is removed by RNA triphosphatase, leaving a bisphosphate group (i.e. 5′(ppN)$[pN]_n$);
2. GTP is added to the terminal bisphosphate by mRNA guanylyltransferase, losing a pyrophosphate from the GTP substrate in the process. This results in the 5′–5′ triphosphate linkage, producing 5′(Gp)(ppN)$[pN]_n$;
3. The 7-nitrogen of guanine is methylated by mRNA (guanine-N7-)-methyltransferase, with S-adenosyl-L-methionine being demethylated to produce S-adenosyl-L-homocysteine, resulting in 5′(m7Gp) (ppN)$[pN]_n$ (cap-0);
4. Cap-adjacent modifications can occur, normally to the first and second nucleotides, producing up to 5′(m7Gp) (ppN*) (pN*)$[pN]_n$ (cap-1 and cap-2)
5. If the nearest cap-adjacent nucleotide is 2′-O-ribose methyl-adenosine (i.e. 5′(m7Gp) (ppAm)$[pN]_n$), it can be further methylated at the N6 methyl position to form N^6-methyladenosine, resulting in 5′(m7Gp) (ppm6Am)$[pN]_n$.

41. $\int_0^1 (x+3)(x+1)dx$

$$= \int_0^1 \left[x^2+4x+3\right]dx$$

$$= \left[\frac{x^3}{3}+\frac{4x^2}{2}+3x\right]_0$$

$$= \left[\frac{1}{3}+2+3\right]$$

$$= 5.33$$

42.

x	1	2	3	4
$p(x)$	0.3	0.4	0.2	0.1

$E(x) = 1 \times p(x=1) + 2 \times p(x=2) + 3 \times p(x=3) + 4 \times p(x=4)$

$= 1 \times 0.3 + 2 \times 0.4 + 3 \times 0.2 + 4 \times 0.1$

$= 0.3 + 0.8 + 0.6 + 0.4$

$= 2.1$

43. The given series 1, r, r^2, r^3, ... ∞ is infinite G.P. series with common ratio $p(r)$

Now $S_n = \frac{a}{1-r}$

$\therefore \quad 1.5 = \frac{1}{1-r}$

$\therefore \quad 1.5 - 1.5r = 1$

$0.5 = 1.5r$

$\therefore \quad r = \frac{1}{3}$

$\Rightarrow \quad 1+\frac{1}{3}+\frac{1}{9}+...\infty = 1.5 \qquad ...(i)$

Now sum of $1 + 2r + 3r^2 + ... + \infty$

$= 1+2\left(\frac{1}{3}\right)+3\left(\frac{1}{9}\right)+...\infty$

$= 1.5 \times 1.5 \qquad (\because$ from eq. $(i))$

$= 2.25$

44. $\ln(N_0/N_t) = K_d t$

Here Given, $N_0 = 10^{10}$; $N_t = 1$; $K_d = 1\ min^{-1}$

Therefore $t = \ln(10^{10}/1) = 23.03$ min.

45. (Separation factor) Extraction factor is one distribution ratio divide by another, Here, partition coefficient of antibiotic is 170, and distribution ratio of water to solvent is 85.

$\therefore$ Extraction factor

$$= \frac{\text{Partition coefficient of antibiotic}}{\text{Distribution coefficient of water to solvent}}$$

$$= \frac{170}{85} = 2$$

46. $q_{o_2} = 0.4\ h^{-1}$

$= 0.4/3600 = 1.111 \times 10^{-4}\ sec^{-1}$

$x = 12$ g/L

$\therefore \quad OUR = q_{o_2} X = 1.333 \times 10^{-3}$

$\therefore \quad \Delta C = 8\ mg/L = 8 \times 10^{-3}\ gm/L$

$\therefore \quad OUR = k_{La}\Delta C$

$\Rightarrow \quad K_{La} = \frac{1.333 \times 10^{-3}}{8 \times 10^{-3}} = 0.167$

47. $$S = \frac{k_s D}{\mu - D}$$

$$D = \text{Dilution rate} = \frac{F}{V} = \frac{100\text{ ml}}{1000\text{ ml}} = 0.1$$

$$\therefore \quad S = \frac{1 \times 0.1}{0.2 - 0.1} = 1$$

48. $$\ln\left(\frac{x_1}{x_o}\right) = n \ln 2,$$

We know that number of cycles $n = t/t_d$

t = time given = 72 hrs

t_d = doubling time

$x_1 = 10^6$ cells/ml

$x_o = 10^5$ cells/ml

So, $$t_d = \frac{t \ln 2}{\ln\left(\frac{x_1}{x_o}\right)} = \frac{72 \times 0.6931}{2.303} = 21.67 \text{ hrs}$$

49. $$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$$

1 mol of glucose produces → 2 mole of ethanol

MW of glucose = 180, MW of ethanol = 46,

As per equation 180 gm glucose produces

= 2 × 46 i.e., 92 gm of ethanol

Hence, 1 gm of glucose will produce 92/180 gm of ethanol

Therefore 200 gm glucose will produce

= 200 × (92/180) gm ethanol

= 102.22 gm ethanol

50. Let x = initial culture volume in ml.

glucose addition = 200 ml/h, so in 2 hr total 400 ml glucose added.

Final culture volume = 1000 = x + 400

$\Rightarrow$ x = 600 ml

51. For, affine gap scoring scheme, penalty is calculated using following formula = n (A + B)

where n = number of gaps

A = gap initiation penalty = 1

B = gap extension penalty = 0.1

Total gap penalty = 2(1 + 0.1) = 2.2

Now in the given structure the total number of matches = 4.

∴ Score of Alignment = Match score – Total gap penalty

= (4 × 1) – 2.2 = 4 – 2.2 = 1.8

52. Data for soluble enzyme :

$K_1 = 0.03$, half life = $t_1^{1/2}$

Data for immobilized enzyme

$K_2 = 0.005$, half life = $t_2^{1/2}$

$$\therefore \quad \frac{\ln c_1/c_0}{\ln c_2/c_0} = \frac{K_1 t_1^{1/2}}{K_2 t_2^{1/2}}$$

but $$\frac{\ln c_1/c_0}{\ln c_2/c_0} = 1$$

Because final concentration is half of the initial concentration for both types of enzymes. Therefore

$$\therefore \quad \frac{t_1^{1/2}}{t_2^{1/2}} = \frac{K_1}{K_2}$$

$$\therefore \quad \frac{t_1^{1/2}}{t_2^{1/2}} = \frac{0.03}{0.005}$$

$$\therefore \quad \frac{t_1^{1/2}}{t_2^{1/2}} = 6$$

53. Slope = K_m/V_m, as V_m = constant

$$K_{m_2} = 3K_{m_1}$$

Now, $$K_{m_2} = K_{m_1}\left[1 + \frac{[I]}{K_I}\right] = 3K_{m_1}$$

$\Rightarrow$ $2K_1 = [I] = 10$ nM

$\Rightarrow$ $K_1 = 5$ nM

54. From the diagram, we observe that one of the enzymes (EcoRI) used is a sticky end enzyme and the other (HaeII) is a blunt end cutter. For incorporation of the radioactive ATP, Klenow fragment needs a free 3'-OH end and a template to read. This will only be provided at the sticky ends generated by EcoRI enzyme. EcoRI will generate two such ends on two different DNA fragments. Hence we will finally see two bands on the X-ray film.

55. $S = 1$ kg/l

$l = 2 \times 10^{-6}$ mt

$D = 1 \times 10^{-6}$ mt

Volume of bacterium = $\frac{\pi}{4} D^2 l$

$= 1.5708 \times 10^{-18}$ m^3

$= 1.5708 \times 10^{-15}$ kg

[$\because$ 1 m^3 = 1000 kg]

$= 1.5708 \times 10^{-12}$ gm

15% of this is proteins = 2.356×10^{-13} gm.

Now, average weight of one protein

= 50 kDa

$= 50 \times 1000 \times 1.6 \times 10^{-24}$

$= 8 \times 10^{-20}$ gm.

$$\therefore \quad \text{No. of proteins} = \frac{2.356 \times 10^{-13}}{8 \times 10^{-20}}$$

$= 2.945 \times 10^6$

= 2945000

SECTION-I: GENERAL ABILITY

Q. No. 1 to 5 Carry One Mark Each

1. Until Iran come along. India had never been _____ in kabaddi.

(a) defeated (b) defeating
(c) defeat (d) defeatist

2. The fishermen ,_____the flood victims owed their lives, were rewarded by the government

(a) whom (b) to which
(c) to whom (d) that

3. The radius as well as the height of a circular cone is increased by 10%. The percentage increase in its volume is _____.

(a) 17.1 (b) 21.0
(c) 33.1 (d) 72.8

4. Five numbers 10, 7, 5, 4, 2 are arranged in a sequence from left to right following the directions given below:

(1) No two odd or even numbers are next to each other.

(2) The second number from left is exactly half of the left -most number.

(3) The middle number is exactly twice the right most number.

Which is the second number from the right ?

(a) 2 (b) 4
(c) 7 (d) 10

5. "Some students were not involved in the strike". If the above statement is true, which of the following conclusions is/are logically necessary ?

1. Some who were involved in strike were students.
2. No student was involved in the strike.
3. At least one student was involved in the strike.
4. Some who were not involved in the strike were students.

(a) 1 and 2 (b) 3
(c) 4 (d) 2 and 3

Q. No. 6 - 10 Carry Two Marks Each

6. "I read somewhere that in ancient times the prestige of a kingdom depended upon the number of taxes that it was able to levy on its people. It was very much like the prestige of a head-hunter in his own community".

Based on the paragraph above, the prestige of a head-hunter depended upon ____

(a) the prestige of the heads
(c) the number of taxes he could levy
(d) the number of head she could gather

7. Two trains started at 7 AM from the same point. The first train travelled towards north at a speed of 80 km/h and the second train travelled south at a speed of 100 km/h. The time at which they were 540 km apart is _____ AM.

(a) 9 (b) 10
(c) 11 (d) 11:30

8. In a country of 1400 million population, 70% own mobile phones. Among the mobile phone owners only 294 million access the internet. Among these Internet users, only half buy goods from e-commerce portals. What is the percentage of these buyers in the country ?

(a) 10.50 (b) 14.70
(c) 15.00 (d) 50.00

9. The nomenclature of Hindustani music has changed over centuries. Since the medieval period *dhrupad* styles were identified as *baanis*. Terms like *gayaki* and *baaj* were used to refer to vocal and instrumental styles, respectively. With institutionalization of music education the term *gharana* became acceptable. *Gharana* originally referred to hereditary musicians from a particular lineage, including disciples and ground disciples. Which one of the following pairings is NOT Correct ?

(a) Dhrupad, baani (b) Gayaki, Vocal
(c) Baaj, institution (d) Gharana, lineage

10. Since the last one year, after a 125 basis point reduction in repo rate by the Reserve Bank of India, banking institutions have been making a demand to reduce interest rates on small savings schemes. Finally, the government announced yesterday a reduction in interest rates on small saving schemes to bring them on par with fixed deposit interest rates.

Which one of the following statements can be inferred from the given passage ?

(a) Whenever the Reserve Bank of India reduces the repo rate, the interest rates on small saving schemes are also reduced.

(b) Interest rates on small saving schemes are always maintained on par with fixed deposit interest rates.

(c) The government sometimes takes into consideration the demands of banking institutions before reducing the interest rates on small saving schemes.

(d) A reduction in interest rates on small saving schemes follow only after a reduction in repo rate by the Reserve Bank of India.

SECTION-II: BIOTECHNOLOGY

Q. No. 1 to 25 Carry One Mark Each

1. The mass of 1 kmol of oxygen molecules is _____ g (rounded off to the nearest integer).

2. Which one of the following is a database of protein sequence motifs?

(a) PROSITE (b) TrEMBL
(c) SWISSPROT (d) PDB

3. The median value for the dataset (12, 10, 16, 8, 90, 50, 30, 24)is _____

4. Which one of the following enzymes is encoded by human immunodeficiency virus (HIV) genome?

(a) Reverse transcriptase
(b) Phospholipase
(c) Phosphatase
(d) ATP synthase

5. Match the human diseases in Group I with the causative agents in Group II

Group I	Group II
P. Amoebiasis	1. Leishmania donovani
Q. African sleeping sickness	2. Trypanosoma cruzi
R. Kala azar	3. Entamoeba histolytica
S. Chagas' disease	4. Trypanosoma gambiense

(a) P – 3, Q – 4, R – 2, S – 1
(b) P – 3, Q – 2, R – 1, S – 4
(c) P – 3, Q – 4, R – 1, S – 2
(d) P – 4, Q – 3, R – 1, S – 2

6. Which one of the following need NOT be conserved in a biochemical reaction?

(a) Total mass
(b) Total moles
(c) Number of atoms of each element
(d) Total energy

7. Which of the following are geometric series?

P. 1,6,11,16,21,26,....
Q. 9,6,3,0,–3,–6,...
R. 1,3,9,27,81,....
S. 4,–8,16,–32,64,....

(a) P and Q only (b) R and S only
(c) Q and S only (d) P, Q and R only

8. DNA synthesis in eukaryotes occurs during which phase of the mitotic cell cycle?

(a) M (b) G_1
(c) S (d) G_0

9. The degree of reduction for acetic acid ($C_2H_4O_2$) is _____.

10. Which one of the following is used as a pH indicator in animal cell culture medium?

(a) Acridine orange (b) Phenol red
(c) Bromophenol blue (d) Coomassie blue

11. Which one of the following is NOT a part of the human nonspecific defense system?

(a) Interferon (b) Mucous
(c) Saliva (d) Antibody

12. The solution of $\lim_{x \to 8} \left(\frac{x^2 - 64}{x - 8} \right)$ is _____

13. Which one of the following is the unit of heat transfer coefficient?

(a) Wm^2K^{-1} (b) $Wm^{-2}K$
(c) $Wm^{-2}K^{-1}$ (d) Wm^2K

14. A mutation in a gene that codes for a polypeptide results in a variant polypeptide that lacks the last three amino acids. What type of mutation is this?

(a) Synonymous mutation
(b) Nonsense mutation
(c) Missense mutation
(d) Silent mutation

15. Which one of the following statements is CORRECT for enzyme catalyzed reactions? (ΔG is Gibbs free energy change, K_{eq} is equilibrium constant)

(a) Enzymes affect ΔG, but not K_{eq}
(b) Enzymes affect K_{eq}, but not ΔG
(c) Enzymes affect ΔG and K_{eq}
(d) Enzymes do not affect ΔG or K_{eq}

16. Protein concentration of a crude enzym preparation was 10mg mL^{-1}. 10µL of this samp gave an activity of 5µ mol min^{-1} under standa assay conditions. The specific activity of this cru enzyme preparation is _____ units mg^{-1}

17. Which one of the following is catabolized duri endogenous metabolism in a batch bacteri cultivation?

(a) Internal reserves
(b) Extracellular substrates
(c) Extracellular products
(d) Toxic substrates

18. The Bt toxin gene from Bacillus thuringiensis used to generate genetically modified crops is
(a) cry (b) cro
(c) cdc (d) cre

19. Which one of the following can NOT be a limiting substrate if Monod's growth kinetics is applicable?
(a) Extracellular carbon source
(b) Extracellular nitrogen source
(c) Dissolved oxygen
(d) Intracellular carbon source

20. Which one of the following equations represents a one-dimensional wave equation?
(a) $\frac{\partial u}{\partial t} = C^2 \frac{\partial^2 u}{\partial x^2}$
(b) $\frac{\partial^2 u}{\partial t^2} = C^2 \frac{\partial^2 u}{\partial x^2}$
(c) $\frac{\partial^2 u}{\partial t^2} = C^2 \frac{\partial u}{\partial x}$
(d) $\frac{\partial^2 u}{\partial t^2} + \frac{\partial^2 u}{\partial x^2} = 0$

21. Which of the following processes can increase genetic diversity of bacteria in nature?
P. Conjugation Q. Transformation
R. Transduction S. Transfection
(a) P only (b) P and Q only
(c) P, Q and R only (d) P,Q,R and S

22. Matrix $A = \begin{bmatrix} 0 & 6 \\ p & 0 \end{bmatrix}$ will be skew-symmetric when p = _____.

23. Tetracycline inhibits the
(a) Interaction between tRNA and mRNA
(b) Translocation of mRNA through ribosome
(c) Peptidyl transferase activity
(d) Binding of amino-acyl tRNA to ribosome

24. The number of possible rooted trees in a phylogeny of three species is _____

25. Which one of the following techniques can be used to compare the expression of a large number of genes in two biological samples in a single experiment?
(a) Polymerase chain reaction
(b) DNA microarray
(c) Northern hybridization
(d) Southern hybridization

Q. No. 26 to 55 Carry Two Marks Each

26. What is the solution of the differential equation $\frac{dy}{dx} = \frac{x}{y}$, with the initial condition, at x = 0, y = 1?
(a) $x^2 = y^2 + 1$ (b) $y^2 = x^2 + 1$
(c) $y^2 = 2x^2 + 1$ (d) $x^2 - y^2 = 0$

27. For site-directed mutagenesis, which one of the following restriction enzymes is used to digest methylated DNA?
(a) KpnI (b) DpnI
(c) XhoI (d) M1uI

28. A new game is being introduced in a casino. A player can lose Rs. 100, break even, win Rs.100 or win Rs. 500. The probabilities (P(X)) of each of these outcomes (X)are given in the following table:

X (in Rs.)	– 100	0	100	500
P (X)	0.25	0.5	0.2	0.05

The standard deviation for the casino payout is Rs. _____ (rounded off to the nearest integer)

29. A UV-visible spectrophotometer has a minimum detectable absorbance of 0.02. The minimum concentration of a protein sample that can be measured reliably in this instrument with a cuvette of 1 cm path length is _____ μM. The molar extinction coefficient of the protein is 10,000 L mol^{-1} cm^{-1}.

30. The molecular mass of a protein is 22 kDA. The size of the cDNA (excluding the untranslated regions)that codes for this protein is _____ kb (rounded off to 1 decimal place).

31. Yeast biomass $C_6H_{10}O_3N$ grown on glucose is described by the stoichiometric equation given below:
$C_6H_{12}O_6 + 0.48\ NH_3 + 3O_2 \rightarrow 0.48\ C_6H_{10}O_3N + 3.12\ CO_2 + 4.32\ H_2O$
The amount of glucose needed for the production of 50gL^{-1} of yeast biomass in a batch reactor with a working volume of 100000 L is _____kg (rounded of to the nearest integer).

32. Match the instruments in Group I with their corresponding measurements in Group II.

Group I	Group II
P. Manometer	1. Agitator speed
Q. Rotameter	2. Pressure difference
R. Tachometer	3. Cell number
S. Haemocytometer	4. Air flow rate

(a) P – 4, Q – 1, R – 2, S – 3
(b) P– 3, Q – 4, R – 1, S – 2
(c) P – 2, Q – 4, R – 1, S – 3
(d) P – 2, Q – 1, R – 4, S – 3

33. Which one of the following is coded by the ABO blood group locus in the human genome?
(a) Acyl transferase
(b) Galactosyltransferase
(c) Transposase
(d) β-Galactosidase

34. Which one of the following covalent linkages exists between 7-Methyl guanosine (m^7G) and mRNAs?
(a) 2′ – 3′ triphosphate
(b) 3′ – 5′ triphosphate
(c) 5′ – 5′ triphosphate
(d) 2′ – 5′ triphosphate

35. Group I lists spectroscopic methods and Group II lists bimolecular applications of these methods. Match the methods in Group I with the applications in Group II

Group I
P. Infrared
Q. Circular Dichroism
R. Nuclear Magnetic Resonance

Group II
1. Identification of functional groups
2. Determination of secondary structure
3. Estimation of molecular weight
4. Determination of 3-D structure

(a) P – 4, Q – 3, R – 1
(b) P – 2, Q – 1, R – 3
(c) P –1 , Q – 2, R – 4
(d) P – 3, Q – 2, R – 4

36. In a cross-flow filtration process, the pressure drop (ΔP)driving the fluid flow is 2 atm, inlet feed pressure (P_i)is 3 atm and filtrate pressure (P_f) is equal to atmospheric pressure. The average transmembrane pressure drop (ΔP_m) is _____ atm.

37. Which one of the following amino acid residues will destabilize an α – helix when inserted in the middle of the helix?
(a) Pro (b) Val
(c) Ile (d) Leu

38. The Laplace transform of the function $f(t) = t^2 + 2t + 1$ is
(a) $\frac{1}{s^3} + \frac{3}{s^2} + \frac{1}{s}$ (b) $\frac{4}{s^3} + \frac{4}{s^2} + \frac{1}{s}$
(c) $\frac{1}{s^3} + \frac{2}{s^2} + \frac{1}{s}$ (d) $\frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s}$

39. In pea plants, purple color of flowers is determined by the dominant allele while white color is determined by the recessive allele. A genetic cross between two purple flower-bearing plants results in an offspring with white flowers. The probability that the third offspring from these parents will have purple flowers is _____ (rounded off to 2 decimal places)

40. Which of the following statements are CORRECT when a protein sequence database is searched using the BLAST algorithm?
P. A larger E-value indicates higher sequence similarity
Q. E-value $< 10^{-10}$ indicates sequence homology
R. A higher BLAST score indicates higher sequence similarity
S. E-value > 10^{10} indicates sequence homology
(a) P, Q and R only (b) Q and R only
(c) P, R and S only (d) P and S only

41. Which of the following statements are CORRECT about the function of fetal bovine serum in animal cell culture?
P. It stimulates cell growth
Q. It enhances cell attachment
R. It provides hormones and minerals
S. It maintains pH at 7.4
(a) P and Q only
(b) P and S only
(c) P, Q and R only
(d) P,Q,R and S

42. $\int_{-1}^{1} f(x)dx$ Calculated using trapezoidal rule for the values given in the table is _____ (rounded off to 2 decimal places)

x	-1	$-\frac{2}{3}$	$-\frac{1}{3}$	0	$\frac{1}{3}$	$\frac{2}{3}$	1
f(x)	0.37	0.51	0.71	1.0	1.40	1.95	2.71

43. The hexapeptide P has an isoelectric point (pI) of 6.9. Hexapeptide Q is a variant of P that contains valine instead of glutamate at position 3. The two peptides are analyzed by polyacrylamide gel electrophoresis at pH 8.0. Which one of the following statements is CORRECT?
(a) P will migrate faster than Q towards the anode
(b) P will migrate faster than Q towards the cathode
(c) Both P and Q will migrate together
(d) Q will migrate faster than P towards the anode

44. Which one of the following statements is CORRECT about proportional controllers?
(a) The initial change in control output signal is relatively slow
(b) The initial corrective action is greater for larger error
(c) They have no offset
(d) There is no corrective action if the error is a constant

45. In general, which one of the following statements is NOT CORRECT?
(a) Hydrogen bonds result from electrostatic interactions
(b) Hydrogen bonds contribute to the folding energy of proteins
(c) Hydrogen bonds are weaker than van der Waals interactions
(d) Hydrogen bonds are directional

46. Phenolic wastewater discharged from an industry was treated with Pseudomonas sp, in an aerobic bioreactor. The influent and effluent concentration of phenol were 10,000 and 10 ppm, respectively. The inlet feed rate of wastewater was 180L h^{-1}. The kinetic properties of the organism are as follows:

Maximum specific growth rate $(\mu_m) = 1h^{-1}$

Saturation constant $(K_s) = 100$ mg L^{-1}

Cell death rate $(k_d) = 0.01$ h^{-1}

Assuming that the bioreactor operates under 'chemostat' mode, the working volume required for this process is _____ L (rounded off to the nearest integer)

47. The difference in concentrations of an uncharged solute between two compartments is 1.6-fold. The energy required for active transport of the solute across the membrane separating the two compartments is_____cal mol-1 (rounded off to the nearest integer). (R = 1.987 cal mol^{-1} K^{-1}, T = 37°C)

48. The dimensions and operating condition of a lab-scale fermentor are as follows:

Volume = 1 L

Diameter = 20 cm

Agitator speed = 600 rpm

Ratio of impeller diameter to fermentor diameter = 0.3

This fermentor needs to be scaled up to 8,000 L for a large scale industrial application. If the scale-up is based on constant impeller tip speed, the speed of the agitator in the larger reactor is _____rpm. Assume that the scale-up factor is the cube root of the ratio of fermentor volumes.

49. Which of the following factors affect the fidelity of DNA polymerase in polymerase chain reaction?
P. Mg^{2+} concentration
Q. pH
R. Annealing temperature
(a) P and Q only (b) P and R only
(c) Q and R only (d) P, Q and R

50. Match the organelles in Group I with their functions in Group II.

Group I	Group II
P. Lysosome	1. Digestion of foreign substances
Q. Smooth ER	2. Protein targeting
R. Golgi apparatus	3. Lipid synthesis
S. Nucleolus	4. Protein synthesis
	5. rRNA synthesis

(a) P – 1, Q – 3, R – 2, S – 5
(b) P – 1, Q – 4, R – 5, S – 3
(c) P – 2, Q – 5, R – 3, S – 4
(d) P – 1, Q – 3, R – 4, S – 5

51. Which of the following statements is ALWAYS CORRECT about an ideal chemostat?
P. Substrate concentration inside the chemostat is equal to that in the exit stream
Q. Optimal dilution rate is lower than critical dilution rate
R. Biomass concentration increases with increase in dilution rate
S. Cell recirculation facilitates operation beyond critical dilution rate
(a) P and Q only (b) P, R and S only
(c) P and S only (d) P, Q and S only

52. Determine the correctness or otherwise of the following Assertion [a] and the Reason [r]
Assertion [a] : It is possible to regenerate a whole plant from a single plant cell.
Reason [r] : It is easier to introduce transgenes in to plants than animals.
(a) Both [a] and [r] are true and [r] is the correct reason for [a]
(b) Both [a] and [r] are true but [r] is not the correct reason for [a]
(c) Both [a] and [r] are false
(d) [a] is true but [r] is false

53. An industrial fermentor containing 10,000 L of medium needs to be sterilized. The initial spore concentration in the medium is 10^6 spores mL^{-1}. The desired probability of contamination after sterilization is 10^{-3}. The death rate of spores at 121° C is $4min^{-1}$. Assume that there is no cell death during heating and cooling phases. The holding time of the sterilization process is_____min (rounded off to the nearest integer).

54. Antibody-producing hybridoma cells are generated by the fusion of a
(a) T cell with a myeloma cell
(b) B cell with a myeloma cell
(c) Macrophage with a myeloma cell
(d) T cell and a B cell

55. Which of the following factors can influence the lag phase of a microbial culture in a batch fermentor?
P. Inoculum size
Q. Inoculum age
R. Medium composition
(a) P and Q only
(b) Q and R only
(c) P and R only
(d) P, Q and R

ANSWERS

Section - A: General Ability

1. (a)	2. (c)	3. (c)	4. (c)	5. (c)	6. (d)	7. (b)
8. (a)	9. (c)	10. (c)				

Section - B: Biotechnlogy

1. (31800 to 32000)		2. (a)	3. (20 to 20)	4. (a)	5. (c)	6. (b)
7. (b)	8. (c)	9. (4 to 4)	10. (b)	11. (d)	12. (16 to 16)	13. (c)
14. (b)	15. (d)	16. (50 to 50)	17. (a)	18. (a)	19. (d)	20. (b)
21. (b)	22. (–6 to –6)	23. (d)	24. (3 to 3)	25. (b)	26. (b)	27. (b)
28. (128 to 130)		29. (2 to 2)	30. (6 to 0.6)	31. (12500 to 14500)		32. (c)
33. (b)	34. (c)	35. (c)	36. (1 to 1)	37. (a)	38. (d)	39. (0.75 to 0
40. (b)	41. (c)	42. (2.2 to 2.5)	43. (a)	44. (b)	45. (c)	46. (970 to 1
47. (280 to 295)		48. (30 to 30)	49. (d)	50. (a)	51. (d)	52. (b)
53. (9 to 10)	54. (b)	55. (d)				